Supervised by:
Dr. Mohammed Sharkawy

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Prepared by: Dr. Omar Naser

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Table of Contents

TOPIC: ANIMAL AND HUMAN BITES 3

TOPIC: CAT SCRATCH DISEASE 10

TOPIC: POST-EXPOSURE PROPHYLAXIS 11

TOPIC: CHICKEN POX 12

TOPIC: SHINGLES 21

TOPIC: Post-herpetic neuralgia: 27

TOPIC: LONG-TERM EFFECT OF CORONAVIRUS 28

TOPIC: INFLUENZA VACCINATION 29

TOPIC: PNEUMOCOCCAL VACCINE 30

TOPIC: GLANDULAR FEVER 'INFECTIOUS MONONUCLEOSIS' 31

TOPIC: NOROVIRUS 36

TOPIC: LEGIONELLA 37

TOPIC: MYCOPLASMA PNEUMONIAE 38

TOPIC: WHOOPING COUGH 39

TOPIC: RESPIRATORY PATHOGENS 44

TOPIC: TRAVELLERS DIARRHEA 45

TOPIC: GASTROENTERITIS 52

TOPIC: TRAVEL ADVICE AND IMMUNIZATION 64

TOPIC: PRE-TRAVEL COUNSELING 65

TOPIC: GONORRHEA 66

TOPIC: CHALYMDIA – UNCOMPLICATED GENITAL 73

TOPIC: SYPHILIS 77

TOPIC: GENITAL WARTS 83

TOPIC: HUMAN PAPILLOMA VIRUS (HPV) VACCINE 85

TOPIC: HIV INFECTION AND AIDS 87

TOPIC: HAND FOOT AND MOUTH 100

TOPIC: PARVOVIRUS B19 INFECTION 104

TOPIC: MUMPS 109
TOPIC: Measles 113
TOPIC: RUBELLA 114
TOPIC: MENINGITIS 118
TOPIC: SCARLET FEVER 125
TOPIC: SEPSIS 129
TOPIC: LYME DISEASE 130
TOPIC: MALARIA 137
TOPIC: MALARIA PROPHYLAXIS 143
TOPIC: LEISHMANIASIS 151
TOPIC: LEPTOSPIROSIS 152
TOPIC: SCHISTOSOMIASIS 153
TOPIC: ORF 155
TOPIC: BED BUGS 156
TOPIC: BODY FLUID SPILLAGES 157
TOPIC: DENGUE FEVER 158
TOPIC: CUTANEOUS LARVA MIGRANS 159
TOPIC: BRUCELLOSIS 160
TOPIC: SPLENECTOMY 161
TOPIC: STRONGYLOIDES STERCORALIS 162
TOPIC: TRYPANOSOMIASIS 163
TOPIC: ZIKA VIRUS 164
TOPIC: ANTIBIOTIC GUIDELINES 165
TOPIC: CLASSIFICATION OF BACTERIA 168

TOPIC: ANIMAL AND HUMAN BITES

- A bite is an injury inflicted by the teeth of a human or animal.
    - Bite wounds can take a number of forms including lacerations, abrasions, puncture wounds, avulsions, and crush or degloving injuries.
- The most common mammalian bites are associated with dogs, cats, and humans.

What are the characteristics of human, dog, and cat bites?

- **Human** bites are either:
    - Occlusal injuries — inflicted by actual biting, or
    - Clenched-fist injuries ('fight bites') — sustained when a clenched fist hits a person's teeth, often during a fight, causing small-looking wounds over the dorsum of the hand or metacarpophalangeal joints, which may have damaged underlying structures.
- Most human bites occur on the hand, most often from a fist fight:
    - Men are most frequently bitten on the hand, arm, and shoulder.
    - Women are most frequently bitten on the breast, genitalia, leg, and arm.
- Human bites may be purposeful or incidental, and occur in a number of situations.
    - Deliberate human bites can occur during fights, child abuse, and sexual crimes. Young children may bite to express stress, aggression, or frustration.
    - Accidental bites are associated with sports, school activities, consensual sexual activity, and occupational injury (dental health professionals).
    - Bites can also be self-inflicted and are sometimes found in people with emotional disturbance or rare genetic conditions such as Lesch-Nyhan syndrome.
- **Dog** bites characteristically involve puncture wounds from the canine teeth, which anchor the victim whilst the other teeth bite, shear, and tear the tissues, causing structural damage:
    - Adult dogs have very powerful jaws. These can inflict high pressures, which, in addition to the shearing forces delivered during a bite, can cause deep, open cuts, crush wounds, devascularization, bone damage, and soft-tissue avulsion. In some cases, a body part may be torn off by a dog bite.
    - The severity of a dog bite depends on the type and size of dog, with breeds such as Rottweilers, German Shepherds, and pit bull terriers being able to cause significant damage. Bites from these breeds may be more ragged and avulsive in nature and are more likely to need reconstructive surgery.
- **Cats** have fine, sharp teeth and, despite having a weaker bite than dogs, inflict narrow deep puncture wounds inoculated with saliva, and are capable of penetrating bone, joints, and tendons.
    - Fluids can only drain from the small skin puncture wounds, therefore infections such as abscesses and osteomyelitis are more common with cat bites.

What are the complications of bites?

- **Bacterial infection**
    - Bacterial infection is a risk if there is a break in the skin. Skin and soft tissue infection is the most common complication following a bite injury. Infective complications resulting from a bite wound include cellulitis, abscesses, tenosynovitis, septic arthritis, osteomyelitis, necrotizing fasciitis, and systemic spread (for example sepsis, meningitis, endocarditis, and organ abscesses). The longer an infected bite wound is untreated, the more likely severe local and systemic complications are to occur.
        - Human bites are most commonly infected by Streptococcus spp, Staphylococcus aureus, Haemophilus spp, Eikenella corrodens, Bacteroides spp, and other anaerobes.

        - Most animal bite wound infections are polymicrobial and contain a mixture of aerobic and anaerobic organisms. Pasteurella multocida is commonly isolated from cat bites and can cause severe, fast-spreading infections in humans. S. aureus, Pasteurella canis, and P. multocida are among the species affecting dog bites. Capnocytophaga canimorsus infection is rare after a dog bite but can have severe effects (particularly in elderly or immunocompromised people) such as sepsis, meningitis, disseminated intravascular coagulation, and endocarditis.
    - The infection risk is related to the type of bite wound, its site, the person's individual risk factors, and the species causing the bite.
        - The risk of infection is high in deep or contaminated wounds; injuries with significant tissue destruction; in areas of poor perfusion; bites affecting the hands, feet, face, and genitals; and where there is bone, joint, or tendon involvement.
        - Human bites are known to be at risk of infection and subsequent complications, with an infection rate of 10 to 50%. Clenched fist injuries or 'fight bites' are particularly prone to infection.
        - Cat bites are around twice as likely as dog bites to become infected (infection rate of 30–50% for cats, 5–25% for dogs) because of the deep inoculation of bacteria into the puncture wounds caused by their small, sharp teeth.
    - **Tetanus** is a disease caused by infection with Clostridium tetani which releases tetanus toxin. The condition is characterized by generalized rigidity and muscle spasms that can be fatal. Tetanus can occur after an animal bite or human bite, especially in puncture wounds or those containing devitalized tissue, dirt, or foreign bodies. However, tetanus after a human bite is extremely rare.
    - **Cat-scratch disease** is caused by Bartonella henselae and may follow a bite or scratch from a cat or dog. Cat-scratch disease presents with a mild infection at the wound site 3–14 days after the injury, followed by lymphadenopathy and symptoms including fever, malaise, headache, and poor appetite. Lymph glands near the scratch become swollen, and swelling may persist for several months. Occasionally the eyes, brain, heart, and other organs can be affected.
    - **Syphilis** transmission via a human bite has also been reported.
- **Viral infections**
    - There is a possibility of viral transmission through human bite wounds (for example hepatitis, HIV, and herpes simplex virus), although the risk is low and depends on the nature of the bite, the status of the biter and the injured person, and the presence of blood in the biter's saliva.
    - **Rabies** is an acute viral encephalomyelitis that is almost always fatal. It usually occurs after a person is bitten or scratched by an animal with rabies (most commonly a dog). The risk of contracting rabies in the UK is extremely low, as there is no indigenous rabies in terrestrial animals, but bats anywhere in the world can carry rabies-like viruses, therefore are considered to be a rabies risk.
- **Structural damage**
    - Disfiguring wounds are often a complication of severe bites or mauling
    - Traumatic amputation of body parts (for example fingers) can result from a severe bite.
    - Tendons and joint capsule damage can occur during a clenched-fist human bite injury.
    - Bite wounds can also cause fractures and neurovascular damage.
        - There are reports of fatal exsanguination from dog bites to the neck.
- **Psychological effects**
    - There may be psychological distress or increased anxiety after a bite injury in both children and adults and permanent scarring and disfigurement may lead to

depression and decreased self-esteem. Bite injuries can lead to avoidance behaviours and post-traumatic stress disorder (PTSD).

What is the prognosis?

- The outcome of a bite wound is dependent on the size and severity of the wound and the person's individual risk factors.
    - Minor bites generally have a good prognosis with appropriate treatment.
    - The potential for infection, scarring, or disfigurement is greater with more severe bite wounds and correlates with a lack of proper medical attention and inadequate wound care.
- Death resulting from dog bites is rare.

Management:

Scenario: Managing a human bite

From age 1 month onwards

How should I assess someone with a human bite?

- **Document how and when the bite occurred.**
    - Who was bitten, and by whom.
    - Whether the skin was broken or blood was involved.
    - The nature of the bite (i.e. occlusal or clenched fist).
- **Monitor vital signs** if the bite is particularly traumatic, or sepsis is suspected.
- **Examine the bite using gloves,** bearing in mind that deep layers of tissue may move with positional changes after the bite injury, disguising the true depth of the wound. Document the following (record both positive and negative findings as there may be future litigation):
    - The location of the wound. Photographs or diagrams may be useful.
    - The size and depth of the injury.
        - An intercanine distance of less than 3 cm suggests a child bite; more than 3 cm suggests an adult bite.
    - The type of wound (for example laceration, puncture, abrasion, crush, haematoma, avulsion, or amputation).
    - The degree of crush injury, devitalized tissue, nerve or tendon damage, and involvement of muscle, bones, joints, or blood vessels.
        - Examine wounds overlying a joint through the full range of motion to detect retracted injuries and tendon rupture.
    - Neurovascular function in the area distal to the bite — check pulses and sensation.
    - The range of movement of any adjacent joints.
    - Any lymphadenopathy.
    - The presence of any foreign bodies (for example teeth) in the wound.
    - Any signs of infection (for example redness, swelling, induration, fluctuance, necrotic tissue, purulent discharge, pain, localized cellulitis, lymphangitis, lymphadenopathy, or fever).
    - Facial bites: perform an intraoral examination to exclude cheek lacerations with an intraoral communication.
- **Determine whether the person is at increased risk of the wound becoming infected,** due to:
    - The nature of the bite (deep, contaminated wounds; puncture or crush wounds; or significant tissue destruction).

    - Site of injury (for example hands, feet, face, or genitals; areas of poor perfusion or lymphatic return; or near a prosthetic joint implant).
    - Wound penetrating bone, joints, tendons, or vascular structures.
    - Delayed presentation (more than 8 hours).
    - Associated medical conditions (for example diabetes mellitus, asplenia, immunocompromised status, chronic liver disease, or prosthetic heart valve or joint).
    - Their age (neonates, infants, and elderly people are at higher risk of infection).
- **Although rare, suspect child maltreatment** if there is a report or appearance of a human bite marks that is thought unlikely to have been caused by a young child. Also consider safeguarding issues if a vulnerable adult receives a bite injury.
    - Refer for further assessment in line with local policy.
- Assess the person's risk of tetanus and blood borne virus infection.

How should I assess the risk of tetanus and blood-borne virus infection?

- **To assess the risk of tetanus:**
    - Enquire about tetanus immunization status.
    - Determine whether the injury is considered to be a tetanus-prone wound; for example, there is:
        - A need for surgical intervention which is delayed for more than 6 hours.
        - A significant amount of devitalized tissue
        - A puncture wound (especially if in contact with soil or manure).
        - Contamination with material likely to contain tetanus spores, such as soil or manure.
        - A foreign body in the wound.
        - A compound-fracture.
        - Systemic sepsis.
- **To assess the risk of acquiring a blood-borne viral infection** (for example hepatitis B or C, or HIV):
    - Assess the status of the person who has been bitten (and the biter if possible):
        - Check their vaccination status for hepatitis B (and surface antibody response if applicable).
        - Ask if they are known to be HIV positive, hepatitis B surface antigen (HBsAg) positive, or hepatitis C positive.
        - Note that if the bite has not broken the skin there is no risk of blood-borne virus transmission.
    - Offer to test for Hepatitis B, Hepatitis C, and HIV.
        - If a child has received a bite, discuss their risk with a specialist (for example a consultant virologist) as it may be possible to avoid testing and the associated anxiety of venepuncture.
    - In most cases the status of the biter will not be known and it is often not practical to obtain a blood sample for testing, but if appropriate, and informed consent is given, the biter should also have their blood tested. Theoretically the biter is at higher risk given they will be exposed to the injured person's blood, whereas the person who has been bitten would usually be exposed to saliva only, unless the biter also has blood in their mouth.

How should I manage a human bite wound?

- **For initial wound management:**
    - If possible, remove any foreign bodies (for example teeth) from the wound.

    - Encourage the wound to bleed (if it has just occurred), unless it is already bleeding freely.
    - Irrigate thoroughly with warm, running water or normal saline.
    - Consider the need for debridement (for example, if the wound is dirty or there is non-viable tissue). Refer to Accident and Emergency if this is required and the skills and resources are not available in primary care.
    - Advise analgesia (paracetamol or ibuprofen) for pain relief, if required.
    - Where body tissue has been torn off as a result of a bite, wrap any torn-off parts (for example, part of an ear) in clean tissue and store in a plastic bag surrounded by ice for transport to hospital.
    - Cover the wound with an appropriate clean dressing.
- **Consider whether closure of the wound is appropriate.**
    - Referral to Accident and Emergency for further assessment and management is usually indicated if wound closure is thought to be necessary. Note the types of bites for which referral is always recommended.
        - If the expertise and resources are available in primary care, seek specialist advice before attempting closure. Types of wounds that may be considered for closure include uncomplicated wounds with no risk factors for infection (for example, those presenting early which are not heavily contaminated, have been adequately irrigated and debrided, and do not involve underlying structures).
    - Allow the following bite wounds to heal without formal closure:
        - Bite wounds over 24 hours.
        - Infected bite wounds.
        - Deep puncture wounds.
        - Crush injuries.
        - Heavy contamination.
        - Uncertain adequacy of debridement.
        - Bites to the limbs, hands, and feet.
- **Do not offer antibiotic prophylaxis to people with a human bite that has not broken the skin.**
- **Offer antibiotic prophylaxis to people with a human bite that has broken the skin and drawn blood.**
- **Consider antibiotic prophylaxis for people with a human bite that has broken the skin but not drawn blood if it:**
    - Involves a high-risk area such as the hands, feet, face, genitals, skin overlying cartilaginous structures, or an area of poor circulation.
    - Is in a person at risk of a serious wound infection because of a comorbidity (such as diabetes, immunosuppression, asplenia, or decompensated liver disease).
- **If antibiotic prophylaxis is indicated,** prescribe co-amoxiclav for 3 days first line for children aged over one month and adults. If there is a penicillin allergy, or co-amoxiclav is unsuitable:
    - For adults and young people aged 12 to 17 years, prescribe metronidazole plus doxycycline for 3 days.
    - For children aged under 12 years, prescribe co-trimoxazole for 3 days.
- **If the wound is infected:**
    - Send pus or a deep wound swab for culture, before cleaning the wound. State on the form that the swab is from an infected human bite.
    - Treat empirically with oral antibiotics if there are symptoms or signs of infection, including increased pain, inflammation, fever, discharge, or an unpleasant smell emanating from the wound.
    - Admit anyone who has a severe infection or who is systemically unwell as intravenous antibiotics may be required.

- **For treatment of an infected bite,** prescribe co-amoxiclav for 5 days first line for children aged over one month and adults. If there is a penicillin allergy, or co-amoxiclav is unsuitable:
    - For adults and young people aged 12 to 17 years, prescribe metronidazole plus doxycycline for 5 days.
    - For children aged under 12 years, prescribe co-trimoxazole for 5 days.
- **Give follow-up advice** if the person has not been referred to hospital.
    - If the bite wound is not infected — advise the person to check for signs of infection and if these develop to attend urgently for review. Consider arranging a routine review within 24 to 48 hours.
    - If the wound is infected — review at 24 and 48 hours to ensure the infection is responding to treatment. Advise the person to attend urgently for review if the infection worsens or if they feel increasingly unwell.
    - For follow-up testing of people who are at risk of a blood-borne virus.
- **Reassess the bite if**:
    - Symptoms or signs of infection develop or worsen rapidly or significantly at any time, or do not start to improve within 24 to 48 hours of starting treatment.
    - The person becomes systemically unwell.
    - The person has severe pain that is out of proportion to the infection.
- Be aware that people who have difficulty communicating may have nonverbal signs of pain, such as a change in behaviour.
- If a skin swab has been sent for microbiological testing, review the choice of antibiotic based on the swab results. If a change of antibiotic is needed, use a narrow-spectrum antibiotic if possible.

How should I manage a person's risk of tetanus and blood-borne viruses?

- **To manage a person's risk of tetanus:**
    - These guide health professionals in the management of tetanus-prone wounds and advise on the need for vaccine booster and/or human tetanus immunoglobulin, which may be required depending on age, vaccination status, and the level of risk of tetanus.
- **To manage a person's risk of blood-borne viruses:**
    - Seek immediate advice from a consultant in infectious diseases, a virologist, or the local Health Protection Team for anyone considered to be at risk of HIV or hepatitis B from a bite wound. Consider all people to be at risk unless the current status of the biter is known (rare).

Tetanus

- The risk of tetanus from a human bite is very low. A reinforcing dose of the tetanus vaccine and possibly further doses may be required if a person is not fully immunized. Usually, tetanus immunoglobulin (TIG) is not required if a person is fully immunized, but the need for TIG will depend on the person's immunization status and whether the wound is high-risk.

Blood-borne viruses

- **Hepatitis B** — post-exposure prophylaxis for hepatitis B may include giving one or more doses of hepatitis B vaccine with or without hepatitis B immunoglobulin depending on the level of exposure, the hepatitis B status of the biter, and the vaccination status of the person with the bite injury.

- **Hepatitis C** — there is no post-exposure prophylaxis available for hepatitis C virus, but serial testing will be required after the bite incident. Only after this can the person be reassured, they have not acquired hepatitis C.
- **HIV** — post-exposure prophylaxis for HIV is not usually required after a human bite because the risk of HIV transmission is very low. It may be considered in some circumstances (after seeking specialist advice) if; for example, either person is known to be HIV positive and the bite draws blood, or is deep.

Scenario: Managing a cat or dog bite

From age 1 month onwards

How should I assess someone with a dog or cat bite? (as above)

How should I manage a person's rabies risk?

- **To manage a person's risk of rabies:**
  - The need for treatment will be determined and organized by a specialist based on the person's rabies risk assessment.
  - Treatment recommendations are grouped into the following categories:
    - No risk, therefore no treatment.
    - Vaccine and human rabies immunoglobulin (HRIG).
    - Vaccine only.
    - Vaccine, HRIG, and blood test with the last dose of vaccine.
    - Observation of animal (domestic cats and dogs only).
  - If treatment is required but has not been started, post-exposure treatment should be commenced as soon as possible, ideally within 24 hours of contact. Initiating post-exposure treatment is not considered a medical emergency, and rabies vaccine/human rabies immunoglobulin (HRIG) can usually be sent out to be given the next day in most cases. However, for high-risk exposures, such as severe and multiple bites to the head and neck bites or from a confirmed rabid animal, treatment should ideally be started within 12 hours of reporting.
  - The mainstay of post-exposure treatment is the rabies vaccine given by intramuscular injection at 0, 3, 7, and 21 days. HRIG may provide short-term immunity in the first week following the initiation of treatment.
  - It is important to note that vaccination and human rabies immunoglobulin should never be given at the same anatomical site. Vaccine doses are given in alternate arms.

Prescribing information:

Co-trimoxazole

What dosage of co-trimoxazole should I prescribe?

- **For treatment of an infected bite wound** as an alternative first-choice oral antibiotic for children under 12 years where co-amoxiclav cannot be used, the dosage of oral co-trimoxazole is:
  - 6 weeks to 5 months: 120 mg(2.5ml)
  - 6 months to 5 years: 240 mg(5ml)
  - 6 years to 11 years: 480 mg(10ml)
- **For prophylaxis** of a bite wound, the dosage is as described above but prescribed for 3 days.

TOPIC: CAT SCRATCH DISEASE

Cat scratch disease is an infection following the scratch of a cat (usually a kitten) with the organism *Bartonella henselae*

Presentation:

- **Erythematous lesion**
- Crusted papules
- After 1-3 weeks, **regional lymphadenopathy**
- Fever, malaise, headache, anorexia (often in immune-compromised patients)
- Rare; parinaud oculogralndualr syndrome, neuroretenitis

Investigation:

- CBC, ESR, CRP
- **Indirect fluorescent antibody (IFA) for** *Bartonella*

Management:

- Supportive therapy with antipyretics and analgesics
- Antibiotics are indicated in immune-compromised patients and atypical cases involving severe or systemic disease.
- In Kuwait all should be treated and given rabies vaccines.
- **Trimethoprim-sulfamethoxazole**, ciprofloxacin or azithromycin are used first-line, with gentamicin being reserved for the severely ill patient.
- Patients should follow up in **2-6 months** for confirmation of symptom resolution.

- Lymphadenitis; azithromycin 500 mg first day then 250 mg OD x 4 days or Clarithromycin 500 mg BD x 7 days.

TOPIC: POST-EXPOSURE PROPHYLAXIS

Hepatitis A

- Human Normal Immunoglobulin (HNIG) or hepatitis A vaccine may be used depending on the clinical situation

Hepatitis B

- HBsAg positive source
  - if the person exposed is a known responder to the HBV vaccine, then a booster dose should be given
  - if they are a non-responder (anti-HBs < 10mIU/ml 1-2 months post-immunization) they need to have hepatitis B immune globulin (HBIG) and a booster vaccine
- unknown source
  - for known responders the HBV vaccine the Green Book advises considering a booster dose of HBV vaccine
  - for known non-responders HBIG + vaccine should be given whilst those in the process of being vaccinated should have an accelerated course of HBV vaccine

Hepatitis C

- Monthly PCR - if seroconversion then interferon +/- ribavirin

HIV

- The risk of HIV transmission depends heavily on the incident (e.g. needle stick, type of sexual intercourse, human bite etc) and the current viral load of the patient
- A combination of oral antiretrovirals (e.g. Tenofovir, emtricitabine, lopinavir and ritonavir) as soon as possible (i.e. Within 1-2 hours, but may be started up to 72 hours following exposure) for 4 weeks
- Serological testing at 12 weeks following completion of post-exposure prophylaxis
- Reduces risk of transmission by 80%

Estimates of transmission risk for single needlestick injury:

Virus	Change of transmission
Hepatitis B	20-30%
Hepatitis C	0.5-2%
HIV	0.3%

TOPIC: CHICKEN POX

- Chickenpox is an acute, infectious disease caused by varicella-zoster virus. It is characterized by an itchy vesicular rash, which is often preceded by fever and malaise.

How is chickenpox transmitted?

- Varicella is very infectious; up to 90% of susceptible close contacts develop the disease.
- Transmission is by direct contact with varicella lesions or droplet spread, with an incubation period (the time from becoming infected until symptoms appear) of 1–3 weeks. This may be prolonged if the person is taking steroids, is immunocompromised, or has received varicella-zoster immunoglobulin (VZIG).
    - Varicella infection of the newborn can occur due to transplacental transmission, ascending infection from the vagina, or from direct contact with lesions during or after delivery.
- Chickenpox is infectious from 24 hours before the rash appears until the vesicles are dry or have crusted over, usually 5 days after the onset of the rash (this period may be longer in people who are immunocompromised).
- Once the infection has subsided, the virus persists in sensory nerve root ganglia. Years or decades later, it can reactivate and cause herpes zoster (shingles).
- It is possible to develop chickenpox after exposure to a person with shingles, but it is not thought possible to develop shingles from exposure to a person with chickenpox.

What are the complications?

- Adults, pregnant women, neonates, and immunocompromised people are more susceptible to serious complications.

Complications in children

- **Chickenpox is usually a self-limiting disease in healthy children. however, complications may occur, including:**
    - Secondary bacterial infection of the skin and soft tissues (for example impetigo, furuncles, cellulitis, erysipelas, necrotizing fasciitis) and scarring.
        - Secondary bacterial skin infections (most often caused by group A Streptococcus (GAS) and *Staphylococcus aureus*) may present with sudden high-grade fever (often after initial improvement), erythema, and tenderness surrounding the original chickenpox lesions.
    - Neurological complications (for example Reye's syndrome, acute cerebellar ataxia, encephalitis, meningoencephalitis, polyradiculitis, myelitis).
    - In rare cases, myocarditis, glomerulonephritis, appendicitis, pancreatitis, Henoch–Schönlein purpura, orchitis, arthritis, vasculopathy, optic neuritis, iritis, and keratitis.

Complications in adults

- **Chickenpox can be more serious in adults than in children,** and adults with varicella are more likely to be admitted to hospital.
- **Complications include pneumonia, hepatitis, and encephalitis.**
    - Primary viral pneumonia is the most common complication in adults.
    - Smokers are particularly at risk of fulminating varicella pneumonia.
- **Shingles can occur from reactivation of latent varicella-zoster infection.**

Complications in pregnancy

- **Complications for the mother:**
  - Varicella in pregnancy can result in severe chickenpox. The mother is at increased risk of varicella pneumonia and other complications (including hepatitis and encephalitis), compared with the general adult population.
- **Complications for the fetus:**
  - Infection with varicella-zoster during the first 28 weeks of pregnancy can lead to intrauterine infection and fetal varicella syndrome (previously known as congenital varicella syndrome), which is characterized by one or more of:
    - Skin scarring in a dermatomal distribution.
    - Eye defects (for example microphthalmia, chorioretinitis, and cataracts).
    - Hypoplasia of the limbs.
    - Neurological abnormalities (microcephaly, cortical atrophy, learning difficulties, dysfunction of bowel and bladder sphincters).

Complications in neonates

- **Neonates are at increased risk of disseminated or haemorrhagic varicella.**

Complications in immunocompromised people

- Severe disseminated chickenpox with haemorrhagic complications is more common in immunocompromised people (including those on high dose corticosteroids) than in other population groups.
- Immunocompromised people with chickenpox are also at increased risk of pneumonia, encephalitis, disseminated intravascular coagulopathy, and hepatitis.

What is the prognosis?

- For most children, chickenpox infection is usually a self-limiting, relatively mild disease without complications.
- Severe disease and complications are more likely to occur in children younger than 1 year of age, adolescents, adults, pregnant women, and immunocompromised people.
- Risk to the fetus and neonate from maternal varicella infection is related to the time of infection in the mother.
- Recovery from primary varicella infection usually leads to life long immunity — recurrence of varicella infection in otherwise healthy people has been reported but is uncommon.
  - Recurrence may be more likely in people who are immunocompromised.

Diagnosis:

When should I suspect chickenpox?

- **In most cases, the diagnosis can be made clinically from the characteristic chickenpox rash.**
  - If there is doubt, a history of recent exposure to chickenpox (or shingles), or cases occurring in close contacts, may help confirm the diagnosis.
- **Take a history asking about:**
  - Typical features of chickenpox, including:

        - A prodrome that includes nausea, myalgia, anorexia, and headache (particularly adolescents and adults).
        - General malaise, loss of appetite, and feeding problems.
        - Rash.
    - Recent exposure to chickenpox (or shingles).
    - Previous chickenpox infection or vaccination.
    - Risk factors for severe disease and complications such as:
        - Pregnancy.
        - Immunosuppression (including high dose corticosteroid use).
        - Chronic skin or respiratory disease.
    - Symptoms suggestive of complications such as shortness of breath, cough, chest pain, persistent or recurrent fever, reduced urine output, confusion or reduced level of consciousness.
- **Examine the person:**
    - Look for fever and signs of severe illness.
    - Look for rash:
        - Small, erythematous macules appear on the scalp, face, trunk, and proximal limbs, which progress over 12–14 hours to papules, clear vesicles (which are intensely itchy), and pustules.
        - Vesicles can also occur on the palms and soles, and mucous membranes can also be affected, with painful and shallow oral or genital ulcers.
        - Vesicles appear in crops; stages of development of the rash can therefore differ on different areas of the body.
    - Look for signs of complications such as secondary bacterial infection of skin lesions, pneumonia, and encephalitis.
        - Prolonged or recurrent fever is suggestive of secondary infection.
    - Be aware that:
        - Adults may experience a more widespread rash and more prolonged fever than children.
        - Immunosuppressed people with chickenpox may present with atypical rash and more extensive lesions (which may be haemorrhagic).
- **Laboratory tests can be used for confirmation, but are rarely required in primary care.**
    - Confirmation of infection may be required, for example, where there are implications for vulnerable contacts — discuss with a specialist if unsure.

What else might it be?

- Chickenpox is usually easy to distinguish from other rashes. However, differential diagnoses include:
    - **Other vesicular viral rashes,** such as:
        - Herpes simplex (not usually disseminated).
        - Herpes zoster (usually unilateral and localized to dermatomes).
        - Hand, foot, and mouth disease (caused by Coxsackie virus).
    - **Other infections,** such as:
        - Impetigo.
        - Scabies.
        - Syphilis.
        - Meningococcaemia (can be confused with haemorrhagic varicella).
        - Toxic shock syndrome.
    - **Skin disorders,** such as:
        - Guttate psoriasis.
        - Drug eruption.
        - Insect bites.

        - Papular urticaria.
        - Erythema multiforme.
        - Stevens–Johnson syndrome.
        - Henoch–Schönlein purpura.
        - Dermatitis herpetiformis.

Management:

From age 2 months onwards

How should I manage an otherwise healthy child or adult with chickenpox?

- If serious complications (such as pneumonia, encephalitis, dehydration, or severe secondary bacterial infection of the skin) are suspected, admit to hospital.
- Consider prescribing oral aciclovir 800 mg 5 times a day for 7 days for an immunocompetent, non-pregnant adult or adolescent (aged 14 years or older) with chickenpox who presents within 24 hours of rash onset, particularly for people with severe chickenpox or those at increased risk of complications, such as smokers.
    - Aciclovir is not recommended for otherwise healthy children with uncomplicated chickenpox.
- Offer symptomatic treatment.
- Give advice about contact with other people and when to seek medical advice.
    - If the person develops a high temperature (particularly after initial improvement) with redness and pain surrounding the chickenpox lesions, consider bacterial superinfection and manage accordingly.

What advice should I give an adult or child with chickenpox?

- **Advise the following simple measures to help alleviate symptoms:**
    - Encourage adequate fluid intake to avoid dehydration.
    - Dress appropriately to avoid overheating or shivering.
    - Wear smooth, cotton fabrics.
    - Keep nails short to minimize damage from scratching and secondary bacterial infection from scratching.
- **Advise that the most infectious period is from 24 hours before the rash appears,** but infectivity continues until all the lesions are dry and have crusted over (usually about 5 days after the onset of the rash).
    - During this time, advise a person with chickenpox to avoid contact with:
        - People who are immunocompromised (for example those receiving cancer treatment or high doses of oral steroids, or those with conditions that reduce immunity).
        - Pregnant women.
        - Infants aged 4 weeks or less.
    - Advise that children with chickenpox should be kept away from school or nursery until all the vesicles have crusted over.
- **Inform the person to seek urgent medical advice if their condition deteriorates or they develop complications.** Parents of young children with chickenpox should be particularly aware of:
    - Bacterial superinfection — manifesting as sudden high-grade pyrexia (often after initial improvement), erythema and tenderness surrounding the original chickenpox lesions.

- Dehydration — encourage and monitor fluid intake and seek medical attention if signs of dehydration develop (for example reduced urine output, lethargy, cool peripheries, reduced skin turgor).

How should I treat symptoms in an adult or child with chickenpox?

- Consider offering:
  - Paracetamol if pain or fever are causing distress (avoid nonsteroidal anti-inflammatory drugs). Note that oral paracetamol is not licensed for use in children less than 2 months of age.
  - Topical calamine lotion to alleviate itch.
  - Chlorphenamine for treating itch associated with chickenpox for people 1 year of age or older.

Avoidance of NSAIDs

- The recommendation on avoidance of nonsteroidal anti-inflammatory drugs (NSAIDs) is based on concerns that use of NSAIDs in varicella may be associated with an increased risk of severe skin and soft tissue infections.

Calamine lotion

- Calamine lotion is thought to relieve itch by evaporating from the skin to induce a cooling effect.

Antihistamines

- There is very limited evidence to support the use of topical or systemic antihistamines in relieving pruritus in those with chickenpox. Piriton is licensed for the symptomatic relief of itching due to chickenpox.

Scenario: Pregnant woman:

From age 12 years to 60 years (Female)

How should I manage a pregnant woman with chickenpox?

- **Admit to hospital (preferably somewhere with access to specialists in obstetrics, infectious diseases, and paediatrics) if a pregnant woman has suspected chickenpox and any of:**
  - Respiratory symptoms.
  - Neurological symptoms.
  - Haemorrhagic rash or bleeding.
  - Severe disease (for example dense rash with or without numerous mucosal lesions).
  - Significant immunosuppression (including recent use of systemic corticosteroids).
- **Consider/discuss the need for admission with a specialist if other risk factors for severe illness and** complications **are present such as:**
  - Pregnancy approaching term.
  - Previous obstetric complications or risk factors.
  - Smoking.
  - Chronic lung disease.

    - Social risk factors.
    - Close monitoring in the community is not possible.
- **For all other pregnant women with chickenpox, seek immediate specialist advice from an obstetrician regarding further management** such as antiviral treatment and outpatient follow up for the fetus.
- **Offer** symptomatic treatment.
- **Give** advice **about contact with other people and when to seek medical advice.**

- Guidance from the UK Health Security Agency (UKHSA) advises that all women exposed to chicken pox should be offered anti-viral post-exposure prophylaxis.
- Intravenous aciclovir is indicated if chickenpox is severe or there are any complications.

What advice should I give a pregnant woman with chickenpox?

How should I treat symptoms in a pregnant woman with chickenpox?

- Consider offering:
    - Paracetamol if pain or fever are causing distress (avoid nonsteroidal anti-inflammatory drugs).
    - Topical calamine lotion to alleviate itch.
- Chlorphenamine is not recommended for the management of the itch of chickenpox in pregnancy.

Scenario: Breastfeeding woman

From age 12 years to 60 years (Female)

How should I manage a breastfeeding woman with chickenpox?

- **Admit the woman to hospital if serious complications (for example pneumonia or encephalitis) are suspected.**
- **For all other breastfeeding women:**
    - Consider prescriping acyclovir if the woman presents within 24 hours of rash onset, particularly if she has severe chickenpox or is at increased risk of complications.
    - Seek urgent specialist advice regarding whether she should continue to breastfeed and whether her baby requires treatment to minimize the risk of complications.
    - Offer symptomatic treatment.
    - Give advice about contact with other people and when to seek medical advice.
    - If a high temperature develops (particularly after initial improvement) with redness and pain surrounding the chickenpox lesions, consider bacterial superinfection and manage accordingly.

Antiviral treatment

    - Aciclovir given within 24 hours of rash onset reduced the time to full crusting of lesions and reduced the number of lesions.
    - Aciclovir given more than 24 hours after rash onset showed no significant difference in time to full crusting of lesions or time to no new lesions.
- The recommended dose, duration, and window period for treatment with aciclovir, if indicated, is based on guidance from PHE; oral aciclovir 800 mg 5 times a day for 7 days for

an immunocompetent, adult or adolescent (aged 14 years or older) with chickenpox who presents within 24 hours of rash onset, particularly for people with severe chickenpox or those at increased risk of complications, such as smokers.

Seeking urgent specialist advice about whether a mother should continue to breastfeed

- Prodigy advises seeking specialist advice regarding using aciclovir when breastfeeding 'with caution'.
- Aciclovir is present at low concentrations in breastmilk and is not expected to adversely affect the infant; although data is limited it is considered to be the antiviral of choice where indicated, for breastfeeding women.

Scenario: Neonate

From birth to 1 months

How should I manage a neonate with chickenpox?

- Seek immediate specialist advice regarding further management.
- Give advice to the parents/carers about contact with other people.

Scenario: Immunocompromised person

From age 1 month onwards

How should I manage an immunocompromised person with chickenpox? As mentioned above

Scenario: Person exposed to chickenpox

From birth onwards

How should I assess someone who has been in contact with chickenpox?

- **For all people with a history of exposure to chickenpox, establish whether:**
  - The diagnosis of chickenpox in the contact is certain.
  - The exposure was significant enough to put the person at risk of infection.
  - The person has had chickenpox in the past.
  - The person is at increased risk of complications of chickenpox (for example pregnant women, immunocompromised people, and neonates).
  - The person is in contact with others at high risk of complications (for example healthcare workers).

Significant exposure

How should I manage a child or adult who has been exposed to chickenpox?

- Perform a general assessment to establish the person's risk of chickenpox on the basis of their history of chickenpox, the certainty of chickenpox in the contact, and the level of exposure.
  - If the person's exposure to chickenpox is not significant, or if they have a history of chickenpox, or if they are known to be immune to chickenpox, reassure.

    - If the person is not immune, advise them that they may develop chickenpox.
- For healthcare workers advise that:
    - If they have a definite history of chickenpox or shingles and have had a significant exposure to the varicella-zoster virus, they can continue working as they are considered to be protected, however, if they develop a rash or fever, or feel unwell they should seek advice from occupational health before patient contact.
    - If they are not vaccinated and do not have a definite history of chickenpox or shingles, they should avoid contact with high-risk patients for 8–21 days from exposure and contact their occupational health department.
- **How should I manage a neonate exposed to chickenpox?**

    - Perform a general assessment to establish the neonate's risk of chickenpox, on the basis of the certainty of chickenpox in the contact, and the level of exposure.
    - If the neonate's mother is the contact, determine when, in relation to delivery, she developed chickenpox. If someone else is the contact, determine the age of the neonate at the time of contact.

How should I manage an immunocompromised person exposed to chickenpox?

- Perform a general assessment to establish the certainty of chickenpox in the contact, the level of exposure, and whether the person fulfils the criteria for immunocompromise.
- Seek same-day specialist advice regarding testing and management.

Prescribing information:

Chlorphenamine(piriton)

Dose

- For children aged:
    - Less than 1 year — chlorphenamine is not recommended as it is not licensed in this group.
    - 1–2 years: give 1 mg twice daily.
    - 2–6 years: give 1 mg every 4–6 hours (maximum 6 mg daily).
    - 6–12 years: give 2 mg every 4–6 hours (maximum 12 mg daily).
    - 12–18 years: give 4 mg every 4–6 hours (maximum 24 mg daily).
- For adults:
    - Give 4 mg every 4–6 hours (maximum 24 mg daily). If the person is elderly, reduce the dose to a maximum of 12 mg daily.

Pregnancy

- Avoid use in pregnancy unless considered essential by a physician:
    - Data on the use of chlorphenamine in pregnant women is lacking and the potential risk is unknown. Use during the third trimester may result in adverse reactions in neonates such as irritability, paradoxical excitability, and tremor.

Breastfeeding

- Avoid use in breastfeeding unless considered essential by a physician:
    - Chlorphenamine maleate may inhibit lactation and may be secreted in breast milk.

Aciclovir

Dose

- **By mouth, for an adult or adolescent (aged 14 years or older):**
  - 800 mg 5 times a day for 7 days.

Adverse effects

- Aciclovir is generally well tolerated.
- Common adverse effects include gastrointestinal adverse effects (such as nausea, vomiting, diarrhoea, and abdominal pain), headache, dizziness, fatigue, fever, and skin rashes (including photosensitivity and urticaria).

Contraindications and cautions

- **Do not prescribe aciclovir** to a person with a known allergy to aciclovir or valaciclovir.
- **Prescribe aciclovir with caution** in a person who:
  - Is elderly.
  - Has underlying neurological abnormalities
  - Has severe liver or electrolyte abnormalities
  - Has significant hypoxia.
  - Has renal impairment — advise the person to maintain adequate hydration. Dose adjustment is required.

Drug interactions

- Aciclovir may increase the plasma concentration of aminophylline and theophylline.
- Use with other nephrotoxic drugs increases the risk of renal impairment.

Pregnancy

- Use of aciclovir should be considered only when potential benefits outweigh possible risks — seek specialist advice.

Breastfeeding

- Aciclovir has been detected in breast milk. Caution is advised if aciclovir is considered in breastfeeding — seek specialist advice.

TOPIC: SHINGLES

- **Shingles (herpes zoster) is a viral infection of an individual nerve and the skin surface that is served by the nerve (dermatome).** It is caused by the reactivation of the varicella-zoster virus, the same virus that causes chicken pox.
    - Following a chicken pox infection, which typically occurs during childhood, the varicella-zoster virus lies dormant in the dorsal root ganglia and reactivates when the immune system is weakened.
    - Unlike chicken pox, which follows the initial infection and causes a generalized rash, shingles occurs years (sometimes decades) after the primary infection and symptoms are usually localized to a specific dermatome.
    - People with active lesions of shingles, particularly if they are immunocompromised, can transmit varicella zoster virus to susceptible people to cause chickenpox. However, there is no evidence that shingles can be acquired from a person who has chickenpox.

What are the risk factors?

- **Risk factors for shingles include:**
    - Increasing age — although shingles can occur at any age, the incidence of the disease (and the risk of complications) increases with age. This is thought to result from an age-related decline in virus-specific, cell-mediated immune responses.
    - Immunocompromise — other conditions that change cell-mediated immunity can increase the risk of shingles (and associated complications). These include HIV infection, lymphoproliferative malignancies, immunosuppressive treatment (such as corticosteroids [long-term use] and chemotherapy), and organ transplantation.
    - Certain comorbidities, including:
        - Rheumatoid arthritis.
        - Asthma and chronic obstructive pulmonary disease.
        - Chronic kidney disease.
        - Depression.
        - Diabetes.
        - Systemic lupus erythematosus.
        - Granulomatosis with polyangiitis previously known as Wegener's granulomatosis.
        - Malignancies.
    - Psychological factors — shingles has been linked to physical, emotional, and sexual abuse. Other contributing factors may include financial stress, inability to work, decreased independence, and an inadequate social support environment.
    - Female sex — there is some evidence that women have a greater risk of developing shingles than men.
    - Race/ethnicity — a systematic review and meta-analysis found that black people had almost half the risk of shingles as white people.
    - Statin use — limited evidence suggests that the use of statins may increase the risk of shingles.
- **In addition to increasing age and immunocompromise, other risk factors for complications of shingles include:**
    - Severe atopic dermatitis/eczema.
    - Herpes zoster of the head and/or neck area.
    - Herpes zoster with:
        - Moderate to severe prodromal or acute zoster-associated pain.
        - Severe rash and/or signs of cutaneous dissemination.

        - Signs of involvement of the central nervous system.
        - Signs of visceral involvement.

What are the complications?

- **Post-herpetic neuralgia (defined as pain persisting for, or appearing more than, 90 days after rash onset) is the most common complication of shingles** and is caused by herpes zoster-induced peripheral nerve damage.
    - It is most common in older people.
    - Post-herpetic neuralgia is rare in people younger than 50 years of age.
- **Other complications of shingles include:**
    - Skin changes — scarring, changes in pigmentation, and keloid formation may occur after a shingles rash has healed.
    - Superinfection of skin lesions — secondary infection of the lesions, usually with staphylococcal or streptococcal bacteria, may occur and can result in cellulitis, osteomyelitis, or life-threatening complications (such as necrotizing fasciitis and sepsis).
    - Herpes zoster oticus (Ramsay Hunt syndrome) — occurs when the virus infects the facial nerve (cranial nerve VII). It is characterized by lesions in the ear, facial paralysis, and associated hearing and vestibular symptoms.
    - Herpes zoster ophthalmicus — occurs when the virus infects the ophthalmic division of the trigeminal nerve. Complications include keratitis, corneal ulceration, conjunctivitis, optic neuritis, retinitis, glaucoma, and blindness if untreated. Hutchinson's sign (a rash on the tip, side, or root of the nose) may be present and indicates nasociliary branch involvement and an increased likelihood of eye inflammation and permanent corneal denervation. Ophthalmic complications occur more frequently and can be more severe in immunocompromised people.
    - Peripheral motor neuropathy — occurs when a motor rather than a sensory nerve is involved, and causes loss of movement in the affected area. It occurs in 5% of cases and is more common in elderly people, people with underlying malignancy, and people with cranial nerve involvement.
    - Central nervous system (CNS) complications — asymptomatic involvement of the CNS is frequently reproducible in people with shingles of the head or neck area. CNS complications, such as encephalitis, meningoencephalitis, myelitis, cerebellitis, cerebrovascular disease, radiculitis, and Guillain–Barré syndrome, have been reported, especially in people who are immunocompromised.
    - Cardiovascular complications — herpes zoster has been shown to be an independent risk factor for vascular disease, particularly stroke, transient ischaemic attack, and myocardial infarction.
    - Negative impact on quality of life — the quality of life of a person with shingles can be substantially diminished by both the acute pain associated with the condition and the longer-term, potentially debilitating pain of post-herpetic neuralgia, both of which can limit the person's productivity and their ability to conduct activities of daily living.
- **Systemic dissemination is a rare but serious complication of shingles.**
    - The reactivated virus can, in some cases, disseminate into the lungs, liver, gut, and brain, leading to pneumonia, hepatitis, encephalitis, and disseminated intravascular coagulopathy.
    - Disseminated disease is more likely to occur in those who are severely immunocompromised, with a case fatality rate reported to be between 5–15% and most deaths being due to pneumonia.

What is the prognosis?

- **Shingles usually is a self-limiting disease.** However, complications may occur if there are risk factors for a complicated a course, such as severe immunocompromise.
- Recurrent herpes zoster is unusual. The incidence is not certain, but it is thought to be higher in immunocompromised people.

Diagnosis:

How should I diagnose shingles?

- **Take a history.**
  - Ask about the symptoms experienced, including when they started and the location, duration, and severity.
  - Identify risk factors for shingles, such as immunocompromise due to long-term corticosteroid use.
- **Examine the person to:**
  - Assess for the clinical features of shingles.
  - Exclude differential diagnosis, such as herpes simplex virus infection and contact dermatitis.
- **Diagnose shingles on the basis of typical clinical features.**
  - There is usually a prodromal phase with abnormal skin sensations and pain in the affected dermatome (area of skin served by an individual nerve). The pain can be described as burning, stabbing, or throbbing; can be intermittent or constant, and may be so severe it interferes with sleep and quality of life. Headache, photophobia, malaise, and fever (less common) may also occur as part of the prodromal phase.
  - Within 2–3 days (more rarely up to 7 days), a rash typically appears in a dermatomal distribution. It starts as maculopapular lesions then develops into clusters of vesicles, with new vesicles continuing to form over 3–5 days. The rash is usually painful, itchy, and/or tingly, and, unlike other rashes, does not cross the midline of the body.
  - The vesicles then burst, releasing varicella-zoster virus, and crust over within 7–10 days.
  - Healing occurs over 2–4 weeks, and often results in scarring and permanent pigmentation in the affected area.
  - **The location of symptoms depends on the affected nerve:**
    - In immunocompetent people, the infection usually occurs on the thorax, with dermatomes T1 to L2 most commonly affected.
    - In immunocompromised people, symptoms can be more widespread and affect multiple dermatomes (disseminated disease).
  - **Note that:**
    - The rash may be atypical in certain groups of people, for example older people (in whom the rash may not be vesicular) and immunocompromised people (in whom the rash may be severe or long-lasting).
    - In rare cases, pain without a rash (zoster sine herpete) can occur.
- **Investigations are not usually required in primary care.** If there is diagnostic uncertainty, refer or seek specialist advice.

- Important differences between shingles and differential diagnoses include the prodrome, the dermatomal distribution, the vesicle groupings, and pain in the area of the rash.
  - Shingles usually occurs on the thorax (53% of cases), with dermatomes T1 to L2 most commonly affected. Other dermatomal distributions include cervical (20%), the ophthalmic (first) division of the trigeminal nerve (10–20%), and lumbosacral (11%).

    - The rash of shingles does not cross the midline, whereas other rashes can.
    - Mucous membranes within the affected dermatome are also involved. Bilateral involvement is rare.
- Investigations are typically not needed but may be considered:
    - In people with recurring lesions that are suspicious for herpes simplex.
    - In people with suspected zoster sine herpete, in which the virus causes pain without lesions.
    - For atypical presentations, such as the widely disseminated lesions that may occur in immunocompromised people.
    - For differentiating herpes zoster from other vesicular dermatoses, such as contact dermatitis.
- Investigations which may be used to diagnose shingles include polymerase chain reaction testing of vesicle or other body fluids to identify varicella-zoster virus DNA, immunohistochemical analysis of skin scrapings, or virus culture.

What else might it be?

- **Differential diagnoses of shingles include:**
    - Herpes simplex virus (HSV) infection — this can also present with a unilateral, red, maculopapular, vesicular rash and acute pain. Features which are more likely to indicate HSV infection include multiple recurrences (around the mouth or genital areas) and lack of chronic pain.
    - Candidal skin infection — the appearance of the skin is variable, depending on the affected area. Soreness and itching is usual. Thin-walled pustules with a red base may be present. Scales may accumulate, producing a white-yellow, curd-like substance over the infected area.
    - Contact dermatitis — localized rash or irritation of the skin caused by contact with a foreign substance. The pain and the rash usually occur simultaneously.
    - Folliculitis — a superficial infection of the hair follicles, which develop into small inflammatory papules or pustules.
    - Primary HIV infection — herpes zoster is an HIV indicator condition.
    - Impetigo — presents with vesicles, which can rupture and crust.
    - Insect bite or sting — there may be local symptoms (such as pain, swelling, and erythema) and symptoms that may indicate a systemic reaction (such as urticaria, rhinitis, wheezing, abdominal pain, vomiting, and dizziness).
    - Scabies — an intensely itchy skin infestation characterized by symmetric erythematous papules, often excoriated, seen in a characteristic pattern.
    - Urticaria — a superficial swelling of the skin (epidermis and mucous membranes) that results in a red, raised, itchy rash.
- **The prodromal pain of shingles can often mimic other conditions, depending on the site of infection, including:**
    - Acute appendicitis — characterized by pain in the right lower abdominal quadrant.
    - Cholecystitis — characterized by pain in the right upper quadrant of the abdomen, accompanied with nausea, vomiting, and fever. On examination the person may have Murphy's sign.
    - Migraine — characterized by attacks of moderate or severe headache and associated symptoms, such as photophobia, phonophobia, nausea, and vomiting.
    - Renal calculi — characterized by severe colicky pain and inability to lie still, and flank pain on examination.

Scenario: Management

From age 12 months onwards

How should I manage a person with shingles?

- **Assess:**
  - The severity of symptoms.
    - Consider assessing pain intensity using a validated pain assessment scale, such as the Visual Analog Scale.
    - Note that in immunocompromised people, rash may be severe, long-lasting, or affect multiple dermatomes.
  - For possible complications of shingles, such as infection and ocular complications.
  - For risk factors for complications, such as long-term corticosteroid use and shingles of the head and/or neck area.
- **Admit to hospital (or seek immediate specialist advice) if:**
  - Serious complications (such as meningitis, encephalitis, or myelitis) are suspected.
  - The person has shingles in the ophthalmic distribution of the trigeminal nerve, especially those with:
    - Hutchinson's sign — a rash on the tip, side, or root of the nose, representing the dermatome of the nasociliary nerve, which is a prognostic factor for subsequent eye inflammation and permanent corneal denervation.
    - Visual symptoms.
    - An unexplained red eye.
  - A severely immunocompromised person has shingles, or an immunocompromised person has shingles where the rash is severe, widespread, or they are systemically unwell.
  - An immunocompromised child has shingles.
- **For all other people with shingles:**
  - Consider the need for oral antiviral treatment.
  - Manage pain.
  - Refer, or seek specialist advice, if:
    - New vesicles are forming after 7 days of oral antiviral treatment, or healing is delayed.
    - A person who is thought to be immunocompetent has had two episodes of shingles.
    - Shingles recurs in an immunocompromised person — long-term prophylaxis to prevent recurrence of uncomplicated shingles is not routinely recommended, but may be useful for some people.
    - Pain is inadequately controlled by oral analgesia, or a strong opioid (such as morphine) is being considered.
- **Use clinical judgment to decide who to consult or refer to (for example ophthalmology, dermatology, obstetrics, immunology, or pain specialist, depending on the presenting problem) and the urgency, depending on the risk to the person and their clinical condition.**

How should I manage the pain of shingles?

- **For adults with mild pain,** offer a trial of paracetamol alone or in combination with codeine or a nonsteroidal anti-inflammatory drug (NSAID), such as ibuprofen.
  - If this is not effective, or the person presents with severe pain, consider offering amitriptyline (off-label use), duloxetine (off-label use), gabapentin, or pregabalin.

        - The choice of drug depends on the relative contraindications, possible drug interactions, and risk of adverse effects for each person.
    - Consider oral corticosteroids in the first 2 weeks following rash onset in immunocompetent adults with localized shingles if pain is severe, but only in combination with antiviral treatment.
        - Use clinical judgments, taking into account the risks and benefits of corticosteroid therapy for each person.
    - Seek specialist advice or refer to the pain team if:
        - Pain is inadequately controlled by oral analgesia.
        - A strong opioid (such as morphine) is being considered.
    - If there is persisting pain after the resolution of skin lesions.
- **For children with pain,** offer a trial of paracetamol (avoid the use of NSAIDs).
    - If this is not effective, seek specialist advice.

What information and advice should I give a person with shingles?

- **Explain that** the person with shingles is infectious until all the vesicles have crusted over (usually 5–7 days after rash onset).
- **Advise people with shingles to:**
    - Avoid contact with people who have not had chickenpox, particularly pregnant women, immunocompromised people, and babies younger than 1 month of age.
    - Avoid sharing clothes and towels.
    - Wash their hands often.
    - Wear loose-fitting clothes to reduce irritation.
    - Cover lesions that are not under clothes while the rash is still weeping.
    - Avoid use of topical creams and adhesive dressings, as they can cause irritation and delay rash healing.
    - Keep the rash clean and dry to reduce the risk of bacterial superinfection. They should seek medical advice if there is an increase in temperature, as this may indicate bacterial infection.
    - Avoid work, school, or day care if the rash is weeping and cannot be covered. If the lesions have dried or the rash is covered, avoidance of these activities is not necessary.

Scenario: Prevention

From age 50 years onwards

How can shingles be prevented?

- Provided there are no contrindications, offer the 2-dose Shingrix® vaccine to:
    - People aged 50 years and above, with a period of 8 to 24 weeks between doses.

TOPIC: Post-herpetic neuralgia:

Management:

- **Offer self-management advice, including information on the nature of post-herpetic neuralgia**. Advise people with post-herpetic neuralgia they may find it helpful to:
  - Wear cotton or silk fabrics, as these may cause the least irritation.
  - Protect sensitive areas by applying a protective layer (such as a firm bandage, compression clothing, cling film, or a plastic wound dressing such as Opsite®).
  - Consider frequent application of cold packs, unless this causes pain (allodynia).
- **When deciding a treatment plan, consider:**
  - The degree and nature of the pain (whether constant or intermittent, and the presence or absence of allodynia).
  - The impact of the pain on participation in daily living activities (including self-care, general tasks and demands, interpersonal interactions and relationships, mobility, and sleeping).
  - The person's mood (in particular, consider whether they have signs of depression or anxiety).
- **Offer analgesia to manage pain:**
  - Offer paracetamol with or without codeine if the person's pain is mild or moderate, and there are no contraindications. Be aware this is unlikely to be entirely effective alone, but may contribute to pain relief.
  - If pain remains uncontrolled, consider offering a drug to treat neuropathic pain, such as amitriptyline, duloxetine, gabapentin, or pregabalin, depending on local prescribing policies, and titrate the dosage according to response and tolerability.
  - Consider prescribing a topical treatment such as capsaicin cream or lidocaine plasters if pain is mild, if allodynia is a prominent feature, in older people if there are concerns about central nervous system side effects of oral medications, or as an adjunct to oral therapy if pain is severe.
  - Tramadol may be considered if acute rescue therapy is required, but should not be prescribed long term without specialist supervision.
- **Arrange early** follow up **to assess the progress made with dose titration, and the tolerability and effectiveness of neuropathic pain treatments.** Use clinical judgement to decide how soon to follow up the person.
- **Advise the person:**
  - On the importance of dosage titration, and the titration process, providing written information if possible. Explain that the treatment does not work immediately; it commonly takes weeks to titrate up to an effective dose.
  - That a daily pain diary may be useful to help people learn to manage their pain.
- **Consider referring the person for cognitive behavioural therapy, and/or to a neurologist or specialist pain service if:**
  - They have severe pain.
  - Their pain significantly limits their participation in daily activities (including self-care, general tasks and demands, interpersonal interactions and relationships, mobility, and sleeping).

  Treatment options available to primary care have been exhausted, and symptoms remain significant. Treatments which may be available from specialist pain clinics or neurologists include more specialist medications, such as the stronger capsaicin 8% patch, nerve blocks, injections or stimulation, sub-cutaneous Botulinum Toxin A injections, transcutaneous electrical nerve stimulation (TENS), and other psychological interventions.

TOPIC: LONG-TERM EFFECT OF CORONAVIRUS

- **Coronavirus disease is an infectious disease caused by a novel betacoronavirus known as severe acute respiratory syndrome-coronavirus 2 (SARS-CoV-2).** The disease associated with it is now referred to as 'COVID-19'.
- **The following clinical case definitions can be used to identify and diagnose the long-term effects of COVID-19:**
    - **Acute COVID-19** — signs and symptoms of infection consistent with COVID-19 for up to 4 weeks.
    - **Ongoing symptomatic COVID-19** — signs and symptoms of COVID-19 infection from 4 weeks up to 12 weeks.
    - **Post-COVID-19 syndrome** — signs and symptoms that develop during or after an infection consistent with COVID-19, continue for more than 12 weeks, and are not explained by an alternative diagnosis. It usually presents with clusters of symptoms which may overlap, fluctuate, change over time, and affect any body system.
- **Long COVID** is a multi-system condition with a range of debilitating symptoms — signs and symptoms continue or develop after acute COVID-19, continue for more than 4 weeks, and are not explained by an alternative diagnosis.
    - It includes both ongoing symptomatic COVID-19 (from 4-12 weeks) and post-COVID-19 syndrome (12 weeks or more).
    - Long COVID may consist of a number of distinct syndromes, which can include post-ICU syndrome, post-viral fatigue syndrome, long-term COVID syndrome, and permanent organ damage.

TOPIC: INFLUENZA VACCINATION

Children:

- Vaccine is given intranasally
- The first dose is given at 2-3 years, then annually after that
- It is a live vaccine
- Children who were traditionally offered the flu vaccine (e.g. asthmatics) will now be given intranasal vaccine unless this is inappropriate, for example if they are immunosuppressed. In this situation the inactivated, injectable vaccine should be given
- Only children aged 2-9 years who have not received an influenza vaccine before need 2 doses
- It is more effective than the injectable vaccine

Contraindications (to live attenuated vaccine):

- Immunocompromised
- Aged < 2 years
- Current febrile illness or blocked nose/rhinorrhoea
- Current wheeze (e.g. ongoing viral-induced wheeze/asthma) or history of severe asthma (BTS step 4)
- Egg allergy (although according to Up-to date complications of those with egg allergy is minimal)
- Pregnancy/breastfeeding
- If the child is taking aspirin (e.g. for Kawasaki disease) due to a risk of Reye's syndrome

Side-effects: blocked nose/rhinorrhea, headache, anorexia, sorethroat

Adults:

The influenza vaccine:

- It is an inactivated vaccine, so cannot cause influenza. A minority of patients however develop fever and malaise which may last 1-2 days
- Should be stored between +2 and +8ºC and shielded from light
- In adults the vaccination is around 75% effective, although this figure decreases in the elderly
- It takes around 10-14 days after immunisation before antibody levels are at protective levels

Contraindications and precautions:

- **Allergy**:
- **Patients on anticoagulation** − Patients receiving an oral anticoagulant may receive the influenza vaccine by intramuscular injection. Measures should be taken to minimize the risk of hematoma like using a small-gauge needle when possible and applying firm pressure to the vaccination site for at least two minutes following vaccination.

Adverse reactions :

- **•Local injection site reaction**
- **•Shoulder bursitis**

TOPIC: PNEUMOCOCCAL VACCINE

There are two types of pneumococcal vaccine currently in use:

- Pneumococcal conjugate vaccine (PCV)
- Pneumococcal polysaccharide vaccine (PPV)

The PCV is given to children as part of their routine immunisations (at 3 and 12-13 months).

The PPV is offered to all adults.

TOPIC: GLANDULAR FEVER 'INFECTIOUS MONONUCLEOSIS'

- **Glandular fever (infectious mononucleosis) is an infection most commonly caused by the Epstein-Barr virus (EBV).**
    - EBV (also known as the human herpesvirus 4) is one of the most common human viruse.
    - It causes approximately 80–90% of cases of glandular fever, with the remainder due largely to cytomegalovirus, human herpesvirus 6, toxoplasmosis, HIV, and adenovirus.

How is it transmitted?

- **Glandular fever is not particularly contagious.**
    - It is spread mainly through contact with saliva, usually from asymptomatic carriers, such as through kissing or sharing food and drink utensils**.** Small children may be infected by chewing toys that have been contaminated with the virus.
    - It can also spread during sexual contact (through blood and semen), blood transfusions, organ transplantations, and by intrauterine transmission.
    - The incubation period of glandular fever is about 4–7 weeks. The disease is contagious during the incubation period and while symptoms are present; some people may be contagious for as long as 18 months after having the infection.
- **Epstein-Barr virus (EBV) infection leads to a lifelong latent carrier state (the virus remains in the body lifelong).**
    - There is persistent low-grade replication, and the virus is shed intermittently into pharyngeal secretions, particularly saliva. These low concentrations of infectious virus account for the low-to-moderate contagiousness of the disease and the apparent requirement of intimate contact for disease transmission.

What are the complications?

- **Complications of glandular fever include:**
    - **Hepatitis** — this is a common and self-limiting early complication of glandular fever. Fulminant hepatitis may rarely occur and can be life threatening.
    - **Upper airways obstruction** — this may be due to gross enlargement of the tonsils or a peritonsillar abscess (quinsy).
    - **Cardiac complications** — pericarditis, myocarditis, and cardiac conduction abnormalities have been reported in people with glandular fever.
    - **Renal complications** — interstitial nephritis, myositis-associated acute kidney injury, haemolytic uraemic syndrome, and jaundice-associated nephropathy have rarely been reported in people with acute symptomatic glandular fever.
    - **Neurological complications** — encephalitis occurs in about 1% of adolescents and adults with glandular fever, usually during the first 2 weeks of infection. Other possible neurological complications include aseptic meningitis, facial nerve palsy, transverse myelitis, Guillain-Barré syndrome, and optic neuritis.
    - **Haematological complications** — mild thrombocytopenia has been reported in 25–50% of people with glandular fever, and autoimmune haemolytic anaemia in 3% of people. Mild neutropenia is common and self limiting. Rarely, severe neutropenia may cause neutropenic sepsis, pneumonia, or death.
    - **Splenic rupture** — this is rare (less than 1% of cases) and usually occurs in the first 3 weeks of the acute illness. About 50% of cases of splenic rupture are spontaneous, and 50% follow trauma (which may be minimal). People with glandular fever are advised to avoid heavy lifting and contact or collision sports for the first month of the

illness to reduce the risk of splenic rupture, after which these activities can be resumed provided the person has no ongoing features of acute EBV infection.
    - **Chronic fatigue** — about 10% of people with glandular fever will experience prolonged fatigue, lasting for several months after the initial infection. Most people with persistent fatigue recover within 2 years.
    - **Cancer** — association between glandular fever and lymphoproliferative cancers, especially Hodgkin's lymphoma and Burkitt lymphoma, are well established.
    - **Multiple sclerosis** — glandular fever as a risk factor for multiple sclerosis has been documented and may be linked to genetic susceptibility.
    - **Chronic active Epstein-Barr virus (EBV) infection** — a very rare condition characterized by severe, chronic, or recurrent infectious mononucleosis-like symptoms after a well-documented primary infection with EBV in a previously healthy person. Chronic active EBV infection is occasionally associated with the development of lymphoma.
    - **Abnormal liver function tests (LFTs)** — LFTs are abnormal in about 90% of people with glandular fever, with aspartate aminotransferase (AST) and alanine aminotransferase (ALT) being elevated to 2–3 times the upper limit of normal. The increase in LFTs typically peaks in the second week of the illness and then normalizes over the next 3–4 weeks.

What is the prognosis?

- **In most people, glandular fever is self limiting and lasts 2–4 weeks.** Occasionally the disease course can be prolonged (lasting for weeks or months).
    - Serious complications are rare and include upper airways obstruction, splenic rupture, and neutropenia. In immunocompromized people, EBV infection may result in malignant disease, such as Hodgkin's lymphoma and nasopharyngeal carcinoma.
    - Fatigue is common and usually lasts for a few weeks, or for several months in a minority of people (about 10%).
- **Epstein-Barr virus (EBV) infection leads to a lifelong latent carrier state (the virus remains in the body lifelong).** The virus may reactivate but does not always cause symptoms. People who are immunocompromized are more likely to develop symptoms if EBV reactivates.

Diagnosis:

When should I suspect glandular fever?

- **Suspect glandular fever if a person has:**
    - Fever.
    - Lymphadenopathy.
        - Bilateral posterior cervical lymphadenopathy is typical. There may also be anterior cervical, submandibular, suboccipital, postauricular, epitrochlear, axillary, and inguinal lymphadenopathy.
        - Lymph nodes are mildly tender and mobile.
    - Sore throat (usually severe).
        - There may be tonsillar enlargement, the tonsils may meet in the mid-line, and there is usually a 'whitewash' exudate on the tonsils.
        - Pharyngeal inflammation and palatal petechiae are also common — petechiae are 1–2 mm in diameter and occur in crops lasting 3–4 days.
        - Sore throat may fail to improve, or become worse, after several days.

- **Other clinical features that may support a diagnosis of glandular fever include:**
  - Prodromal symptoms — this usually lasts for several days and includes general malaise, fatigue, myalgia, chills, sweats, anorexia, and retro-orbital headache.
  - Non-specific rash — this may be macular, petechial, urticarial, or erythema multiforme-like, or a maculopapular rash that presents after being treated with amoxicillin.
  - Splenomegaly — the spleen reaches maximum size at the beginning of the second week of illness and regresses over 7–10 days.
  - Hepatomegaly — although more often there is mild tenderness in the right hypochondrium.
- **Be aware that:**
  - Glandular fever is most common in people aged 15–24 years and is spread mainly through contact with saliva, such as through kissing or sharing food and drink utensils.
  - In people older than 40 years of age, glandular fever is rare and may present atypically:
    - Without sore throat and lymphadenopathy (in more than 50% of people).
    - With unexplained fever of more than 2 weeks' duration.
    - With jaundice (in about 20% of older adults).
  - In a child (under 3 years of age), infection with Epstein-Barr virus (EBV) is usually asymptomatic, or the symptoms are indistinguishable from other childhood viral illnesses.
  - Glandular fever is difficult to distinguish from other causes of sore throat (especially streptococcal sore throat).

How should I investigate a person with suspected glandular fever?

- **In children younger than 12 years of age and in people who are immunocompromized at any age:**
  - Arrange blood tests for Epstein-Barr virus (EBV) viral serology after the person has been ill for at least 7 days.
- **In children older than 12 years of age and in immunocompetent adults:**
  - Arrange a full blood count (FBC) with differential white cell count and a monospot test (heterophile antibodies) in the second week of the illness.
    - Glandular fever is likely if the monospot test is positive or the FBC has more than 20% atypical or 'reactive' lymphocytes, *or* more than 10% atypical lymphocytes and the lymphocyte count is more than 50% of the total white cell count.
  - If the monospot test is negative/not supportive of infectious mononucleosis due to EBV, repeat the test in 5–7 days.
    - If the person has clinical features of glandular fever and a negative monospot test, or if a rapid diagnosis is needed (for example if an urgent return to contact sports is desired), arrange blood tests for EBV viral serology.
  - If a second monospot test is negative:
    - Consider testing for cytomegalovirus and toxoplasmosis if the person is pregnant or immunocompromized.
    - Consider testing for HIV.

- **Consider checking liver function tests (LFTs).**
    - Usually aspartate aminotransferase (AST) and alanine aminotransferase (ALT) are elevated to 2–3 times the upper limit of normal.

Monospot test (heterophile antibodies)

- The monospot test is not recommended for general use by the CDC because:
    - The antibodies detected by monospot can be caused by conditions other than glandular fever.
    - The monospot produces both false-positive and false-negative results. For example, the heterophile antibodies detected by monospot are often not present in children with glandular fever.
    - At best, the monospot test may indicate that a person has a typical case of glandular fever, but does not confirm the presence of Epstein-Barr virus (EBV) infection.

Testing for cytomegalovirus, toxoplasmosis, and HIV

- A large research study concluded that:
    - HIV testing may be appropriate, as primary HIV infection may present as a glandular fever-like illness.

Liver function tests (LFTs)

- Expert opinion in review articles is that LFTs may provide additional support to the diagnosis of glandular fever by showing a hepatitic picture. If liver dysfunction presents it usually develops early in the illness, and normalizes after 20 days.

What else might it be?

- **The differential diagnosis of glandular fever includes:**
    - **Other causes of sore throat,** such as streptococcal sore throat — lymphadenopathy is usually anterior cervical and submandibular (unlike with glandular fever, where posterior cervical lymphadenopathy is most common). Splenomegaly or hepatomegaly is not typical in adults. Fatigue is less prominent.
    - **Other causes of lymphadenopathy,** such as:
        - Local infection or inflammation.
        - Lymphoma or metastatic solid tumour.
        - Leukaemia (rarely presents with lymphadenopathy).
    - **Other viral infections that may present with a glandular fever-like illness with an atypical lymphocytosis,** such as:
        - Cytomegalovirus primary infection — may present with splenomegaly, hepatomegaly, and a negative monospot test. Sore throat and lymphadenopathy are rare.
        - Acute toxoplasmosis — lymphocytosis is mild, but there may be hepatomegaly or splenomegaly.
        - Acute viral hepatitis — may cause malaise, fever, lymphadenopathy, and atypical lymphocytosis. usually, the lymphocytes are less than 10% of the total white cell count and liver function tests (LFTs) are markedly elevated.
        - Primary HIV infection — may present with pharyngitis, fever, and lymphadenopathy, rash, diarrhoea, weight loss, nausea, and vomiting. Mononucleosis-like illness is an HIV indicator condition.

        - Rubella — may rarely present with fever, lymphadenopathy, and atypical lymphocytosis.
        - Roseola (human herpes virus-6 primary infection) — is usually mild with a modest atypical lymphocytosis.
        - Mumps — typically presents with parotitis and there may be low-grade fever and malaise.
        - Herpes simplex virus-1 — typically presents with acute onset of symptoms and high fever with myalgia, arthralgia, and cough. Symptoms typically last less than 2 weeks; posterior cervical lymphadenopathy is absent.

Scenario: Management of glandular fever

From age 1 month onwards

How should I manage a person with suspected or confirmed glandular fever?

- **Arrange hospital admission if the person has:**
    - Stridor.
    - Dehydration or difficulty swallowing fluids.
    - A suspected potentially serious complication of glandular fever, such as splenic rupture.
- **If hospital admission is not indicated:**
    - Advise on the use of paracetamol or ibuprofen to relieve pain and fever symptoms.
    - Explain the expected course of the illness. Advise that:
        - Symptoms usually last for 2–4 weeks.
        - Tiredness is common and is often the last symptom to resolve.
    - Advise the person:
        - That exclusion from work or school is not necessary.
        - To return to normal activities as soon as possible. If they are tired, they should tailor their activities to what they can manage comfortably. Bed rest is not normally needed.
        - To limit spread of the disease by avoiding kissing and sharing eating or drinking utensils, and to thoroughly clean all items that may have been contaminated by saliva.
        - To avoid heavy lifting and contact or collision sports for the first month of the illness (to reduce the risk of splenic rupture).
    - Advise the person to seek urgent medical advice if they:
        - Develop stridor or respiratory difficulty.
        - Have difficulty swallowing fluids or have signs of dehydration, such as reduced urine output.
        - Become systemically very unwell.
        - Develop abdominal pain (may indicate splenic rupture).

Treatments not recommended in primary care

- **Antiviral treatment**
    - Antiviral drugs have been used to treat glandular fever, but the use of antivirals for this indication is controversial. They may be effective by preventing viral replication that helps to keep the virus inactive. However, there are no guidelines for the use of antivirals in glandular fever.
- **Corticosteroid treatment**
    - Cortisosteroids have been used for their anti-inflammatory effects, but there are no universal criteria for their use.

TOPIC: NOROVIRUS

Norovirus, also known as the winter vomiting bug, is one of the most common causes of gastroenteritis in the UK according to NHS England. It is a genus that encompasses a range of non encapsulated RNA virus species.

Symptoms:

- Develop within 15 - 50 hours of infection (quoted from the European Centre of Disease Control and Prevention), with patients experiencing nausea, vomiting, and diarrhoea, which may be accompanied by headaches, low-grade fevers, and myalgia.
- The majority of patients experience both vomiting and diarrhoea

Transmission:

- Faecal-oral route, with the virus becoming aerosolized when the patient vomits or when a toilet containing infected bodily fluids (vomit or faeces) is flushed.
- Viral particles are transmitted directly to surrounding potential hosts and to surrounding surfaces, from which they can be transmitted by cross-contamination (for example a patient visitor touching the patient environment, then eating without first washing their hands).
- Infection may also be transmitted from an infected individual in the process of direct physical contact, or contact with food preparation.
- The virus enters the cell via host receptor-mediated endocytosis and replicates in the small intestine. Norovirus is highly contagious.

Limiting transmission:

- Isolation of infected strongly recommended where possible, particularly in areas where the risk of person to person, transmission is high, for example, in hospitals schools, and care facilities
- Good hand hygiene using soap and water is essential
- Alcohol hand gel has been shown to be less effective in reducing transmission

Diagnosis:

- Clinical history and stool culture viral PCR (polymerase chain reaction).

Resolution and Potential Complications:

- The infection is self-limiting in immunocompetent patients and symptoms generally resolve within 72 hours
- Dehydration and electrolyte imbalances may arise as a result of vomiting and diarrhoea, leading to significant morbidity and mortality and patients should be managed supportively with rehydration and electrolyte supplementation where necessary.

TOPIC: LEGIONELLA

Legionnaire's disease is caused by the intracellular bacterium Legionella pneumophilia. It typically colonizes water tanks and hence questions may hint at air-conditioning systems or foreign holidays. Person-to-person transmission is not seen

Features:

- Flu-like symptoms including fever (present in > 95% of patients)
- Dry cough
- Gastrointestinal symptoms such as nausea, vomiting, and diarrhea
- Relative bradycardia
- Confusion
- Lymphopaenia
- Hyponatraemia
- Elevated hepatic transaminases
- Pleural effusion: seen in around 30% of patients
- C-reactive protein levels >100 mg/L

Management:
Reassurance and Explanation:
Legionnaires' disease is a serious lung infection (pneumonia) with chills, high temperature (fever) and muscle pains. You get the infection by breathing in droplets of water which contain legionella germs (bacteria) from contaminated water systems. The earlier the illness is treated with antibiotic medication, the better the likely outcome.
Advice:

The illness is serious and can be fatal. However, it cannot be passed from person to person (it is not contagious)

Prescription:
According to up-to date:

- Levofloxacin and azithromycin are the preferred agents for the treatment of Legionnaires' disease (up-to date)
- Alternatives include ciprofloxacin, moxifloxacin, clarithromycin, and roxithromycin
- minimum of 5 days and do not stop therapy until the patient is clinically stable and afebrile for at least 48 hours. Patients with severe pneumonia or chronic comorbidities may be slow to respond to therapy and often require 7 to 10 days of treatment.
- immunocompromised patients are at risk for both severe infection and relapse. Thus, we generally treat for a minimum of 14 days
- Patients with confirmed or suspected *Legionella* infection do not require isolation or contact precaution

Investigation:

- Polymerase chain reaction
  - PCR is the preferred initial test for the diagnosis of Legionnaires' disease.
- Urine Legionella antigen testing
  - If PCR is not available or if sputum cannot be obtained
- Culture — Culture on special media is considered the gold standard for diagnosis of *Legionella* infections.
- Chest x-ray features of legionella pnuemonia are non-specific but includes a mid-to-lower zone predominance of patchy consolidation. Pleural effusions are seen in around 30%.

TOPIC: MYCOPLASMA PNEUMONIAE

Mycoplasma pneumoniae is a cause of atypical pneumonia which often affects younger patients. It is associated with a number of characteristic complications such as erythema multiforme and cold autoimmune haemolytic anaemia. Epidemics of Mycoplasma pneumoniae classically occur every 4 years. It is important to recognise atypical pneumonia as it may not respond to penicillins or cephalosporins due to it lacking a peptidoglycan cell wall.

Features:

- The disease typically has a prolonged and gradual onset
- Flu-like symptoms classically precede a dry cough
- Bilateral consolidation on x-ray
- Complications may occur as below

Complications:

- Cold agglutins (IgM): may cause a haemolytic anaemia, thrombocytopenia
- Erythema multiforme, erythema nodosum
- Meningoencephalitis, Guillain-Barre syndrome and other immune-mediated neurological diseases
- Bullous myringitis: painful vesicles on the tympanic membrane
- Pericarditis/myocarditis
- Gastrointestinal: hepatitis, pancreatitis
- Renal: acute glomerulonephritis

Investigations:

- Diagnosis is generally by Mycoplasma serology
- Positive cold agglutination test

Management:

- Doxycycline or a macrolide (e.g. erythromycin/clarithromycin)

TOPIC: WHOOPING COUGH

- **Whooping cough, also known as pertussis, is a highly infectious disease usually caused by the bacterium *Bordetella pertussis*,** an exclusively human pathogen that can affect people of all ages.
- **The infection is transmitted via respiratory secretions generated by coughing and sneezing, or via objects contaminated with respiratory secretions.**
    - The incubation period is approximately 7–10 days (this ranges from 4–21 days).
    - If untreated, people typically remain infectious for 21 days from the onset of symptoms.
- **Prevention is mainly by vaccination.**
    - The maternal whooping cough vaccination programme was introduced to protect infants (through the intrauterine transfer of maternal antibodies) from birth until they can be actively protected by the routine infant vaccination programme. The programme offers vaccination to pregnant women, ideally between 16–32 weeks (but up to 38 weeks) of pregnancy.

What is the prognosis?

- **In infants, whooping cough may be atypical and is associated with high rates of hospitalization, severe illness, and death.**
- **In older children and adults, the prognosis is generally very good, although the cough may persist for months.**
    - Morbidity and mortality are increased in older adults and in people with pre-existing pulmonary disease.
- **Immunity (from childhood vaccination or natural disease) wanes with time.**
    - People with immunity may experience shorter and milder symptoms but do not have lifelong protection from the disease.

What are the complications?

- **Serious complications include:**
    - Apnoea.
    - Pneumonia (usually caused by secondary bacterial infection).
    - Seizures.
    - Cerebral hypoxia, with a resulting risk of brain damage.
    - Encephalopathy (rare in adults).
- **Minor complications include:**
    - Otitis media in children (caused by secondary bacterial infection).
    - Subconjunctival haemorrhage.
    - Epistaxis.
    - Facial oedema.
    - Ulceration of the tongue or surrounding area.
    - Unilateral hearing loss (rare).
- **Increased intra-thoracic and intra-abdominal pressure due to violent and/or prolonged coughing can cause:**
    - Pneumothorax.
    - Umbilical and inguinal hernias, and rectal prolapse.
    - Rib fracture and herniation of lumbar intervertebral discs.
    - Urinary incontinence.

    - Subconjunctival or scleral haemorrhage, and facial and truncal petechiae.
- **Frequent post-tussive vomiting can lead to:**
    - Severe dehydration.
    - Malnutrition.
    - Weight loss.

Diagnosis:

What are the clinical features of whooping cough?

Whooping cough has 3 phases of symptoms: catarrhal, paroxysmal, and convalescent.

- **The catarrhal phase** typically begins 7–10 days (and ranges from 4–21 days) after exposure and lasts for 1–2 weeks.
    - Symptoms are often difficult to distinguish from those of other upper respiratory tract infections and include rhinorrhoea, malaise, mild cough, sore throat, and conjunctivitis.
    - Fever is uncommon and low-grade.
    - People are most infectious in this stage.
- **The paroxysmal phase** usually lasts 1–6 weeks but can last up to 10 weeks.
    - It is characterized by rapid, violent, and uncontrolled coughing fits (paroxysms) due to difficulty expelling thick mucus from the tracheobronchial tree.
    - The coughing fits:
        - Typically consist of a short expiratory burst followed by an inspiratory gasp (causing the characteristic 'whoop' sound). The ‘whoop’ is less common in adults, and in children younger than 3 months of age (who may present with apnoea alone).
        - Occur frequently at night, with an average of 15 attacks per 24 hours.
        - May be triggered by external stimuli, such as cold or noise.
        - Are frequently associated with post-tussive vomiting and may be severe enough to cause cyanosis in children. Adults may experience sweating attacks with facial flushing, and rarely, cough syncope.
        - Increase in frequency during the first 1–2 weeks, remain at the same frequency for 2–3 weeks, and then gradually decrease.
    - Fever is absent or minimal.
    - Most diagnosis are made during this stage.
- **The convalescent phase** usually lasts 2–3 weeks, during which there is a gradual improvement in cough frequency and severity. However, paroxysms can recur with subsequent respiratory infections for many months after the initial infection.

How should I confirm a diagnosis of whooping cough?

- **Suspect whooping cough if:**
    - A person presents with an acute cough lasting for 14 days or more without an apparent cause *plus* one or more of the following:
        - Paroxysmal cough.
        - Inspiratory 'whoop'.
        - Post-tussive vomiting.
        - Undiagnosed apnoeic attacks (in young infants).
    - A person has clinical features consistent with whooping cough and:

        - Has been in contact with a confirmed case in the previous 21 days.
        - Is known to be part of an ongoing outbreak investigation in a specific group of people; for example, children attending a school where whooping cough is known to be circulating.
- Whooping cough is a notifiable disease.
- **Consider an** alternative cause **if the** clinical features **are atypical.**

Laboratory tests for whooping cough

- **Laboratory confirmation of clinically suspected cases of whooping cough can be made by culture, polymerase chain reaction (PCR), serological testing, or oral fluid testing (OFT).**
- The local health protection team will advise on appropriate tests for confirmation and surveillance, this will depend on the person's age, duration of symptoms, and local laboratory facilities. The following tests are available from Public Health England (PHE) laboratories:
    - **Culture** and isolation of *Bordetella pertussis* **(from** nasopharyngeal aspirate [NPA] or nasopharyngeal swab [NPS]/pernasal swab [PNS]) is suitable for people of all age groups with a cough duration of less than 21 days.
    - **PCR, a molecular technique used to detect DNA sequences of** *B. pertussis***,** is suitable for people of all age groups with a cough duration of less than 21 days. PCR is usually more sensitive than culture as the organism does not need to be viable; however, PCR is less likely to be positive in people with a symptom duration of 21 days or more.
    - **Serology** is suitable for older children (aged over 16 years old) and adults with more than 14 days history of cough and at least one year after the most recent dose of whooping cough vaccine (including any dose administered in pregnancy). Serology detects antibodies to the whooping cough toxin. It may be suitable in people with a cough duration of 21 days or more, when culture and PCR are unlikely to yield positive results.
    - **OFT** is suitable for people aged between 2–16 years (that is, aged 2 to under 17 years) with a history of more than 14 days of cough and at least one year after the most recent dose of whooping cough vaccine. It is tested for anti-pertussis toxin immunoglobulin G (IgG).
- **Whooping cough is confirmed by the presence of clinical features plus one of the following:**
    - *B. pertussis* isolated from a respiratory sample, typically an NPA or NPS/PNS (or throat swab).
    - *B. pertussis* PCR positive in a respiratory clinical specimen.
    - IgG titre greater than 70 International Units per millilitre (IU/ml) from a serum specimen, or greater than 70 arbitrary units (aU) from an oral fluid specimen (in the absence of vaccination within the past year).

What else could it be?

- **Other infectious causes of cough include:**
    - *Bordetella parapertussis.*
    - *Mycoplasma pneumoniae.*
    - *Chlamydia trachomatis*.
    - *Chlamydia pneumoniae*.
    - Adenoviruses.
    - Respiratory syncytial virus (especially in children).

- Cough due to these causes is usually less severe as pertussis toxin is not produced. For more information.
- **Other non-infectious causes of cough include:**
  - Asthma.
  - Chronic obstructive pulmonary disease (in adults).
  - Post-infectious cough.
  - Upper-airway cough syndrome.
  - Gastro-oesophageal reflux disease.
  - Underlying lung malignancy.

Management:

Scenario: Management of whooping cough

From birth onwards

How should I manage a person with whooping cough?

- **Arrange hospital admission for people** with significant breathing difficulties (for example apnoea episodes, severe paroxysms, or cyanosis) or significant complications (for example seizures or pneumonia).
  - Have a low threshold for admitting children aged 6 months or younger, as they are at increased risk of severe illness and death.
  - Inform the hospital of the need for appropriate isolation before the person is admitted.
- **If admission is not necessary and the onset of the cough is within the previous 21 days, prescribe antibiotic treatment.**
  - Prescribe a macrolide first line:
    - For infants aged under 1 month, clarithromycin is preferred. Azithromycin may be used although there are limited data in this age group.
    - For children aged over 1 year, prescribe azithromycin or clarithromycin.
    - For non-pregnant adults, prescribe azithromycin or clarithromycin.
    - For pregnant women, prescribe erythromycin.
  - If macrolides are contraindicated or not tolerated, prescribe co-trimoxazole.
    - **Do not prescribe co-trimoxazole to pregnant women or infants younger than six weeks old.**
- **Advise the person and/or their family/carers:**
  - On self-care measures including rest, adequate fluid intake, and the use of paracetamol or ibuprofen for symptomatic relief.
  - That even with antibiotic treatment, whooping cough is likely to cause a protracted non-infectious cough that may take several weeks to completely resolve.
    - Symptoms are likely to be less severe and resolve more quickly if the person has been immunized or has had whooping cough before.
  - That they should seek medical advice if they develop complications.
  - That exclusion from nursery, school, or work is required.
    - Children and healthcare workers who have suspected or confirmed whooping cough should stay off nursery, school, or work until they have completed 48 hours of appropriate antibiotic treatment, or for 21 days from the onset of symptoms if not treated.

How should I manage close contacts of a person with suspected or confirmed whooping cough?

- **Offer antibiotic prophylaxis to close contacts of the 'index case' when the symptoms in the 'index case' occurred within the previous 21 days *and* the close contact is in a priority group for public health action (Group 1 or 2 as described below):**
    - **Group 1:** people at increased risk of severe complications from whooping cough ('vulnerable'). This includes:
        - Unimmunized infants (born after 32 weeks gestation) aged younger than 2 months old whose mothers did not receive the whooping cough vaccine after 16 weeks of pregnancy and at least 2 weeks prior to delivery.
        - Unimmunized infants (born before 32 weeks gestation) aged younger than 2 months old regardless of maternal vaccine status.
        - Unimmunized and partially immunized infants (less than 3 doses of whooping cough-containing vaccine) aged 2 months and older regardless of maternal vaccine status.
    - **Group 2:** people at increased risk of transmitting to 'vulnerable' people in 'group 1' who have not received a whooping cough-containing vaccine more than 1 week and less than 5 years ago. This includes:

        - Pregnant women at 32 weeks gestation or more.
        - Healthcare workers who work with infants and pregnant women.
        - People whose work involves regular close or prolonged contact with infants too young to be fully vaccinated.
        - People who share a household with an infant too young to be fully vaccinated.
    - **Close contacts** are defined as family members or people living in the same household (close 'household contacts') and contacts in institutional settings with an overnight stay in the same room (such as boarding school dormitories) during the infectious period.
        - Other types of contact, for example, contact at work or school, would generally not be considered close contact although each situation would need to be assessed on an individual basis where vulnerable contacts are involved.
        - **Prescribe a macrolide antibiotic first line (as mentioned above)**
        - **Advise the person and/or their family/carers that** exclusion from nursery, school, or work for asymptomatic contacts is not required.
        - **Offer immunization to all people who have been offered antibiotic prophylaxis.**
    - Unimmunized and partly immunized contacts up to 10 years of age should complete the schedule of the Childhood Immunization Programme with the appropriate vaccination.
    - A booster dose of pertussis containing vaccine is recommended for people aged 10 years or older who have not received a dose of a whooping cough-containing vaccine in the last five years and no tetanus-diphtheria-inactivated polio vaccine (Td-IPV) in the preceding month.
    - Post-exposure vaccination in pregnancy is important and specifically recommended in the following individuals who have not received a pertussis containing vaccine in the previous 5 years:
        - Women exposed to whooping cough after 32 weeks, or
        - Women exposed to whooping cough at any stage of pregnancy if they are at risk of transmitting to 'vulnerable' people in 'Group 1' (for example a healthcare worker).

TOPIC: RESPIRATORY PATHOGENS

The table below lists the more common respiratory pathogens:

Pathogen	Associated condition
Respiratory syncytial virus	Bronchiolitis
Parainfluenza virus	Croup
Rhinovirus	Common cold
Influenza virus	Flu
Streptococcus pneumoniae	The most common cause of community-acquired pneumonia
Haemophilus influenzae	Community-acquired pneumonia Most common cause of bronchiectasis exacerbations Acute epiglottitis
Staphylococcus aureus	Pneumonia, particularly following influenza
Mycoplasma pneumoniae	Atypical pneumonia Flu-like symptoms classically precede a dry cough. Complications include haemolytic anaemia and erythema multiforme
Legionella pneumophilia	Atypical pneumonia Classically spread by air-conditioning systems, causes dry cough. Lymphopenia, deranged liver function tests and hyponatraemia may be seen
Pneumocystis jiroveci	Common cause of pneumonia in HIV patients. Typically patients have few chest signs and develop exertional dyspnoea
Mycobacterium tuberculosis	Causes tuberculosis. A wide range of presentations from asymptomatic to disseminated disease are possible. Cough, night sweats and weight loss may be seen

TOPIC: TRAVELLERS DIARRHEA

- **Travellers' diarrhoea** is defined as passing three or more unformed stools in a 24-hour period with at least one additional symptom, such as abdominal pain or cramps, nausea, vomiting, fever, or blood in the stools. It occurs during, or within, 10 days of travel.
    - Stool passage in healthy young children can vary, so in this group, diarrhoea may be more appropriately defined as stooling that is more frequent than normal for the individual.
- **Travellers' diarrhoea can be described as:**
    - **Mild** (acute) when it is tolerable, not distressing, and does not interfere with planned activities.
    - **Moderate** (acute) when it is distressing or interferes with planned activity.
    - **Severe** (acute) when it is incapacitating or completely prevents planned activities. Note that all dysentery (passage of bloody stools) is considered severe.

What causes it?

- **Travellers' diarrhoea is caused by an infection with one or more bacteria, viruses, or parasites.**
    - **Bacterial infection** is thought to be responsible for 80% to 90% of cases.
        - *Escherichia coli* is the most common bacterial pathogen (associated with acute watery diarrhoea), followed by *Campylobacter* (more severe symptoms including fever and abdominal pain), *Shigella*, and Salmonella. Other potential bacterial causes include *Vibrio cholerae*, *Aeromonas* spp, *Plesiomonas* spp, and the newly recognised *Acrobacter*, *Larobacter*, and enterotoxigenic *Bacteroides fragilis*.
        - Enterotoxigenic *E. coli* (ETEC) is responsible for the majority of cases of travellers' diarrhoea worldwide and is predominant in Central and South America. Enteroaggregative *E. coli* (EAEC) and enteropathogenic *E. coli* (EPEC) are also significant causes of travellers' diarrhoea.
        - *Campylobacter* infection is more prevalent in people visiting South East Asia, and a high number of these cases are resistant to fluoroquinolones. It is less common in East and West Africa, South America and the Caribbean.
    - **Intestinal viruses** are thought to contribute 5% to 15% of cases.
        - Astrovirus, norovirus, and rotavirus are most commonly implicated, with norovirus often associated with outbreaks of diarrhoea on cruise ships.
    - **Protozoal pathogens** account for approximately 10% of cases in long-term travellers, with absolute risk dependent on the area of travel.
        - Giardia intestinalis is the most common protozoal cause of travellers' diarrhoea*,* with both *Entamoeba histolytica,* and *Cryptosporidium* being relatively uncommon.
        - Other causes include *Cyclospora* (which is seasonal and geographical with the most well-documented risk in Nepal, Peru, Haiti, and Guatemala) and *Dientamoeba fragilis*.
        - Protozoal pathogens tend to cause more prolonged symptoms.
    - **Helminth infection** is an uncommon cause of travellers' diarrhoea, although it may be caused by acute heavy hookworm infection or schistosomiasis.

What are the risk factors for travellers' diarrhoea?

- **The risk of travellers' diarrhoea is influenced by a number of factors, including:**
    - **Destination**:

        - Low-risk areas (fewer than 7% of travellers experience travellers' diarrhoea) include Western Europe, the USA, Canada, Japan, Australia, and New Zealand.
        - Intermediate risk areas (8–20% of travellers experience travellers' diarrhoea) include southern Europe, Israel, South Africa, and some Caribbean and Pacific islands.
        - High-risk areas (more than 20% of travellers experience travellers' diarrhoea) include Africa, Latin America, the Middle East, and most parts of Asia.
    - **Risk of dietary exposure:**
        - Choice of eating establishment is an important factor in avoidance. Backpackers have around double the risk of travellers' diarrhoea compared with business travellers.
        - Visiting family and friends — immunity tends to be short-lived but people who have previously acquired resistance to travellers' diarrhoea in a developing country may be less discriminating about the safety of food and water.
        - All-inclusive holidays such as cruise ship travel have been associated with outbreaks of viral and bacterial gastroenteritis.
    - **Increased susceptibility:**
        - Age less than 6 years — infants and toddlers are more likely to acquire severe travellers' diarrhoea and require hospitalisation.
        - Older or frail people.
        - Use of proton pump inhibitors or H2 receptor antagonists.
        - Altered upper gastrointestinal anatomy.
        - People with inflammatory bowel disease — incidence is higher in this group and episodes may last for longer.
        - Genetic factors (for example, blood group O predisposes to Shigellosis and severe cholera infection).
        - Immunosuppression or HIV infection with low CD4 cell counts.
        - Travellers originating from a low-risk area.

What are the complications of travellers' diarrhoea?

- **In addition to changes to plans, incapacitation due to illness, cost of overseas medical treatment, and/or hospitalisation, complications of travellers' diarrhoea include:**
    - Dehydration.
    - Intestinal perforation and bacteraemia as a result of infection with invasive bacterial pathogens (for example *Shigella, Salmonella and Campylobacter*).
    - Sepsis (in severe cases).
- **Longer-term consequences of travellers' diarrhoea can include**:
    - **Post-infectious irritable bowel syndrome** (occurs in 3–17% of people after travellers' diarrhoea) – this refers to the new onset of irritable bowel syndrome symptoms following an episode of gastroenteritis or travellers' diarrhoea where investigations do not show chronic enteric infection and there is no underlying organic gastrointestinal disease.
    - **Reactive arthritis** — linked with several pathogens, for example, *E. Coli, Campylobacter, Salmonella,* and *Shigella*. Joint symptoms typically appear 1–4 weeks after the episode of travellers' diarrhoea and are commonly polyarticular and affect the lower extremities, with presentation varying from transient arthralgia to debilitating arthritis. Extra-articular features (affecting the mucosa, urethra, skin, or eyes) may occur.
    - **Guillain-Barre syndrome** — this is a rare complication of *Campylobacter* infection with around 1 in 1000 people affected. An autoimmune response against peripheral nerves results in peripheral neuropathy and acute neuromuscular failure. Guillain-

Barre syndrome presents with global weakness of proximal and distal limbs, and the cranial nerves and respiratory function may be affected. Symptoms usually become evident 1–3 weeks post-infection and progress quickly. Most people will make a full recovery, but a small number will remain disabled or die as a result of the disease.
    - **Haemolytic-uremic syndrome** — a rare complication following infection with *Shigella dysenteriae* type 1.

What is the prognosis?

- **Travellers' diarrhoea is, for most people, a short-lived, self-limiting illness with recovery in a few days**. It is rarely life-threatening.
    - The severity of travellers' diarrhoea depends on the cause and the individual person. Most people do not have severe enough symptoms to be confined to bed or need medical attention.
    - The effects of travellers' diarrhoea are greater in very young children, elderly people and people living with frailty.
    - Most cases of travellers' diarrhoea are self-limiting, but children are more likely to experience dehydration and electrolyte disturbance as a consequence of fluid loss.
- **In terms of duration:**
    - Most episodes last up to seven days, with 8 out of 10 episodes resolving within a week.

Diagnosis:

Scenario: Diarrhoea - prevention and advice for travellers

From age 1 month onwards

How should I assess the risk of acquiring travellers' diarrhoea before travelling?

- **Ask which country the person is going to visit**.
- **For up-to-date risk assessment of individual countries and the requirement for strict food, water, and personal hygiene precautions**.
- **Assess for other risk factors known to further increase the risk of travellers' diarrhoea such as:**
    - Local amenities and sanitation — in areas with low standards of hygiene and sanitation and poor control over the safety of food, drink, and drinking water, the risk of acquiring travellers' diarrhoea is high.
    - How the person will be travelling — backpackers, campers, adventurers, and passengers on cruise ships are at increased risk.
    - When the person will be travelling — the peak incidence of travellers' diarrhoea is in the warmer months. Lower rates occur in winter.
    - What the person eats — for example, risks are associated with food from buffets, restaurants, and street vendors, and certain types of food such as raw fish and seafood, salads, and meat and poultry that have been inadequately cooked.
- **Review the person's susceptibility to traveller's diarrhoea and their risk of complications.** For more information on which groups of people are considered to be at higher risk.
- **Enquire about access to medical amenities** — this is particularly important for those travelling to remote areas with few or no medical amenities (especially trekking and camping).

- **Discuss whether visits to high-risk areas are essential** (particularly if the person is at increased risk of complications) and whether they are suitably prepared (for example if backpacking through remote areas).

How should I manage someone at low or intermediate risk of travellers' diarrhoea?

- **For people at low or intermediate risk of travellers' diarrhoea:**
  - Advise the person that most travellers' diarrhoea is caused by the consumption of contaminated food or water.
  - Reassure the person that no special precautions (other than basic hygiene measures) are required when travelling to locations with a high standard of hygiene.
  - Provide information on food hygiene and safe drinking water if the person is travelling to locations with lower standards of hygiene and sanitation.
  - Offer advice regarding self management and when to seek medical advice if the person develops diarrhoea during their trip.

How should I manage someone at high risk of travellers' diarrhoea?

- **For people at** high risk**:**
  - **Emphasize the importance of personal hygiene, as well as food hygiene and safe drinking water, to reduce the risk of travellers' diarrhoea.**
  - **Do not routinely offer prophylactic treatment for prevention of travellers' diarrhoea, as it is not indicated for most travellers.**
    - Antibiotic prophylaxis may be appropriate for certain high-risk travellers.
    - The use of bismuth subsalicylate and probiotics for prophylaxis is not recommended.
    - Antidiarrhoeal drugs (such as loperamide) should not be taken prophylactically.
  - **Consider whether 'stand by' antibiotic treatment (to use if affected) is appropriate.**
  - **Inform the person that there are no universal vaccines to cover all the infections which may cause travellers' diarrhoea.** However, travellers to risky areas must seek advice about appropriate vaccination against other infections that cause gastrointestinal effects, such as cholera, hepatitis A, and typhoid. For further information on vaccines recommended for overseas travel, extended holidays, or working overseas.
  - **Offer advice regarding self-management and when to seek medical advice if the person develops diarrhoea during their trip.**

What food and drink measures are recommended for preventing travellers' diarrhoea?

- **General hygiene measures**
  - Hands should always be washed with soap and water before handling and consuming food (particularly after contact with raw meat and uncooked food), using the toilet, or caring for an ill person.
  - If soap and water are not available, an alcohol-based hand sanitiser can be used as an alternative until it is possible to wash the hands. Note that hand sanitiser is not effective when hands are greasy or visibly dirty and has limited effectiveness against *Cryptosporidium* and norovirus.
- **Advice on food and drink**

- Food must have been thoroughly cooked and must remain steaming hot prior to serving.
    - Risks are associated with food from buffets, restaurants, and street vendors. Food from these sources should be avoided, if it has neither been kept hot, nor refrigerated or kept on ice.
    - Cooked food that has been in contact with raw food, or contains raw or uncooked eggs (such as homemade mayonnaise), should be avoided.
- If hygiene and sanitation are likely to be inadequate, avoid salads, uncooked vegetables, and unpasteurised fruit juices. Fruits and vegetables that can be peeled by the end-user are suitable choices (fruits and vegetables with damaged skins should be avoided).
- Ice should be avoided, unless it has been made from safe water.
- Unpasteurised (raw) milk or cheese should be avoided.
- Commercially prepared carbonated drinks, pasteurised drinks, fruit juices, and alcoholic beverages are usually safe provided they are in factory-sealed containers. The surface of the container should be wiped clean and dried if the drink is to be consumed directly from the container.

- **Advice on drinking water**
    - Tap water should be avoided for drinking, preparing food, making ice, cooking, and brushing teeth, unless there is reasonable certainty it is safe.
    - Bottled water is the safer choice for drinking water. The seal must not have been tampered with.
    - Water should be boiled (for at least 1 minute) if its safety for drinking is in doubt.
    - Other options include micropore filtering and the use of disinfectant preparations. These might be more relevant for people travelling to areas where there is little or no access to safe drinking water.

What should I advise about self-management of travellers' diarrhoea?

- **Advise the person:**
    - That most episodes of travellers' diarrhoea are short-lived and self-limiting, and last a few days.
    - To consider purchasing sachets of oral rehydration salts before travelling as they may not be available at the destination.
- **Advise that during an episode of diarrhoea:**
    - It is important to prevent dehydration — particularly for young children, pregnant women, elderly people, and those with pre-existing illnesses.
        - For infants, breastfeeding should not be interrupted.
        - Hydration must be maintained, for example with boiled, treated, or sealed bottled water. Most otherwise healthy adults can stay hydrated by eating and drinking as normal.
        - Alcohol and other drinks and beverages with a diuretic effect (such as coffee and tea) and overly sweet drinks (in large quantities) should be avoided.
        - For more severe symptoms, or people prone to complications from dehydration, oral rehydration powders diluted into clean drinking water can help to correct electrolyte imbalances.
        - Immediate medical attention is required if children show signs of dehydration such as restlessness or irritability, great thirst, sunken eyes, and dry skin with reduced elasticity.
    - Symptomatic treatment with loperamide or bismuth subsalicylate (for example Pepto-Bismol®) may be considered for *adults* to relieve mild-to-moderate diarrhoea for a maximum of two days.

        - Loperamide is usually considered the standard treatment when rapid control of symptoms is required — for example during travelling where toilet amenities are limited or unavailable.
        - Bismuth subsalicylate is more suitable for mild diarrhoea; loperamide has a faster onset of action and is more effective than bismuth subsalicylate for controlling diarrhoea and cramping.
        - Bismuth subsalicylate is not suitable for people with aspirin allergy, renal insufficiency, gout, severe enteric disease or HIV (risk of bismuth absorption), or people who are taking an anticoagulant such as warfarin, or pregnant or breastfeeding women. Darkened tongue and stools are common adverse effects.
- **Advise the person *not* to use loperamide or bismuth subsalicylate (for example Pepto-Bismol®) if they have blood or mucous in the stool and/or high fever or severe abdominal pain.**
- **Neither loperamide nor bismuth subsalicylate (for example Pepto-Bismol®) are recommended for children to relieve diarrhoeal symptoms.**
    - Loperamide should not be used in children younger than 12 years of age because of concerns that it may cause intestinal obstruction.
    - Bismuth subsalicylate must not be used in children younger than 16 years of age because of the possible association between salicylates and Reye's syndrome.
- **Medical assistance must be sought if:**
    - Stools are blood-stained, profuse and watery, or there is high or persistent fever or severe abdominal pain.
    - Severe diarrhoea occurs in infants or elderly people, or a child is unwell with travellers' diarrhoea and symptomatic treatment is needed.
    - The person becomes dehydrated — particularly in infants and children (restlessness or irritability, very thirsty, sunken eyes, and dry skin with reduced elasticity); or it is difficult to maintain adequate hydration.
    - Diarrhoea does not begin to improve within 24–36 hours despite self-treatment.
    - There is a medical comorbidity (for example immunosuppression or gastrointestinal disorder).

When should I consider prescribing a prophylactic antibiotic for travellers' diarrhoea?

- **Do not routinely prescribe prophylactic antibiotic treatments for travellers' diarrhoea.**
- **The decision to prescribe prophylactic antibiotic treatments depends on an assessment of:**
    - The risk of the person developing travellers' diarrhoea.
    - The potential harm to the person if they were to develop travellers' diarrhoea.
    - The benefits and risks of antibiotic treatment.
    - Access to medical assistance while travelling.
- **Consider prophylactic antibiotic treatment only for people who are at high risk of travellers' diarrhoea and its consequences — seek specialist advice before prescribing.** Groups who may benefit from prophylactic treatment include those who:
    - Have increased susceptibility to infection or are immunocompromised.
    - Have a high risk of complications if they were to develop travellers' diarrhoea, for example:
        - People with chronic gastrointestinal disease (such as Crohn's disease, or ulcerative colitis).
        - People with an ileostomy or colostomy.
        - People with other diseases (such as malignancy, type 1 diabetes mellitus, renal disease, stroke, congestive heart failure, or sickle cell disease) in whom a diarrhoeal illness might severely impact on their health.

    - Have a history of significant long-term morbidity after an enteric infection (for example, reactive arthritis).
    - Are undertaking critical trips in which a short bout of diarrhoea could severely impact the purpose of the trip (for example, athlete, politician, professional musician, lecturer).
    - May not have access to adequate medical treatment if they experience complicated travellers' diarrhoea.

When should I prescribe a 'stand-by' antibiotic for treatment of travellers' diarrhoea?

Do not routinely prescribe 'stand-by' (to be used if affected) treatments for travellers' diarrhoea.

- **Consider a 'stand-by' (to be used if affected) antibiotic for people who are:**
    - Travelling to high-risk locations (especially where access to medical assistance is poor or not available, or there is doubt about the safety and purity of antibiotics available at their destination).
    - At high risk of severe illness.
- **Seek specialist advice if considering prescribing a 'stand-by' antibiotic.**

TOPIC: GASTROENTERITIS

- **Gastroenteritis** is a transient disorder due to enteric infection with viruses, bacteria, or parasites. It is characterized by the sudden onset of diarrhoea, with or without vomiting. Similarly, it can be defined as acute diarrhoeal disease of rapid onset, with or without nausea, vomiting, fever, or abdominal pain.
- **Food poisoning** is defined as 'an illness caused by the consumption of food or water contaminated with bacteria and/or their toxins, or with parasites, viruses, or chemicals'.
- **Dysentery** is an acute infectious gastroenteritis characterized by diarrhoea with blood and mucus, often with fever and abdominal pain.
- **An outbreak** is defined as 'two or more cases associated in time and place'.
- **Antibiotic-associated diarrhoea** usually refers to clinical infection when the normal gut flora is disturbed by antibiotic use, enabling certain strains of *Clostridium difficile* to grow and produce toxins.

Which organisms can cause gastroenteritis?

Most infectious diarrhoea is a self-limiting illness, caused by virus **(rather than** bacteria **or** parasite**), with nearly half of episodes lasting less than 1 day.**

- Transmission of gastrointestinal infection from person-to-person may occur through one or more of a variety of different pathways, including faecal-oral, foodborne, environmental, and airborne routes.
    - The natural history of infection and risk of transmission vary depending on the underlying pathogen, the age of the person, associated co-morbidities, immune status, and environmental factors.
- Food poisoning may be primarily caused by enterotoxins produced by the microorganism (rather than the microorganism itself), and diarrhoea and vomiting usually have a rapid onset and last for less than 24 hours. Cases are often under-reported and under-detected, and are most commonly caused by:
    - *Clostridium perfringens* — usually caused by inadequate storage and insufficient reheating of contaminated meat dishes or cooked meats or meat products, for example in institutional catering settings.
    - *Bacillus cereus* — caused by contaminated cooked foods subjected to inadequate post-cooking temperature control that has allowed bacterial growth (such as reheated rice, pasta, meat or vegetable dishes, and dairy products).
    - *Staphylococcus aureus* — usually found in cooked meats and cream products.

Viruses

- **Rotavirus**
    - This is the most common cause of viral gastroenteritis in children, however, there has been a reduction in disease prevalence since the introduction of the rotavirus vaccine, as part of the UK national childhood immunization programme.
- **Noroviruses** (formerly known as 'Norwalk-like viruses')
- **Adenoviruses**
    - These commonly cause respiratory tract infections, but can also cause gastroenteritis, particularly in children.

bacteria

- ***Campylobacter jejuni* and *Campylobacter coli***
  - Most cases are self-limiting within 2–3 days and usually resolve within 1 week.
  - It is usually associated with the consumption of contaminated food and drink, such as undercooked meat (especially poultry), unpasteurised milk, or untreated water, but the source of infection is often not found.
- ***Escherichia coli* (including Shiga toxin-producing *E. coli* [STEC] or verocytotoxin-producing *E. coli* [VTEC])**
  - *E. coli* O157 is the most common serogroup of STEC causing infections in the UK.
  - Infection with *E. coli* O157:H7 may be asymptomatic, or cause diarrhoea (which may be bloody), fever, abdominal cramps, and vomiting. Illness is usually self-limiting and resolves within 10 days.
  - *E. coli* is transmitted through contaminated food, particularly meat, salad products, water, and unpasteurised milk. It can also be transmitted person-to-person by direct contact (faecal-oral route), particularly in households, schools and childcare settings, and care homes; by contact with infected animals (especially cattle, sheep, goats, and other ruminants); or environmental exposure to contaminated water.
- **Salmonellosis (excluding *Salmonella typhi* and *Salmonella paratyphi* causing enteric fever)**
  - The majority of cases are sporadic, but outbreaks may occur in the general population and in institutions.
  - Ingestion of contaminated food is the most common source, such as red and white meats, raw eggs, milk, and dairy products. Person-to-person spread and contact with infected animals can also occur.
  - Typical features are watery and sometimes bloody diarrhoea, abdominal pain, headache, nausea, vomiting, and fever. The illness usually lasts for 4–7 days, and people usually recover spontaneously.
- ***Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei***
  - Shigellosis is most commonly transmitted person-to-person by the faecal-oral route, particularly in households, nurseries, and schools. More rarely, it can be transmitted through contaminated food, or sexually transmitted (particularly in men who have sex with men).
  - Typically, 1–3 days after infection, there is diarrhoea (may have blood and mucus), fever, and abdominal cramps, with or without nausea and vomiting, headache, and malaise. Shigellosis usually resolves in 5–7 days.
- ***Yersinia enterocolitica***
  - This is a rare infection and occurs most commonly in children.
  - Common symptoms are watery diarrhoea (which is often bloody), fever, and abdominal pain. In older children and adults, right-sided abdominal pain and fever may occur.
  - Symptoms typically develop 4–7 days after exposure and may last 2 days to 6 weeks.
  - *Y. enterocolitica* is transmitted by direct contact with infected animals and person-to-person (faecal-oral route), and through contaminated food (especially raw pork and pork products) and water.

Parasites

- **Cryptosporidiosis**
  - Infection is transmitted by animal-to-human or human-to-human contact, by occupational or recreational exposure to contaminated land or water, or by consuming contaminated water or food.
  - It typically causes profuse watery diarrhoea associated with abdominal cramps or pain, nausea, vomiting, fever, and loss of appetite. Symptoms usually last for 1–2 weeks, and recurrence of symptoms is reported in around one-third of cases.
- ***Entamoeba histolytica* (amoebiasis)**
  - Transmission occurs through the ingestion of contaminated food or water. Person-to-person transmission may also occur between household and sexual contacts via the faecal-oral route.
  - Most cases in the UK are imported by travellers to endemic areas.
  - 90% of cases are asymptomatic. Diarrhoea with or without dysentery occurs in intestinal disease.
  - Symptoms are often mild diarrhoea and abdominal pain, but severe disease (amoebic dysentery) can occur, causing fever, severe abdominal pain, and blood and mucus in the faeces.
- ***Giardia intestinalis* or *Giardia lamblia***
  - Giardiasis can be transmitted by person-to-person spread by the faecal-oral route; by contact with the faeces of infected animals; by consumption of contaminated food or drink; waterborne including swimming in contaminated water; or by sexual transmission, particularly among men who have sex with men. Many cases are associated with recent foreign travel, particularly from South Asia, and it is the most commonly identified pathogen in returning travellers with prolonged diarrhoea.
  - Symptoms include diarrhoea, malaise, abdominal pain, loss of appetite, flatulence, bloating, and rarely nausea. Malabsorption, weight loss, and faltering growth may occur in children.

What are the complications?

- Dehydration, electrolyte disturbance, and acute kidney injury (AKI)
- Haemorrhagic colitis
  - This can be a complication from Shiga toxin-producing *Escherichia coli* (STEC) that typically presents with acute bloody diarrhoea and severe abdominal pain in children lasting for 2 weeks.
- Haemolytic uraemic syndrome (HUS)
  - This is a rare but potentially life-threatening complication of acute infectious gastroenteritis that occurs mostly in children aged less than 5 years.
  - About 10% of STEC 0157 cases develop HUS after an initial prodrome of gastroenteritis or haemorrhagic colitis. It may also be associated with *Shigella dysenteriae* and *Campylobacter* spp. infection.
  - It is characterized by AKI, thrombocytopenia, and microangiopathic haemolytic anaemia, usually 1 week after the onset of bloody diarrhoea. About 50% of those affected develop chronic renal complications. Mortality rates are between 3–5%.
- Thrombotic thrombocytopaenic purpura (TTP)
  - This primarily affects adults infected with STEC, and can present with fever, flu-like symptoms, petechial haemorrhages on the lower limbs, haematuria, anaemia, renal dysfunction, and possible neurological deficits.
- Reactive arthritis including Reiter's syndrome

    - This may be associated with *Campylobacter* spp., *Shigella flexneri*, and *Yersinia enterocolitica* infection. Reiter's syndrome describes a combination of urethritis, inflammatory arthritis, and uveitis.
- Aortitis, osteomyelitis
    - These may be associated with *Salmonella* spp. and *Yersinia* spp. infection.
- Sepsis
    - Invasive infection with *Salmonella* spp. and *Yersinia enterocolitica* may lead to sepsis.
- Toxic megacolon
    - This can rarely occur in people with rotavirus gastroenteritis or severe travellers' diarrhoea.
- Pancreatitis, sclerosing cholangitis, and liver cirrhosis
    - This may complicate *Cryptosporidium* spp. infection, particularly in the immunocompromised.
- Chronic diarrhoea
    - Chronic intractable diarrhoea for weeks, months, or longer may occur with *Giardia* spp. and *Cryptosporidium* spp. infection and viral gastroenteritis, particularly in people who are immunocompromised.
- Irritable bowel syndrome (IBS)
- Inflammatory bowel disease (IBD)
    - There is an increased risk of incident IBD following non-typhoidal salmonella, *Campylobacter* spp., and *Clostridium difficile* infections.
- Acquired or secondary lactose intolerance
    - Lactase deficiency occurs in up to 40% of people with giardiasis, and can persist for several weeks after parasite eradication.
- Guillain-Barré syndrome
- Meningitis
    - Infants under 3 months of age with *Salmonella* spp. infection are at particularly high risk.
- Invasive entamoeba infection
    - Rarely, extraintestinal disease occurs when *Entamoeba histolytica* invades the liver causing abscesses, which can be life threatening.

Diagnosis:

How should I assess a person with suspected gastroenteritis?

- **Ask about:**
    - Symptoms such as diarrhoea (including blood, mucus, and/or pus), nausea, vomiting, abdominal pain, fever, and general malaise.
    - The onset, frequency, duration, and severity of symptoms.
    - Risk factors for developing dehydration, current fluid intake (including breastfeeding in children), food intake, and urinary output.
    - Any features of complications.
    - Any co-morbid conditions including history of immunosuppression.
    - Occupation, if appropriate, such as food handler, healthcare or social care worker, or nursery worker.
    - Recent food intake (for example barbecue or restaurant, and types of food eaten) that may suggest food poisoning as a cause.
    - Recent exposure to untreated or potentially contaminated water.
    - Contact with other affected individuals or outbreaks.
    - Recent foreign travel (which country and region visited).

    - Recent antibiotic or proton pump inhibitor treatment, or recent hospital admission (may increase the risk of *Clostridium difficile* infection).
    - Current drug treatment.
        - Drugs such as diuretics and angiotensin-converting enzyme (ACE) inhibitors may need to be stopped during acute illness, to reduce the risk of dehydration and acute kidney injury.
        - Be aware that the efficacy of some drugs (such as warfarin, anticonvulsants, and the oral contraceptive pill) may be affected by severe diarrhoea or vomiting.
    - Sexual history (particularly in men who have sex with men) to exclude sexually transmitted enteric infection.
    - Features suggesting an alternative diagnosis, including potentially severe or life-threatening conditions.
- **Examine the person (if necessary and appropriate):**
    - Assess for clinical features of dehydration or shock in children and adults, including general appearance (including irritability/lethargy), temperature, pulse, blood pressure (if possible), and breathing pattern.
    - Assess for possible sepsis.
    - Examine the abdomen for distension, tenderness, masses, and bowel sounds.
    - Assess weight and for signs of malnutrition.
    - Assess for features suggesting an alternative diagnosis.
- **Arrange for stool culture and sensitivity testing if indicated. This is *not* routinely needed for children and adults presenting with acute diarrhoea, but should be considered, depending on clinical judgement, if:**
    - The person is systemically unwell or immunocompromised.
    - There is acute painful diarrhoea or blood, mucus and/or pus in the stool (suggesting possible dysentery, such as caused by Shiga toxin-producing *E coli* [STEC] infection 0157, particularly in children).
    - The person has had recent antibiotic or proton pump inhibitor treatment, or recent hospital admission (to exclude *Clostridium difficile* infection).
    - Diarrhoea has not resolved by day 7.
    - There is suspected food poisoning.
    - The person has recently travelled abroad to anywhere other than western Europe, North America, Australia, or New Zealand.
        - Send an additional three specimens (5 mL each) for ova, cysts, and parasites 2–3 days apart to exclude parasite infection
    - Diarrhoea is recurrent or prolonged (lasts over 14 days).
        - Send an additional three specimens (5 mL each) for ova, cysts, and parasites 2–3 days apart to exclude parasite infection.
    - There is diarrhoea in a person at risk of transmission of infection, such as:
        - A food handler (work involves preparing or serving unwrapped ready to eat food and drink).
        - Clinical, social care, or nursery staff who work with young children, the elderly, or any other particularly vulnerable people.
        - Any person who is unable to perform adequate personal hygiene due to lack of capacity or ability to comply.
        - All children aged 5 years old or under (up to the sixth birthday) who attend school, pre-school, nursery, or other similar childcare or childminding groups (due to the increased risk of *E. coli* infection 0157).
    - There is uncertainty about the diagnosis, to help exclude alternative causes for symptoms.
- **Arrange for stool culture and sensitivity testing if there has been *contact with* an affected person or outbreak of diarrhoea:**

    - In a care home (increased risk of norovirus infection).
    - In a household or community, where isolating the microorganism will help pinpoint the cause.
    - With confirmed STEC infection, where there may be serious complications—symptomatic contacts and asymptomatic contacts at increased risk of transmission of infection, particularly all children aged 5 years old or under who attend childcare settings should be tested. Contacts may need to be excluded from work or childcare settings until a negative result is known.
    - With confirmed Shigella dysentery — asymptomatic contacts at increased risk of transmission of infection such as food handlers; clinical, social care, or nursery staff; and all children aged 5 years old or under who attend childcare settings may need to be tested, depending on specialist advice.
    - In a household with giadia spp or entameba histolytica
- **If there is any uncertainty regarding the need for stool testing, or follow-up testing for microbiological clearance after confirmed infection, seek advice from the** local health protection team
    - Advise that when collecting a stool sample.
- **Do not routinely arrange blood tests for acute gastroenteritis in primary care.**

Assessing for dehydration in children

Be aware that risk factors for developing dehydration in children include:

- Infants younger than 1 year of age, particularly younger than 6 months.
- Infants who were of low birthweight.
- Children who have passed six or more diarrhoeal stools in the previous 24 hours.
- Children who have vomited three or more times in the previous 24 hours.
- Children who have not been offered, or have not been able to tolerate, supplementary fluids before presentation.
- Infants who have stopped breastfeeding during the illness.
- Children with signs of malnutrition.

What else might it be?

Other conditions that may present similarly to acute gastroenteritis include:

- **Other infective causes, including:**
    - Urinary tract infection.
    - Pyelonephritis.
    - Chest infection.
    - Otitis media.
    - Sepsis.
- **Other causes of diarrhoea, including:**
    - Irritable bowel syndrome.
    - Inflammatory bowel disease.
    - Colorectal cancer.
    - Hirschsprung's enterocolitis.
    - Short bowel syndrome.
    - Food-sensitive enteropathy (such as lactose intolerance).
    - Coeliac disease.
    - Constipation with overflow.

    - Diabetes mellitus.
    - Hyperthyroidism.
    - Addison's disease.
    - HIV/AIDS.
    - Secretory tumours, for example, carcinoid tumours, medullary tumour of the thyroid, and vasoactive intestinal peptide (VIP)-secreting adenomas.
    - Drugs, for example, antibiotics, antimalarials, magnesium-containing antacids, antimotility drugs, and laxative misuse (the condition normally reverses on withdrawal of the drug).
- **Other causes of abdominal pain, including:**
    - Acute appendicitis.
    - Intussusception (may present with right lower quadrant tenderness, a sausage-shaped mass in the abdomen, and redcurrant jelly stool).
    - Malrotation of the midgut (may present with bilious vomiting, abdominal pain, diarrhoea, and bloody stools).
    - Biliary colic and acute cholecystitis.
    - Perforated peptic ulcer.
    - Diverticulitis.
    - Pancreatitis.
    - Testicular tortion (may present with abdominal pain and nausea or vomiting in boys).
    - Pelvic inflammatory disease, salpingitis, or tubo-ovarian abscess.
    - Ectopic pregnancy.

Management:

Scenario: Child gastroenteritis

From age 1 month to 15 years

When should I admit a child with gastroenteritis?

If a child presents with suspected gastroenteritis:

- **Arrange emergency hospital admission if:**
    - The child is systemically unwell and/or there are clinical features suggesting severe dehydration and/or progression to shock.
    - There is intractable or bilious vomiting.
    - There is acute-onset painful, bloody diarrhoea in previously healthy children, or confirmed Shiga toxin-producing Escherichia coli (STEC) infection 0157.
    - There is a suspected serious complication, such as haemolytic uraemic syndrome or sepsis.
- **Consider arranging hospital assessment, the urgency depending on clinical judgement, if:**
    - There are clinical features suggesting a potentially life-threatening or serious alternative diagnosis
    - There is an inadequate response to oral rehydration salt (ORS) solution, or the child is unable to take ORS solution orally.
    - There are red flag symptoms or signs suggesting increased risk of progression to shock, and the child is clinically deteriorating despite use of ORS solution.
    - There are risk factors for developing dehydration.
    - The parents/carers are unable to monitor the child's condition or provide appropriate management safely at home.
- **Liaise with a microbiologist or consultant in communicable disease control if:**

- There is uncertainty about the need for admission in a child with a confirmed microbiological cause on stool culture and sensitivity testing.
- There is uncertainty about the need for management in a child with moderate or severe suspected travellars diarrhea.

How should I manage a child with gastroenteritis in primary care?

If a child does not need immediate hospital admission, advise on management and follow up in primary care:

- **Advise on how to monitor fluid intake and prevent/treat dehydration.**
  - In children *without* clinical features of dehydration, advise the parents/carers to:
    - Continue with usual feeds, including breastfeeding (if applicable) and other milk feeds.
    - Encourage regular fluid intake.
    - Offer low-osmolarity oral rehydration salt (ORS) solution as supplemental fluid if the child is at increased risk of dehydration.
    - Discourage fruit juices and carbonated drinks, especially in children at increased risk of dehydration.
  - In children *with* clinical features of dehydration who can safely be managed at home, advise the parents/carers to:
    - Give the child low-osmolarity ORS solution frequently and in small amounts, to rehydrate the child.
    - Consider supplementation with usual fluids if the child refuses to take sufficient quantities of ORS solution, provided there are no red flag symptoms and signs (suggesting increased risk of progression to shock).
    - Discourage fruit juices and carbonated drinks.
    - Avoid giving solid food until the child is rehydrated.
  - After rehydration, advise the parents/carers to:
    - Encourage the child to drink plenty of usual fluids, including milk feeds (if these were stopped).
    - Reintroduce the child's usual diet.
    - Discourage fruit juices and carbonated drinks until the diarrhoea has stopped.
- **Advise that drug treatment with antidiarrhoeal drugs, antiemetics, zinc supplements, and probiotics is *not* recommended** for use in children in primary care.
- **Do not routinely prescribe antibiotics to children with gastroenteritis.**
  - Arrange treatment of confirmed microbial pathogens, if appropriate, following stool culture and sensitivity testing.
- **Advise on methods to prevent transmission of infection:**
  - Wash hands thoroughly with liquid soap in warm running water, and dry carefully.
  - Always use a flush toilet, if possible.
    - If a potty must be used, it should be handled with gloves, the contents disposed of into the toilet, and then washed with hot water and detergent and allowed to dry.
    - Wash hands after going to the toilet and changing nappies; before preparing, serving, or eating food; and after handling pets or non-domestic animals.
    - Toilet seats, flush handles, wash-hand basin taps, surfaces, and toilet door handles should be cleaned at least once daily with hot water and detergent. A disinfectant and disposable cloth (or one dedicated for toilet use) should be used to clean toilets.

    - Do not share towels/flannels and bathwater used by infected children.
    - Wash soiled clothing and bed linen separately from other clothes and at a high temperature (for example 60°C or higher for cotton), after removal of excess faecal matter or vomitus into the toilet. Soaking in disinfectant is not necessary. The washing machine should not be more than half full to allow for adequate washing and rinsing.
- **Advise that children should not attend school or other childcare/social settings until at least 48 hours after the last episode of diarrhoea or vomiting.**
    - Note: for some gastrointestinal infections, longer periods of exclusion are needed, and 'microbiological clearance' with negative stool testing may be required before return to specific settings.
    - Note: if cryptosporidiosis or giardiasis is suspected or confirmed, children should not go swimming for 2 weeks after the last episode of diarrhoea.
- **Advise parents/carers on the need for follow up and to seek urgent medical advice if a child has:**
    - Any features requiring hospital admission or assessment.
    - Any new features requiring stool culture testing, such as blood, mucus, and/or pus in the stool (suggesting possible dysentery).
    - Symptoms that do not resolve within the expected timeframe — arrange stool culture testing and consider whether an alternative diagnosis is possible.
        - Diarrhoea usually lasts 5–7 days, and in most children stops within 2 weeks.
        - Vomiting usually lasts 1–2 days, and in most children stops within 3 days.
- **Notify the** local health protection team **immediately by completing a** notification form.

- **Fluid volume requirements**

Fluid volume requirements to maintain hydration in children are based on body weight:

- Weight 0–10 kg: child needs 100 mL/kg fluids per day.
    - For example, if a child weighs 5 kg, the maintenance fluid volume is 500 mL/day.
- Weight 10–20 kg: child needs 1000 mL plus 50 mL/kg for each kg over 10 kg per day.
    - For example, if a child weighs 15 kg, the maintenance fluid volume is 1250 mL/day.
- Weight more than 20 kg: child needs 1500 mL plus 20 mL/kg for each kg over 20 kg per day.
    - For example, if a child weighs 25 kg, the maintenance fluid volume is 1600 mL/day.

If using low-osmolarity oral rehydration salt (ORS) solution to rehydrate a child (240–250 mOsm/L):

- Age 5 years or younger: give 50 mL/kg body weight for fluid deficit replacement, as well as maintenance volume of ORS solution, over 4 hours.
    - Note: breastfeeding can continue, but do not routinely give oral fluids other than ORS solution.
- Age 5–11 years: give 200 mL ORS solution after each loose stool, in addition to the child's normal fluid intake.
- Age 12–16 years: give 200–400 mL ORS solution after every loose stool, dose according to fluid loss.
- After rehydration, for children at increased risk of dehydration: give 5 mL/kg body weight after each large watery stool, to prevent recurrence of dehydration.

How should I manage gastroenteritis due to a confirmed microbiological cause?

If a child with gastroenteritis has a specific microbiological cause confirmed after stool culture and sensitivity testing:

- **Seek advice from the** local health protection team **regarding the need for antibiotic treatment or stool testing for microbiological clearance, particularly in a child:**
  - Younger than 6 months of age with salmonella infection, or who is malnourished or immunocompromised with salmonella infection.
  - With Shiga toxin-producing Escherichia coli (STEC) infection.
    - There is no effective antibiotic treatment available for STEC infection, and advise *not* to give antibiotic treatment.
    - Specialist advice is needed regarding monitoring for the complication of hemolytic ureamic syndrome.
    - Specialist advice is also needed regarding stool testing for microbiological clearance in children at increase risk of transmission of infection — two consecutive negative stool samples are usually needed taken at least 24 hours apart, once the child is symptom-free for at least 48 hours, before the child can return to school or other childcare settings.
  - With giardiasis
    - Drug treatment with tinidazole may be recommended following specialist advice.
  - With dysenteric shigellosis.
    - Antibiotic treatment is not usually needed for children with mild symptoms of shigellosis, but if symptoms are severe (high fever, bloody and/or high-output diarrhoea) or the child is immunocompromised, seek specialist advice on the need for antibiotic treatment.
    - Specialist advice is needed regarding exclusion of cases from childcare settings, and the need for microbiological clearance stool testing, which varies depending on the *Shigella* species.
  - With dysenteric amobeasis.
    - Drug treatment is usually recommended for all confirmed cases after specialist advice.
    - Specialist advice is needed regarding the need for microbiological clearance to confirm treatment success, 1 week after completing treatment.
  - With camplobacteriosis.
    - Antibiotic treatment is not usually needed for children with mild symptoms, as infection is usually self-limiting.
    - If symptoms are severe (high fever, bloody and/or high-output diarrhoea), the child is immunocompromised, or symptoms are worsening, seek specialist advice regarding the need for antibiotic treatment.
  - With crypyosporidiosis.
    - There is no specific treatment licensed in the UK.
    - Seek specialist advice if the child is severely immunocompromised, owing to the risk of serious and life-threatening complications.

Scenario: Adult gastroenteritis:

How should I manage gastroenteritis due to a confirmed microbiological cause?

If an adult with gastroenteritis has a specific microbiological cause confirmed after stool culture **and sensitivity testing:**

- **Seek advice from the** local health protection team **regarding the need for antibiotic treatment or stool testing for microbiological clearance, if there is any uncertainty:**
  - cambylobacteriosis
    - Antibiotic treatment is not usually needed for people with mild symptoms, as infection is usually self-limiting.
    - If symptoms are severe (high fever, bloody and/or high-output diarrhoea) or the person is immunocompromised, consider early prescribing with clarithromycin 250–500 mg twice daily for 5–7 days, within 3 days of onset of illness.
  - cryptosporidiosis
    - There is no specific treatment licensed in the UK.
    - Seek specialist advice if the person is severely immunocompromised, owing to the risk of serious and life-threatening complications.
  - amobiasis
    - Drug treatment is usually recommended for all confirmed cases after specialist advice, such as metronidazole followed by the anti-protozoal drug diloxanide.
    - Seek specialist advice regarding the need for microbiological clearance to confirm treatment success, 1 week after completing treatment.
  - E coli
    - There is no effective antibiotic treatment available for Shiga toxin-producing *E. coli* (STEC) infection.
    - Seek specialist advice on the need for stool testing for microbiological clearance in people at increased risk of transmission of infection, if there is confirmed STEC infection — two consecutive negative stool samples are usually needed taken at least 24 hours apart, once the person is symptom free for at least 48 hours, before the person can return to work or other institutional/social settings.
    - Seek specialist advice if there is confirmed STEC infection regarding monitoring for the complication of hemolytic ureamic syndrome.
    - Seek specialist advice regarding the need for exclusion and microbiological clearance in non-STEC infection in people at increased risk of transmission of infection, such as food handlers — two consecutive negative stool samples are usually needed, the second sample taken 48 hours after the person is symptom free, before the person can return to work, depending on a risk assessment.
  - Giardiasis
    - Prescribe tinidazole 2 g as a single dose.
  - Salmonellosis (non typhoid)
    - Antibiotic treatment is not usually needed.
  - Shigellosis
    - Antibiotic treatment is not usually needed for people with mild symptoms.
    - If symptoms are severe (high fever, bloody and/or high-output diarrhoea) or the person is immunocompromised, seek specialist advice on the need for antibiotic treatment.
    - Advise on safe sexual practices if transmission amongst men who have sex with men has resulted in an outbreak.

- Seek specialist advice on the need for exclusion of cases from childcare or work settings, and the need for microbiological clearance stool testing, which varies depending on the *Shigella* species.

Prescribing information:

Tinidazole

Contraindications and cautions

- **Do not prescribe tinidazole to:**
  - Pregnant women in the first trimester — the manufacturer advises to avoid.
- **Prescribe tinidazole with caution to:**
  - Women who are breastfeeding — the manufacturer advises to avoid breastfeeding during and for 3 days after stopping treatment.

Adverse effects

Possible adverse effects of tinidazole include:

- Abdominal pain, decreased appetite, diarrhoea, headache, nausea, skin reactions, vertigo, and vomiting.

Drug interactions

Possible drug interactions of tinidazole include:

- Alcohol — alcohol potentially causes a disulfiram-like reaction when given with tinidazole. The manufacturer advises to avoid for 72 hours after stopping treatment.
- Warfarin — tinidazole is predicted to increase the anticoagulant effect of warfarin. The manufacturer advises to monitor the international normalised ratio (INR) and adjust the dose accordingly.

TOPIC: TRAVEL ADVICE AND IMMUNIZATION

Advise that the most common infectious diseases contracted abroad cannot effectively be prevented by vaccination.

- Travellers' diarrhoea may occur in up to 50% of visitors abroad and can generally be **prevented by avoiding contaminated food and drink**.
- Malaria is one of the more common serious acute infectious diseases that is contracted abroad and can be prevented with appropriate **chemoprophylaxis**.
- Zika virus is spread by day-biting mosquitos, is present in Africa, Asia, the Pacific Islands, Central and South America and the Caribbean, and can pose a risk to a developing baby if contracted during pregnancy. The risk of Zika virus can be reduced by using **insect bite avoidance measures**.
- Sexually transmitted infections are commonly contracted abroad. Some areas, such as Thailand, sub-Saharan Africa, and the Indian subcontinent, are endemic with HIV.
- Parasitic infestation (such as onchocerciasis, loiasis, schistosomiasis) may present long after the person has returned.

The following diseases are endemic in many regions of the world and travellers to certain areas may require vaccination against some or all the following illnesses:

- Hepatitis A.
- Meningococcal meningitis

Meningococcal ACWY vaccine is recommended for travelers to the meningitis belt in Africa, especially during the dry season (from December through June) and year-round for health care workers. Travelers to Saudi Arabia during the Hajj are required to have a certificate of vaccination with quadrivalent meningococcal vaccine before entering, issued not more than 5 years and not less than 10 days before arrival.

- Poliomyelitis.
- Tetanus.
- Typhoid fever
- Yellow fever.

YF is a mosquito-borne viral infection endemic in equatorial Africa and in areas of South America but not in Asia. Fatalities due to YF acquired by unvaccinated tourists have occurred. YF vaccination is legally required for entrance into specific countries

The following diseases are endemic in parts of the world but do not usually pose a serious risk to travellers overseas. Vaccination against these diseases may be recommended in specific circumstances:

- Cholera.
- Hepatitis B.
- Japanese encephalitis.
- Rabies.
- Tick-borne encephalitis.

Pretravel visit provides an opportunity to ensure that routine immunizations including:

- SARS-CoV-2 vaccine
- Influenza vaccine
- Tetanus, diphtheria, and pertussis vaccine
- Measles, mumps, and rubella
- Poliovirus vaccine
- Varicella vaccine

TOPIC: PRE-TRAVEL COUNSELING

Histoty

-**Trip:** Where? When? How long? Transportation? Trip program? Activities? Presence of children, old or pregnant?
- **Medical:** chronic disease, drugs, allergies, CVD, seizure, psychiatric, thromboembolic + check vaccination status + ICE

Vaccination

Check CDC for common diseases in destination & prophylaxis:
- **Hepatitis A** (2 doses IM, 6:36 mo apart, protection 2-4 wks after 1st dose, up to 25 yrs)
- **Meningococcal** (IM, 2 doses, > 8 wks apart)
- **Typhoid** (4 capsules, one every other day, protection after 2 wks, up to 5 yrs)
- **Yellow fever** (single dose, IM/SC, protection after 10d, lifelong)
- **Japanese encephalitis** (SC, 2 doses, 4 wks apart, protection 1 wk after last dose)
- **Malaria prophylaxis:** Chloroquine 500mg tab weekly, 1-2 wks before, up to 4 wks after leaving - or Doxycycline 100 mg OD (if chloroquine-resistant area) 1-2 days before, up to 4 wks after

Protection

DEET Insect repellant on skin (Off® spray/lotion) & permethrin on clothes and bed nets (EctomethrinR solution) - conditioned room - avoid outdoor activities in dark/dusk - wear long sleeves & tight pants - inspect skin & clothes for ticks daily

Injuries

- **Solar injury** (sunblock SPF≥ 15 - shade- umbrella/wide hat)
- **Food safety** (well cooked food from trusted restaurant - don't reheat only bottled water - avoid ice - hand hygiene)
- **Accidental injuries** (RTAs, drowning, animal bite)
- **Safe sex**
- **Transportation related** (DVT exercise & compression stockings motion sickness → control breathing - avoid heavy /spicy/acidic meal, alcohol, nicotine & caffeine, dehydration - choose center or front of the vehicle support head - Avoid noxious stimuli- look forward at a fixed point - scopolamine patch 4 hrs before travel, then every 72 hrs as needed)
- **Altitude sickness:** Slow ascent/rapid descent - slowly acclimatize + Acetazolamide 125mg BID, for 3d start 1d before

Medical kit:

Alcohol swabs - tweezer - cure aid - scissors - gauze/crepe bandage - analgesic - antihistaminic - Mebo cream- enough stock of prescribed drugs - medical bracelet

Pregnancy

Air travel allowed up to 36 wks - avoid scuba diving, bicycling & skiing - avoid countries with Zika virus - avoid live vaccines

TOPIC: GONORRHEA

- **Gonorrhoea is a sexually transmitted infection (STI) caused by the Gram-negative bacterium *Neisseria gonorrhoeae.***
- Uncomplicated gonorrhoea infection primarily affects the mucous membranes of the urethra, endocervix, rectum, pharynx, and conjunctiva.
- Gonorrhoea is transmitted by direct inoculation of secretions from one mucous membrane to another.
    - Infection of the eye most commonly results from autoinoculation.
    - Gonococcal infection among infants usually results from exposure to infected cervical exudates at birth.

What are the risk factors?

Risk factors for gonorrhoea infection include:

- Young age (15–24 years).
- New sexual contact in the last year, or more than one partner in the last year.
- Inconsistent condom use.
- Certain sexual activities, for example men who have sex with men (MSM).
- Current or prior history of sexually transmitted infection (risk factor for repeat infections).
- History of sexual or physical abuse.
- Previous incarceration.
- Deprivation.

What is the prognosis?

- In most women, gonorrhoea infections usually resolve spontaneously. However, in both men and women, untreated infections may result in complications.
- Appropriate treatment with recommended antibiotics should resolve gonorrhoea infections.

What are the complications?

- **Complications of untreated gonorrhoea include:**
    - **In men:**
        - Epididymitis or orchitis.
        - Prostatitis.
        - Infertility.
        - Urethral stricture.
        - Infection of Mullerian or Cowper glands.
    - **In women:**
        - Pelvic inflammatory disease — this occurs in up to 33% of women with gonorrhoea and can result in chronic pelvic pain, tubal infertility, or ectopic pregnancy.
        - Rarely, peritoneal spread, including perihepatic abscesses (Fitz-Hugh-Curtis syndrome).
        - Pregnancy complications, including spontaneous abortion, premature labour, early rupture of fetal membranes, perinatal mortality, and gonococcal conjunctivitis in the newborn.

- **Disseminated gonorrhoea is a potentially serious complication that is thought to occur in 0.5–3% of untreated gonorrhoea cases.** It occurs with bacteraemia and spreads, leading to septic arthritis, polyarthralgia, tenosynovitis, petechial/pustular skin lesions, or, on rare occasions, endocarditis, or meningitis.

Diagnosis:

What are the clinical features of a gonorrhoea infection?

- **Gonorrhoea is primarily associated with uncomplicated infection of the lower genital tract, which is symptomatic in most men (over 90%) and approximately 50% of women.**
- **In men:**
  - Urethral infection causes mucopurulent or purulent urethral discharge (in more than 80% of cases) and/or dysuria (in more than 50% of cases), appearing 2–8 days after exposure. Frequency and urgency are usually absent. Rarely, the person may complain of testicular and epididymal pain. Uretheral infection is asymptomatic in less than 10% of men.
  - Rectal infection is usually asymptomatic but may cause anal discharge, acute proctitis, perianal/anal pain or discomfort, tenesmus, or rectal bleeding. Rectal pain is more common in men who have sex with men.
  - Pharyngeal infection is asymptomatic in most cases but is occasionally associated with a sore throat.
- **In women:**
  - Urethral infection may present with dysuria without urinary frequency.
  - Endocervical infection may present with:
    - Increased or altered vaginal discharge (up to 50% of cases).
    - Lower abdominal pain (up to 25% of cases).
    - Intermenstrual bleeding or menorrhagia (rarely).
    - Pain during intercourse (dyspareunia) if the infection spreads from the endocervix.
  - Rectal infection is usually asymptomatic but may cause anal discharge and perianal/anal pain or discomfort. Rectal infection in cisgender women is present in up to 30% of cases of urogenital infection, and individuals may not report a history of anal sex. Limited evidence suggests that rectal infection in the absence of urogenital infection is uncommon.
  - Pharyngeal infection is asymptomatic in most cases but is occasionally associated with a sore throat.

How should I assess a person with suspected gonorrhoea?

- **Take a history.**
  - Ask about the clinical features of gonorrhoea.
  - Take a sexual history. Ask about:
    - Partners (including sex of partners and the number of partners in the last year).
    - Practices or types of sexual activities.
    - Previous sexually transmitted infections (STIs).
    - Protection from STIs/HIV (including the use of barrier contraception and intravenous drug use).

        - Prevention of pregnancy (including contraceptive use or whether they are trying to conceive).
- **Examine the person.**
    - **In men:**
        - Inspect and palpate the testes, epididymis, and spermatic cord — epididymitis can present as unilateral testicular pain (without discharge or dysuria), fever, and a swollen and tender epididymis on palpation.
        - Examine the penis shaft, glans, and meatus — in urethral infection, a mucopurulent or purulent urethral discharge is often present on examination. Rarely, the person may complain of testicular and epididymal pain, with tenderness and swelling present on examination.
        - Examine the prostate if symptoms of prostatitis are present. These include pain in the lower back and genital area, urinary frequency and urgency (often at night), and burning or painful urination.
    - **In women:**
        - Inspect the labia and clitoris, then carry out speculum examination of the cervix and vagina — in most women, no abnormal findings are present on examination. However, a mucopurulent discharge may be evident from the cervix, sometimes accompanied with hyperaemia and contact bleeding of the endocervix.
        - Conduct a bimanual pelvic examination for cervical motion tenderness, uterine tenderness, and adnexal tenderness, to assess possible ascending infection which may result in pelvic inflammatory disease (PID).
        - Pelvic and lower abdominal tenderness is an uncommon examination finding in the absence of co-infection with *Chlamydia trachomatis*.
    - **Assess for** complications **of gonorrhoea infection,** including epididymitis and orchitis in men, and PID in women.
    - **Assess for extra-genital infection:**
        - Rectal gonorrhoea may cause mucopurulent discharge from the anus.
        - Pharyngeal gonorrhoea can present with erythema and exudate. Anterior cervical lymphadenopathy may be present.
        - Gonococcal conjunctivitis can present with a thick white/yellow discharge. Examine the eyes with a slit lamp (where available) to exclude corneal infection.
- **Consider screening for other STIs** (chlamydia, HIV, and syphilis as a minimum).
- **Exclude differential diagnoses,** such as candida or genital herpes simplex infections.
- **In children and young people who present with gonorrhoea infection:**
    - Consider the possibility of sexual abuse, particularly in the following circumstances:
        - The child is younger than 13 years of age, unless there is clear evidence of mother-to-child transmission during birth, or of blood contamination.
        - The young person is aged 13–15 years, unless there is clear evidence of mother-to-child transmission during birth, blood contamination, or that the STI was acquired from consensual sexual activity with a peer.
        - The young person is aged 16–17 years and there is no clear evidence of blood contamination or that the STI was acquired from consensual sexual activity, and there is a clear difference in power or mental capacity between the young person and their sexual partner, in particular when the relationship is incestuous or with a person in a position of trust (such as a teacher, sports coach, minister of religion) or there is concern that the young person is being exploited.
    - If child maltreatment is suspected, refer the young person to children's social care, following Local Safeguarding Children Board procedures.

How should I confirm a diagnosis of gonorrhoea?

All people with suspected gonorrhoea should be referred to a genito-urinary medicine (GUM) clinic or other local specialist sexual health service for confirmation of the diagnosis. If the person is unwilling or unable to attend a GUM clinic or other local specialist sexual health service, this can be done in primary care if the appropriate expertise is available and in line with local procedures and protocols.

- **The diagnosis of gonorrhoea is established by the detection of *Neisseria gonorrhoea* at an infected site, either by nucleic acid amplification tests (NAATs) or by culture.**
  - The approach and method used to test for gonorrhoea will be influenced by the clinical setting, storage and transport system to the laboratory, local prevalence of infection, and the range of tests available in the laboratory.
  - No test for gonorrhoea offers 100% sensitivity and specificity.
  - The use of dual NAATs, detecting both chlamydia and gonorrhoea, could lead to inappropriate testing for gonorrhoea.
- **If the person is unwilling, or unable, to attend a GUM clinic or other local specialist sexual health service,** a NAAT should be arranged in line with local procedures and protocols.
  - In women, a vulvovaginal swab (which may be self-taken) should be used.
    - For those who have had a hysterectomy, there is no evidence of an optimal sampling site. Consider urine and vulvovaginal swab for NAAT with subsequent culture from that site if positive.
  - In men, a first pass urine specimen should be used.
  - Rectal and pharyngeal sampling should be:
    - Routine in all men who have sex with men (MSM).
    - Considered in women who are sexual contacts of gonorrhoea.
    - Guided by an assessment of risk and symptoms in everyone else.
  - Oropharyngeal infection is more difficult to treat. Therefore, all people with genital gonorrhoea (regardless of gender or reported sexual behaviour) should have pharyngeal sampling if either of the following apply:
    - Susceptibility results are not available and the infection may have been acquired in the Asia-Pacific region. This is because of high levels of antimicrobial resistance in that region which may lead to treatment failure.
    - Genital infection with a confirmed ceftriaxone-resistant strain.
- **Specimens for culture (urethral, endocervical, neovaginal, anorectal, and pharyngeal swabs) should be taken alongside NAATs from people suspected clinically of having gonorrhoea (and from their sexual contacts).**
  - All people with gonorrhoea diagnosed by NAAT should have cultures taken for susceptibility testing prior to treatment.

Nucleic acid amplification tests (NAATs)

- NAATs amplify and detect nucleic acid sequences that are specific for the organism being detected.
  - They are extremely sensitive and can detect very small amounts of DNA or ribosomal RNA (rRNA). However, detection does not mean the organism is viable.
  - NAATs are approved for most anatomical sites and show high sensitivity (more than 96%) in both symptomatic and asymptomatic infection, which is higher than for

culture (85–95%). Therefore, although NAATs are not licensed for use at extra-genital sites, their use is recommended.
    - NAATs do not provide information on antimicrobial susceptibility. Culture is essential for monitoring antimicrobial susceptibility and to detect emerging resistance to therapy.

What else might it be?

- **Differential diagnoses in men include:**
    - Non-gonococcal urethritis caused by *Chlamydia trachomatis*, *Ureaplasma urealyticum*, *Mycoplasma genitalium*, or *Trichomonas vaginalis.*
    - Acute prostatitis.
    - Genital herpes simplex infection.
    - *Candida* infection.
- **Differential diagnoses in women include:**
    - Chlamydia — it is not possible to distinguish gonorrhoeal infection from chlamydia in women by clinical features alone.
    - Candida infection.
    - Bacterial vaginosis.
    - Trichomoniasis.
    - Pelvic inflammatory disease.
    - Genital herpes simplex infection.

Management:

Scenario: Management

From age 13 years onwards

When should I admit or refer a person with gonorrhoea?

All people with gonorrhoea should be referred to a genito-urinary medicine (GUM) clinic or other local specialist sexual health service for management. If the person is unwilling or unable to attend a GUM clinic or other local specialist sexual health service, management can be undertaken in primary care if the appropriate expertise is available and in line with local procedures and protocols.

- **Arrange hospital admission for:**
    - People with suspected disseminated gonorrhoea (systemic symptoms may be present, such as fever, malaise, joint pain and swelling, and rash).
    - Women with severe or complicated pelvic inflammatory disease.
- **Refer the following people to the appropriate speciality:**
    - Women with suspected ascending infection.
    - People with complications
    - People with conjunctival gonorrhoea.
    - People who do not respond (or are allergic) to recommended antibiotics.

How should I manage a person with gonorrhoea infection?

If the person is unwilling, or unable, to attend a GUM clinic or other local specialist sexual health service despite receiving appropriate information and advice:

- **Prescribe antibiotic treatment.** Ideally, a culture should be taken before prescribing antibiotics, to test for susceptibility and identify resistant strains.
    - **For people with uncomplicated anogenital or pharyngeal infection:**
        - When the antimicrobial susceptibility is known prior to treatment, prescribe ciprofloxacin 500 mg orally as a single dose.
        - When antimicrobial susceptibility is not known prior to treatment, prescribe ceftriaxone 1 g intramuscular (IM) injection as a single dose.
    - **Alternative treatments are recommended for people with an allergy, needle phobia, or other contraindications.**
        - These include gentamicin 240 mg IM as a single dose plus azithromycin 2 g orally OR cefixime 400 mg orally as a single dose plus azithromycin 2 g orally (advisable only if an IM injection is contraindicated or refused by the person).
        - When using alternative regimens without antibiotic susceptibility data, it is recommended to regularly review local and national trends in gonococcal antimicrobial resistance.
    - **For people with penicillin allergy,** ceftriaxone and cefixime are suitable treatment options unless there is a history of severe hypersensitivity (for example, anaphylactic reaction) to any beta-lactam antibacterial agent (penicillins, cephalosporins, monobactams, or carbapenems).
    - **For pregnant or breastfeeding women,** prescribe ceftriaxone 1 g IM injection as a single dose.
        - Azithromycin 2 g as a single oral dose can be used if adequate alternatives are not available and the isolate is known to be susceptible. Otherwise, seek specialist advice.
        - **Do not** prescribe ciprofloxacin.
- **Offer screening for other sexually transmitted infections (STIs) and HIV.**
- **Encourage patient-led partner notification.**
    - Advise the person on:
        - The importance and benefits of partner notification.
        - The possibility of sexual partners being infected even if asymptomatic.
        - The risk of reinfection.
- **Provide appropriate information and advice.**
    - Discuss the possible complications of untreated gonorrhoea.
    - Advise the person to abstain from sex until 7 days after they and their partner(s) have completed treatment.
    - Advise the person on using safer sexual practice in the future.
- **Follow up the person after about 1 week.**
    - Test of cure is recommended for all people who have been treated for gonorrhoea.

Treatment of gonorrhoea

- BASHH recommends treatment in the following cases**:**
    - A confirmed positive nucleic acid amplification tests (NAATs) for *Neisseria gonorrhoeae*.
    - A positive culture for *N. gonorrhoea*e.
    - A laboratory result where there has been identification of intracellular Gram-negative diplococci on microscopy.
    - Sexual partner of confirmed case of gonococcal infection.

How should I manage the sexual contacts of a person with gonorrhoea?

- **The following partners should be notified:**
  - For men with symptomatic urethral infection, all sexual partners within the preceding 2 weeks (1mth in chlamydia), or their most recent partner if this was longer than 2 weeks ago.
  - For all other people (that is, women and men with asymptomatic gonorrhoea or gonorrhoea at other sites), all partners within the preceding 3 months (6mth in chlamydia)
- **Empirical treatment is *not* needed for all sexual contacts.**
  - For those presenting after 14 days of exposure, empirical treatment is recommended only following a positive test for gonorrhoea.
  - For those presenting within 14 days of exposure, empirical treatment should be considered based on a clinical risk assessment and following a discussion with the person. In asymptomatic individuals, it may be appropriate to not give epidemiological treatment, and to repeat testing 2 weeks after exposure.

What follow up is required for a person with gonorrhoea?

- **Follow up all people with gonorrhoea about 1 week after treatment to:**
  - Confirm that the person has adhered to treatment and symptoms have resolved.
  - Ask about adverse reactions.
  - Confirm that partner notification has been carried out.
  - Ask about recent sexual history (and the possibility of re-infection).
  - Reinforce advice on safe sexual practice.
- **A test of cure (TOC) is recommended for all people who have been treated for gonorrhoea.**
  - A positive TOC could be due to treatment failure, reinfection, or residual non-viable organism, and should be interpreted in the clinical context:
    - If signs or symptoms persist, test with culture performed at least 3 days after completion of treatment. Consider additional testing with NAAT (for increased sensitivity) after 1 week if culture is negative.
    - If the person is asymptomatic, test with nucleic acid amplification testing (NAAT) where available (followed by culture if positive), at least 2 weeks after completion of treatment.

TOPIC: CHALYMDIA – UNCOMPLICATED GENITAL

- Genital chlamydial infection is the most commonly reported, curable, bacterial sexually transmitted infection (STI) in the UK. It is caused by the obligate intracellular bacterium *Chlamydia trachomatis*.
    - At least 70% of women and 50% of men infected with *C. trachomatis* are asymptomatic.
    - Infection of the urogenital tract typically causes inflammation of the:
        - Urethra in men.
        - Cervix and/or urethra in women.
    - *C. trachomatis* can also infect the conjunctiva, rectum, and nasopharynx.
- Chlamydial infection is termed:
    - 'Uncomplicated' when the infection has not ascended to the upper genital tract.
    - 'Complicated' when the infection has spread to the upper genital tract, causing pelvic inflammatory disease (PID) in women and epididymo-orchitis in men.

What are the risk factors?

- Risk factors for chlamydia infection include:
    - Age under 25 years.
    - A new sexual partner.
    - More than one sexual partner in the last year.
    - Lack of consistent condom use.
- Infection is primarily through penetrative sex, but can also occur via autoinoculation or splash from genital fluids.

What is the prognosis?

- Untreated infection may persist or resolve spontaneously.
- Clearance increases with the duration of untreated infection, with up to 50% of infections resolving within 12 months of diagnosis.
    - However in some people, untreated genital infection may result in serious complications — risk of developing complications increases with repeated infection.

What complications are associated with chlamydia infection?

Chlamydia infection can lead to complications such as:

- **Pelvic inflammatory disease (PID), including endometritis and salpingitis** — occurs in up to 16% of women with untreated chlamydia infection. PID increases the risk of tubal infertility, ectopic pregnancy, and chronic pelvic pain.
- **Epididymo-orchitis** (pain, swelling, or inflammation of the epididymis and/or testicles) can result from untreated chlamydia infection in men.
- **Lymphogranuloma venereum (LGV).**
    - LGV is most common in men who have sex with men (MSM), with highest rates in men with HIV.
- **Sexually acquired reactive arthritis (SARA).**
    - SARA is a seronegative spondyloarthropathy.
- **Perihepatitis (Fitz-Hugh-Curtis syndrome).**

    - Inflammation of the hepatic capsule associated with right upper quadrant pain that may be referred to right shoulder.
- **Adverse outcomes in pregnancy** — chlamydia infection in pregnancy is associated with adverse outcomes including:
    - Increased risk of premature rupture of membranes, pre-term delivery, and low birth weight in the infant.
    - Increased risk of intra-partum pyrexia and late post-partum endometritis.
    - Infections of the eyes, lungs, nasopharynx, and genitals in the neonate, due to exposure in the birth canal during delivery.
- **Conjunctivitis** — can occur via autoinoculation or splash from genital fluids.
- **Anxiety and psychological distress.**

Diagnosis:

When should I screen for chlamydia in primary care?

- **Asymptomatic people who should be screened for chlamydia include:**
    - Sexual partners of those with proven or suspected chlamydial infection.
    - All sexually active people younger than 25 years of age, annually, or more frequently if they have changed their partner.
    - All people with concerns about a sexual exposure.
        - If the exposure was within the last 2 weeks, a test should be carried out at presentation and if negative, repeated 2 weeks after the exposure.
    - People under the age of 25 years who have been treated for chlamydia in the previous 3 months.
    - People who have had two or more sexual partners in the previous 12 months.
    - All women seeking termination of pregnancy (TOP).
    - All men and women attending genito-urinary medicine clinics.

When should I suspect and test for chlamydia?

- **Be aware that chlamydia is asymptomatic in around 70% of women and 50% of men, and that symptoms in men can be very mild.**
    - Suspect chlamydia in sexually active women with:
        - Increased vaginal discharge.
        - Post-coital or intermenstrual bleeding.
        - Purulent vaginal discharge.
        - Mucopurulent cervical discharge.
        - Deep dyspareunia.
        - Dysuria.
        - Pelvic pain and tenderness.
        - Cervical motion tenderness.
        - Inflamed or friable cervix (which may bleed on contact).
    - Suspect chlamydia in sexually active men with:
        - Dysuria.
        - Mucoid or mucopurulent urethral discharge.
        - Urethral discomfort/urethritis.
        - Epididymo-orchitis.
        - Reactive arthritis.
- **Symptoms of lymphogranuloma venereum (LGV) include:**
    - Tenesmus.
    - Anorectal discharge (often bloody) and discomfort.

    - Diarrhoea or altered bowel habit.
- **Symptoms of rectal chlamydia include:**
    - Anal discharge and anorectal discomfort, although rectal infection is usually asymptomatic.
- **Symptoms of adult chlamydial conjunctivitis include:**
    - Unilateral chronic low grade conjunctival irritation (may be bilateral).
- **Symptoms of oropharyngeal infection include:**
    - Pharyngitis and sore throat but oropharyngeal infection is usually asymptomatic.

How should I test for chlamydia?

If chlamydia is suspected, samples should be taken for nucleic acid amplification tests (NAATs) to confirm the diagnosis.

- **In women:**
    - **A vulvo-vaginal or endocervical swab can be taken** — a vulvo-vaginal swab is the sample of choice in women.
        - A vulvo-vaginal swab is collected by inserting the swab about 5 cm into the vagina and gently rotating for 10–30 seconds.
        - An endocervical swab is taken using a speculum, with the swab rotated 360° inside the cervical os.
    - **Alternatively, a first-catch urine (FCU) sample can be collected** if the woman prefers this.
        - To collect FCU, urine should have been held in the bladder for at least 1 hour before being tested.
        - The first 20 mL of the urinary stream should be captured.
    - Kits for self-taken vulvo-vaginal swab or FCU are available.
- **In men:**
    - **A FCU is the specimen of choice.**
        - Kits for self-taken FCU are available.
    - **A urethral swab is an alternative.**
        - This is collected by inserting a swab about 2–4 cm into the urethra and rotating once before removal.
- **Extra-genital samples**
    - Rectal swabs in symptomatic men and women can be taken 'blind' by the person or a clinician.
    - All people with proctitis should have rectal swabs taken to test for lymphogranuloma venereum (LGV).
    - All HIV-positive men who have sex with men (with or without symptoms) with *Chlamydia trachomatis* at any site should have rectal swabs taken to test for LGV.

What else might it be?

- **Chlamydia can present with similar clinical features to:**
    - **Other sexually transmitted infections such as:**
        - Gonorrhoea.
        - Bacterial vaginosis.
        - Vaginal candidiasis.
        - Trichomonas vaginitis.
    - **Pelvic inflammatory disease.**
    - **Urinary tract infection.**

Management:

Scenario: Management of uncomplicated genital chlamydia

From age 13 years onwards

How should I manage a person with suspected or confirmed chlamydia?

- **If chlamydia infection is suspected or confirmed, strongly recommend referral to a genito-urinary medicine (GUM) clinic for management.**
  - If the person declines, or is unable to attend a GUM clinic, manage uncomplicated genital chlamydia infection in primary care.
- **If a sexually transmitted infection (STI) is suspected or diagnosed in a child or young person:**
  - Have a low threshold for discussing with a GUM specialist or paediatrician — local child safeguarding procedures should be followed.
- **Advise the person that:**
  - Their current partner(s) must also be treated for chlamydia to reduce the risk of re-infection and onward transmission.
  - Sexual intercourse (including oral sex) should be avoided until they and their partner(s) have completed treatment (or waited 7 days after treatment with azithromycin).
- **Treat the infection:**
  - For non-pregnant adults and children over the age of 13 years:
    - First-line treatment is doxycycline 100 mg twice daily for 7 days (contraindicated in pregnancy and breastfeeding).
    - If doxycycline is contraindicated or not tolerated consider azithromycin 1 g orally as a single dose for 1 day, followed by 500 mg orally once daily for 2 days.
    - If doxycycline or azithromycin are contraindicated consider erythromycin 500 mg twice daily for 10–14 days. Ofloxacin 200 mg twice daily for 7 days, or 400 mg once daily for 7 days is a possible alternative, but it is contraindicated in pregnancy, children, and growing adolescents.
  - For pregnant women — discuss management with other healthcare professionals involved in the woman's care (such as the woman's midwife and obstetrician) and GUM. Options that may be considered for treatment include:
    - Azithromycin, 1 g orally for 1 day, then 500 mg orally once daily for 2 days, *or*
    - Erythromycin 500 mg four times daily for 7 days, or erythromycin 500 mg twice daily for 14 days, *or*
    - Amoxicillin 500 mg three times a day for 7 days.
  - If there is any uncertainty, seek specialist advice.
- **Refer the person to GUM:**
  - Urgently if there is no response to first-line treatment.
  - If pelvic inflammatory disease is suspected.
- **Arrange follow up:**
  - Ask about adherence to treatment and resolution of symptoms.
  - Check partner notification has been carried out and reinforce health education.
  - Offer repeat testing to all people under the age of 25 years diagnosed with chlamydia 3–6 months after completion of treatment to check for re-infection. Offering repeat testing to people over the age of 25 years who are at high risk of re-infection.

TOPIC: SYPHILIS

- **Syphilis is an infectious disease caused by the spirochete bacterium *Treponema pallidum*** — untreated infection can persist for years and result in severe cardiovascular, ocular and neurological complications.
- **Most cases are sexually transmitted during direct contact with an infectious lesion** (usually involving the genital, rectal, or oral mucous surfaces).
    - The incubation period is around 21 days but can range from 9 to 90 days depending on the infectious dose.
- **Transmission can also occur:**
    - Vertically from mother to child during pregnancy — serological screening for syphilis is part of routine antenatal care in the UK.
    - Through sharing of needles in injecting drug users.
    - More rarely, through blood products or organ transplant — screening of donations is mandatory in the UK.
- **Early syphilis** (defined as the first 2 years after infection) includes 3 stages:
    - Primary syphyilis— characterized by a painless ulcer (chancre) at the site of infection (most commonly the genital area) and local lymphadenopathy.
    - Secondary syphyilis— systemic involvement which can present with a wide range of clinical features including:
        - Skin and hair changes such as a maculopapular rash, condylomata lata (moist wart-like lesions) and patchy alopecia.
        - Oral lesions ('snail tract lesions').
        - Generalized lymphadenopathy.
        - Low-grade fever, headache, and malaise.
    - Early latent syphyilis — confirmed infection in the absence of any current clinical features.
- **Late syphilis** (defined as more than 2 years after infection) includes 2 stages:
    - Late latent syphyilis— confirmed infection in the absence of any current clinical features.
    - Tertiary syphilis — rare, but may present with cardiac disease (for example aortic aneurysm), cutaneous disease (for example gummatous lesions), and/or neurological disease (for example tabes dorsalis, and general paresis).

What are the risk factors?

- **Factors associated with increased risk of acquiring syphilis infection include**:
    - Unprotected sex.
    - Multiple or anonymous sexual partners.
    - Substance use (including alcohol and illicit drugs [‘chemsex’]) before sex — associated with higher levels of unprotected sex.
    - Transactional sex (for example commercial sex work or exchange of sex for drugs).
    - Previous or current diagnosis of any sexually transmitted infection.
    - Social vulnerability such as poverty; homelessness; migrant or refugee status.
    - Needle-sharing contact with a known syphilis case.
- **Additional factors which have been cited in recent outbreaks of syphilis in men who have sex with men (MSM).**
- Transmission rates of syphilis through sexual contact (genital or extra-genital) range from 10–60%. Risk of transmission is highest in primary and secondary syphilis.

What is the prognosis?

- Syphilis can be cured if treated with appropriate antibiotics before complications develop.
- People with HIV who are co-infected with syphilis may be at increased risk of neurologic complications.

What are the complications?

Untreated syphilis infection can result in serious complications, including:

- **Neurosyphilis** — neurological involvement can occur at any stage of syphilis infection and affect the meninges, arteries, cranial nerves, eyes, brain and spinal cord parenchyma.
    - **Early neurosyphilis** usually involves the meninges and blood vessels and can result in:
        - Meningitis, cranial nerve defects, and infarction.
        - Auditory or ocular abnormalities such as hearing loss, uveitis, optic neuropathy, interstitial keratitis and retinal problems — permanent visual loss may occur.
    - **Late neurosyphilis** usually involves the brain and spinal cord and includes:
        - General paresis — cortical neuronal loss with a gradual decline in memory and cognitive function; hemiparesis and seizures may also occur.
        - Tabes dorsalis — inflammation of the spinal dorsal column/nerve roots leading to ataxia, pain, paraesthesia, Charcot's joints and pupillary changes.
    - **Cardiovascular syphilis** — chronic syphilis infection can lead to aortic aneurysm, aortic regurgitation, angina and heart failure.
    - **Gummatous syphilis** (granulomatous lesions with a necrotic centre) — can occur anywhere but most often affect the skin and bones.
- **Psychological and social complications**
    - Sexually transmitted infections (STIs) have been associated with stigma, shame, loss of self-worth, relationship disruption and domestic violence.
- **Syphilis infection in pregnancy**
    - Untreated syphilis infection in pregnancy is associated with multiple adverse outcomes including hydrops, preterm labour, low birth weight, foetal loss and congenital syphilis in the newborn.
- **HIV infection**
    - Untreated syphilis infection can facilitate HIV transmission.

Diagnosis:

When should I suspect syphilis?

- **Syphilis can present with a wide range of nonspecific symptoms — in some people infection may be asymptomatic. As a result, diagnosis can be delayed or missed**.
- **Suspect syphilis if there is a history or presence of any of the following, especially if there are any** risk factors:
    - **Genital lesion(s)** — the characteristic feature of primary syphilis is a solitary, painless, indurated, genital ulcer (chancre) with a clean base and sharp border.
        - Associated regional lymphadenopathy is common.
    - **A non-pruritic maculopapular rash** — rash typically involves the palms of the hands and soles of the feet.

    - **Moist wart-like lesions** (condylomata lata) — usually develop at sites of skin friction such as perianal and vulval regions, under the breasts, and axillae.
    - **Patchy lesions on the oral mucosa** ('snail tract' lesions).
    - **Generalized lymphadenopathy**.
    - **Unexplained neurological or ophthalmological symptoms**.
- **The presence of:**
    - Chancre may indicate primary syphilis.
    - A maculopapular rash, condylomata lata, oral lesions, generalized lymphadenopathy, unexplained neurological or ophthalmological symptoms may indicate secondary syphilis.
- **Tertiary syphilis** rarely presents in primary care — this may be partly due to widespread use of treponemicidal antibiotics for other infections.
    - Clinical features of tertiary syphilis are varied, non-specific and can include cardiovascular, neurological, ocular and cutaneous symptoms and signs.

- It is widely agreed that diagnosis of syphilis can be easily delayed or missed as presentation of both early and late syphilis is highly variable, non-specific and clinical features can be subtle.
    - In addition, the chancre of primary syphilis can develop on sites which are not visible (for example, the rectum) and if painless, missed.

How should I assess a person with suspected syphilis?

Take a history and ask about:

- symptoms of syphilis.
- Risk factors, including a full sexual history.
- Past medical history including previously diagnosed sexually transmitted infections and treatment.

Examine the person

- **In early disease, examination may include:**
    - Genital examination — looking for chancre.
    - Skin examination including mouth, scalp, palms and soles — looking for skin lesions such as macupapular rash, mucous patches or condylamata lata.
    - Neurological examination — looking for neurological signs (including meningitis, stroke and cranial nerve lesions).
    - Eye examination — looking for signs of ocular disease.
- **In symptomatic late disease, examination may include:**
    - Skin examination — looking for signs such as gumma.
    - Cardiovascular system — looking for signs of cardiovascular disease including aortic regurgitation.
    - Nervous system — looking for signs of tabes dorsalis or general paresis.
    - Eye examination — looking for signs of ocular disease.

Ocular syphilis

- Ocular syphilis may present with blurred vision, visual loss, eye pain or redness — additional systemic features may not be present.

Referral to GUM for investigations

- Referral to GUM is widely recommended to ensure that appropriate diagnostic testing, full sexual health screening (including HIV testing) and disease staging are carried out.

What else might it be?

Many conditions can present with features similar to those associated with the various stages of syphilis:

- **Primary syphilis**
  - **Chancre — the differential diagnosis for ulceration affecting the:**
    - **Genital area** includes genital herpes, balanitis, secondary infected traumatic lesions, lymphogranuloma venereum (*Chlamydia trachomatis*), Behçet's syndrome (especially in the presence of mouth ulcers), genital cancer, and chancroid (*Haemophilus ducreyi*).
    - **Peri-anal area** includes herpes simplex infection, anal fissure, Crohn's disease, and anal cancer.
    - **Cervix** includes cervical herpes, cervical erosions, and cervical cancer.
    - **Oral mucosa** includes herpes simplex infection, aphthous stomatitis, Behçet's syndrome (especially in the presence of genital ulcers), secondary infected traumatic lesions, pemphigus, pemphigoid, drug reactions and oral cancer.
  - **Regional lymphadenopathy:**
    - The differential diagnosis of regional and generalized lymphadenopathy is broad and can include infectious causes, systemic diseases, and malignancy.
    - Lymphadenopathy in syphilis is typically non-tender.

- **Secondary syphilis**
  - **Maculopapular rash — the differential diagnosis includes:**
    - Other infections such as primary HIV infection, rubella, scabies or measles.
    - Dermatological conditions such as guttate psoriasis, pityriasis rosea, and eczema.
    - Drug eruptions.
  - **Condylomata lata — the differential diagnosis includes:**
    - Human papillomavirus, molluscum contagiosum, or haemorrhoids (in the anal area).
  - **Patchy alopecia — the differential diagnosis includes:**
    - Conditions such as alopecia areata, scarring alopecia, and tinea capitis.
  - **Patchy lesions on the oral mucosa ('snail tract lesions'):**
    - The differential diagnosis includes conditions such as oral cancer or aphthous ulcers.
    - If ulcers are present on the tonsils. It can be difficult to differentiate from infectious mononucleosis, especially in the presence of a rash precipitated by administration of ampicillin or amoxicillin.
- **Tertiary syphilis**
  - Symptoms associated with tertiary syphilis are highly variable and non-specific.
  - The differential diagnosis is broad and includes dementia, psychiatric conditions, diseases that affect mobility, and chronic granulomatous lesions of tuberculosis, sarcoidosis, or leprosy.

Management:

From age 16 years onwards

How should I manage a person with suspected syphilis in primary care?

- **Refer all people with suspected syphilis to a genito-urinary medicine (GUM) clinic or other specialist local sexual health service.**
- **Explain the condition, risk factors, and potential complications; discuss prevention.**
  - Preventative measures include:
    - Increased use of condoms.
    - Avoidance of drugs and alcohol when having sex.
    - Risk-reduction counselling and regular syphilis screening for people at high risk — available through GUM.
    - Early recognition of the clinical features of syphilis, prompt treatment and prophylactic treatment of exposed contacts.
- **Advise the person to avoid all sexual contact and exposure of other people to active lesions until:**
  - The diagnosis is excluded, *or*
  - Successful treatment of the condition has been confirmed.
- **For people who are HIV positive who decline referral to a GUM clinic**:
  - Offer referral to an infectious disease clinic or HIV centre if they are routinely reviewed there for HIV monitoring.

Laboratory diagnosis

Specific tests included in a full syphilis screen may vary locally, but usually include a combination of treponemal and non-treponemal tests which should be interpreted in the context of clinical and geographical background.

- **Direct tests for syphilis** are used to demonstrate *T. pallidum* from swabs taken of primary lesions and include:
  - **Dark-field microscopy**:
    - Requires specialist equipment and training.
    - Has a high specificity but low sensitivity (less than 50%) — a negative result does not exclude syphilis.
    - Is not suitable for oral lesions due to presence of commensal treponemes.
  - **Polymerase chain reaction (PCR)**.
    - Is available at reference laboratories.
    - Can be used for oral or other lesions where commensal treponemes may be present.
- **Serological tests for syphilis include:**
  - **Specific or treponemal tests** such as treponemal enzyme immunoassay (EIA), or treponemal chemiluminescent assay (CLIA); Treponema pallidum haemagglutination assay (TPHA); Treponema pallidum particle agglutination assay (TPPA), fluorescent treponemal antibody absorption test (FTA-abs), and Treponema pallidum immunoblot.
    - Most treponemal tests detect treponemal IgG and IgM — they cannot differentiate syphilis from non-venereal treponemal infections (for example yaws [*T. pallidum pertenue*], endemic syphilis [*T. pallidum endemicum*] and pinta [*T. pallidum subspecies carateum*]) which are endemic in some tropical countries.

        - The sensitivity of treponemal tests is variable in early primary syphilis and high in secondary, early latent and late latent stages.
        - Treponemal tests are not useful for monitoring effectiveness of treatment or disease activity as in most people they remain positive for life regardless of treatment.
        - The British Association for Sexual Health and HIV (BASHH) recommend EIA/CLIA as the screening test of choice for syphilis.
    - **Non-specific or non-treponemal tests** (such as Venereal Diseases Research Laboratory (VDRL) carbon antigen test/RPR test):
        - Have low sensitivity in early primary syphilis, high sensitivity in secondary and early latent stages and low sensitivity in late-stage syphilis.
        - Without treatment titres peak after 1–2 years and remain positive at low levels in late-stage disease.
        - Following treatment titres usually decline and can be used to monitor effectiveness of treatment.
- **Repeat testing**
    - All positive screening tests must be confirmed with a different serological test.
    - Negative serology tests should be repeated at 6 and 12 weeks after an isolated high-risk exposure or 2 weeks after possible chancres that are dark-field microscopy and/or PCR negative.
- **Examination of cerebrospinal fluid (in addition to serology)** is required for diagnosis of neurosyphilis or ocular syphilis.

TOPIC: GENITAL WARTS

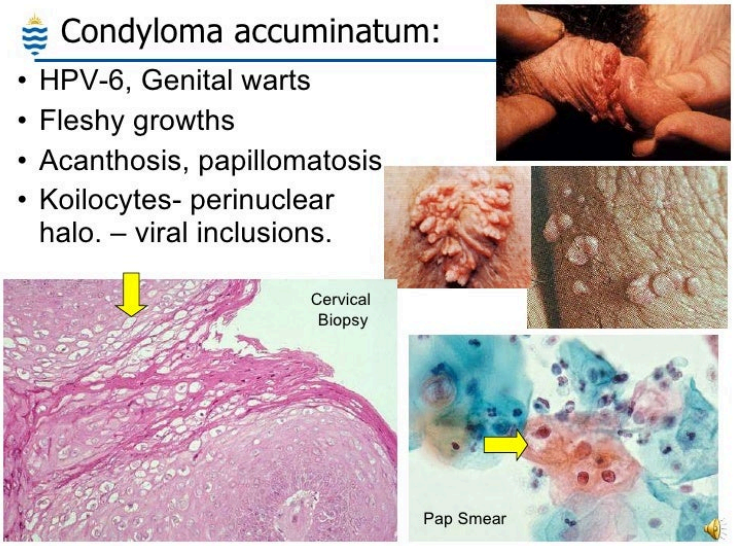
It is, also known as condylomata accuminata or anogenital wart, a common cause of attendance at genitourinary clinics.

Etiology:

- They are caused by the many varieties of the human papillomavirus HPV, especially types 6 & 11.
- It has been established that HPV (primarily types 16,18 & 33) predisposes to cervical cancer.

Clinical Features:

- Develops over vulva, penis, groin, perineum, perianal and/or suprapubic skin
- small (2 - 5 mm) or might be more than few centimeters
- Shape: flat, cauliflower, fungating, dome-shaped or lobulated
- Color: skin-like, white, brown, violaceous or erythematous
- which are slightly pigmented
- may bleed or itch



Diagnosis:

- The diagnosis is based on physical examination unless uncertain might proceed with biopsy
- HPV testing for warts does not confirm or influence the management (UpToDate)
- Assess for coexisting sign of sexual transmitted infection (STI)

Clinical Course:

- Incubation: 3 weeks to 8 months
- After the initial appearance, warts may increase in size/number or regress
- Up to 30% of warts resolve spontaneously within 4 months
- NOTE: resolution does not mean eradication of the HPV infection

Red Flags:

- Single of multiple soft, smooth/papillated papules/plaques limited to anogenital area

Association:

- HPV 6 and 11 are considered of low-risk types; however, they are responsible for most cases of cancer coexisting with high-risk HPV genotypes
- Transformation is more likely to develop in immunocompromised patients

Differential Diagnosis:

- Molluscum contagiosum
- Keratotic plugs
- Seborrheric keratosis
- Lichen Planus
- Condylomata lata of syphilis
- Malignancy

Management:

- There is no single best approach nor superiority in treatment. Therefore, the treatment should be individualized.
- Topical podophyllum or cryotherapy are commonly used as first-line treatments depending on the location and type of lesion
  - multiple, non-keratinized warts are generally best treated with topical agents
  - solitary, keratinized warts respond better to cryotherapy
- Imiquimod is a topical cream that is generally used second line
- Genital warts are often resistant to treatment and recurrence is common although the majority of anogenital infections with HPV clear without intervention within 1-2 years

Therapies preferred for pregnant women (vulva)
Without anesthesia
Trichloroacetic acid or bichloracetic acid 80 to 90%
Cryotherapy with liquid nitrogen or cryoprobe
With anesthesia
Surgical removal (sharp, electrocautery, curettage, laser)

TOPIC: HUMAN PAPILLOMA VIRUS (HPV) VACCINE

It has been established that the human papillomavirus (HPV), which infects the keratinocytes of the skin and mucous membranes, is carcinogenic.

HPV Strains

There are various strains of HPV and the most important to remember are:

- 6 & 11: causes genital warts
- 16 & 18: linked to a variety of cancers, most commonly cervical cancer

Association with HPV Infection:

- Over 99.7% of cervical cancers
  - HPV testing is now integral to cervical cancer screening
- Around 85% of anal cancers
- Around 50% of vulval and vaginal cancers
- Around 20-30% of mouth and throat cancers

Risk Factors:

There are risk factors important in developing cervical cancer such as smoking, combined oral contraceptive pill, multiple partners and high parity.

Immunization:

Human papillomavirus vaccination

ADMINISTRATION > Immunization schedule

- **Individuals initiating the vaccine series at 9 to 14 years of age** – Two doses of HPV vaccine should be given at 0 and at 6 to 12 months.

  If the second dose was administered less than five months after the first, the dose should be repeated a minimum of 12 weeks after the second dose and a minimum of five months after the initial vaccine dose.

- **Individuals initiating the vaccine series at 15 years of age or older** – Three doses of HPV vaccine should be given at 0, 1 to 2 (typically 2), and 6 months.

  The minimum intervals between the first two doses is four weeks, between the second and third doses is 12 weeks, and between the first and third dose is five months. If a dose was administered at a shorter interval, it should be repeated once the minimum recommended interval since the most recent dose has passed.

- **Immunocompromised patients** – Three doses of HPV vaccine should be given at 0, 1 to 2, and 6

TOPIC: HIV INFECTION AND AIDS

- **Human immunodeficiency virus (HIV) is a lentivirus from the subfamily of retroviruses.**
  - Lentiviruses have a long latent phase between infection and the development of symptoms.
  - HIV causes immunodeficiency by preferentially infecting and destroying cells of the immune system, in particular CD4 cells, which are a class of T lymphocytes.
- **Primary HIV infection (or HIV seroconversion illness).**
  - In the first few weeks following infection with HIV (usually between 10 days and 6 weeks), a flu-like illness (which can be mild or severe) is common — this phase is known as 'primary HIV infection' or 'HIV seroconversion illness'.
  - Viral load is high in this stage and the person is very infectious.
- **Asymptomatic phase.**
  - Once symptoms of primary HIV infection resolve, an asymptomatic stage of infection begins.
  - The duration of this varies widely between people with some progressing to advanced HIV disease (also known as acquired immunodeficiency syndrome [AIDS]) within 1–2 years ('rapid progressors') and others maintaining effective immune function more than 10 years later ('slow progressors').
- **Advanced HIV disease (also known as AIDS).**
  - A person living with HIV is said to have advanced HIV disease or AIDS when the number of CD4 cells is very low (less than 200 cells per microlitre) and certain opportunistic infections (such as pneumocystis pneumonia) or malignancies (such as Kaposi's sarcoma) develop — these conditions are known as AIDS-defining illnesses.
  - In recent years the term 'AIDS' is less often used and it is more common to refer to 'advanced HIV disease'.

How is HIV infection transmitted?

- **In a person infected with HIV, the virus is present in cell-containing bodily fluids,** such as blood, semen, vaginal secretions, breast milk, amniotic fluid, pleural effusions, and cerebrospinal fluid.
- **HIV can be transmitted from infected body fluids:**
  - **Through sexual activity** — through vaginal, anal, or oral sex (especially in the presence of oral disease such as ulceration or gingivitis).
  - **Vertically from mother to child** — during pregnancy, childbirth, or through breastfeeding.
  - **By inoculation** — via a contaminated needle, instrument, blood, or blood product; through direct exposure of mucous membranes or an open wound to infected bodily fluids; or by a human bite that breaks the skin.
- **HIV is not transmitted by:**
  - Social interactions, such as shaking hands, hugging, or kissing.
  - Routine sharing of household items, such as eating utensils, glasses, and lavatory seats.
  - Insects such as mosquitoes and lice.

What are the risk factors?

People at higher risk of HIV infection include those who:

- Have a current or former partner who is infected with HIV.
- Are from an area with high HIV prevalence.
- Are men who have sex with men (MSM).
- Are female sexual contacts of MSM.
- Are transwomen.
- Have had multiple sexual partners, engage in high-risk sexual practices such as 'chemsex', or have a history of other sexually transmitted infection (such as syphilis, chlamydia, and gonorrhoea).
- Have a history of injecting drug use.
- Are current or previous sex workers.
- Have been raped.
- Have had blood transfusions, transplants, or other risk-prone procedures in countries without rigorous procedures for HIV screening.
- Have had an occupational exposure such as a needle stick injury.

What is the prognosis?

- Advances in anti-retroviral therapy (ART) have meant that HIV infection is now a manageable chronic disease.
    - People living with HIV who are adherent and clinically responding to ART can expect a normal or near-normal life span.
- The most important factor in reducing mortality and morbidity in HIV infection is early diagnosis as this allows prompt initiation of ART.
- Without ART, CD4 count slowly declines eventually resulting in the development of the constitutional symptoms, opportunistic infections, and malignancies associated with advanced HIV disease (AIDS).

Complications

Complications associated with HIV infection include:

- **Advanced HIV disease or AIDS** – if untreated or non-responsive to therapy.
- **Wasting and diarrhoea.**
    - May be a result of advanced disease or opportunistic infection.
- **Neurological problems.**
    - Neuropathies and neurocognitive dysfunction including HIV-associated dementia may occur as a direct result of HIV infection itself or due to antiretroviral therapy (ART).
- **Mental health problems.**
- **Metabolic abnormalities (such as dyslipidaemia and insulin resistance) and cardiovascular disease.**
- **Renal disease (acute or chronic).**
    - Renal disease may be due to HIV infection itself or to adverse effects associated with ART.
- **Bone disease.**
    - The risk of osteoporosis is about four times higher in people living with HIV.

Diagnosis:

When should I suspect HIV infection?

- **Consider the possibility of HIV infection in a person presenting with one or more of the following:**
  - Common symptoms or infections that are unusually severe, prolonged, recurrent, or unexplained.
  - Conditions related to immunosuppression, such as oral candidiasis or shingles.
  - Glandular fever-like illness.
  - Lymphadenopathy of unknown origin.
  - Pyrexia of unknown origin.
  - Weight loss of more than 10 kg.
  - Risk factors for HIV, such as living in, working in, or coming from a high prevalence area; sex between men; and injecting drug use.
- **Be aware that infants most often present with:**
  - Failure to thrive, *and/or*
  - Pneumocystis pneumonia, *or*
  - Cytomegalovirus disease, *or*
  - HIV encephalopathy.
- **After the first year of life, consider HIV infection in children presenting with:**
  - Recurrent infections such as upper respiratory tract infections.
  - Severe presentations of common childhood infections.
  - Chronic diarrhoea.
  - Herpes zoster.
  - Severe pneumonia.
  - Tuberculosis.
  - Lymphadenopathy.
  - Any AIDS defining condition.

Primary HIV infection

- **Clinical features associated with primary HIV infection** develop in over 60% of people, can be non-specific, and include:
  - Fever, sore throat, maculopapular rash, malaise, lethargy, arthralgia, myalgia, lymphadenopathy, and oral, genital or perianal ulcers.
  - Less commonly, headache, meningitis, cranial nerve palsies, diarrhoea, and weight loss.

Longstanding HIV infection

Conditions associated with longstanding HIV infection can be subtle and people may remain well for some time before developing another condition. Consider HIV if a person has had more than one of the following in the past 3 years, or has had conditions that are severe or resistant to treatment.

Constitutional symptoms

- Constitutional symptoms may be due to HIV infection itself, a related opportunistic infection (such as tuberculosis), or malignancy (such as lymphoma), and include:
  - Flu-like symptoms.

- Fever.
- Weight loss.
- Sweats.
- Lymphadenopathy.

Respiratory conditions

- Opportunistic respiratory infections such as pneumocystis pneumonia (PCP), tuberculosis, bacterial lower respiratory tract infections; and occasionally Kaposi's sarcoma or lymphomas can result in symptoms including cough, sweats, breathlessness, and weight loss.

Neurological and visual conditions

- **HIV infection is associated with many neurological and visual problems including:**
  - **Cryptococcal meningitis** — may present as subacute meningitis with headache and fever, acute confusional state, or cranial nerve palsies.
    - Typical symptoms and signs of meningism may be absent.
  - **Cerebral toxoplasmosis** — occurs owing to reactivation of previous toxoplasmosis infection.
    - May present with confusion, headache, personality change, fever, fits, and focal neurological signs.
  - **Cerebral lymphoma** (in advanced HIV disease).
    - May present with confusion, lethargy, personality change, headache, fever, fits, and focal neurological signs.
  - **Cytomegalovirus (CMV) retinitis** occurs owing to reactivation of previously acquired infection and can lead to loss of vision.
    - Prognosis improves with earlier diagnosis.
    - Symptoms include floaters (due to inflammatory cells in the vitreous), reduced vision, flashing lights, and scotomas.

Cancers

- **HIV infection is associated with increased risk of malignancy including:**
  - **Lymphoma**
    - Lymphoma can present with weight loss, fevers, night sweats, lymphadenopathy, and abdominal masses (lymph nodes, hepatomegaly, or splenomegaly).
  - **Kaposi's sarcoma**
    - Kaposi's sarcoma is the most common tumour in people with HIV infection and typically presents with dark purple or brown intradermal skin lesions that look like bruises but feel firmer.
    - Tumours may occur anywhere on the skin or oropharynx — infiltration of the lungs or gut is rare but can be life threatening.
  - **Cervical cancer**
    - Cervical cancer can present with vaginal bleeding or discharge.
    - Abnormal cervical cytology may be a marker for underlying HIV infection.

Skin conditions

- In addition to Kaposi's sarcoma, HIV infection is associated with many common skin conditions. These are often unusually severe, recurrent, or resistant to treatment, and include:
    - **Fungal skin and nail infections**
    - **Viral infections** such as herpes simplex (especially if the ulcers are unusually large or persistent), molluscum contagiosum, shingles (especially if more than one dermatome is affected), and warts.
    - **Bacterial infections** such as impetigo and folliculitis.
    - **Seborrhoeic dermatitis and psoriasis**

- **HIV infection is associated with:**
    - **Oral candidiasis**
        - Typically presents with thick white plaques covering the buccal mucosa that can be scraped off. Soreness of the mouth or pain on swallowing (odynophagia) may also be present.
    - **Aphthous ulcers**
    - **Oral hairy leukoplakia**
    - **Kaposi's sarcoma**
        - Most often affects the hard and soft palate, gingiva, and dorsal tongue with plaques or tumours of variable colour from non-pigmented to brownish-red or purple.
    - **Gingivitis**
        - Causes swelling and erythema of the gum margins that are usually painful and bleed on tooth brushing or flossing.
    - **Dental abscess**
        - Typically presents with rapid onset of severe dental pain and may be associated with an unpleasant taste in the mouth, fever, and malaise.

Gastrointestinal conditions

- **HIV infection is associated with:**
    - **Oesophageal candidiasis**
        - Symptoms include retrosternal pain or dysphagia. Oral candidiasis is also likely to be present.
    - **Diarrhoea**
        - Diarrhoea is very common and may be persistent and mild or acute and severe.
        - Infection due to bacteria, protozoa (such as *Cryptosporidium*), mycobacteria, or viruses; malignancy (such as non-Hodgkin's lymphoma or Kaposi's sarcoma), or the direct effects of HIV can all result in diarrhoea.
        - Often no specific causative organism is identified.
    - **Hepatitis B and hepatitis C infection**
        - Co-infection with hepatitis B or C is common and increases the risk of cirrhosis and hepatocellular carcinoma.

Genital conditions

- Consider HIV infection in anyone presenting with severe or difficult-to-treat genital candida, genital herpes, or genital warts.

Haematological conditions

- Consider HIV infection in anyone with unexplained abnormalities on full blood count such as neutropenia, anaemia, thrombocytopenia, or other blood dyscrasias.

How should I assess a person with suspected HIV infection?

- **Take a history.**
- **Examine the person for:**
  - Rash — in particular a generalized maculopapular rash.
  - Lymphadenopathy, particularly in extra-inguinal sites.
  - Signs of immunocompromise such as oral candidiasis.
  - Signs of HIV-related conditions, being alert for those that are serious or life threatening, such as:
    - PNEUMOCYSTIS PNEUMONIA
    - Tuberculosis.
    - CMV RETINITIS
    - Lymphoma.
    - Kaposi's sarcoma
- **Review the person's medical records for evidence of previous HIV-associated problems.**

How should I test for HIV?

- **Check the persons understanding of HIV.**
- **Decide with the person whether urgent assessment by a specialist is required and whether HIV testing should be carried out in primary care or elsewhere.**

    - Arrange addmision for urgent specialist assessment if there is concern about the possibility of a serious HIV-related condition such as PNEUMOCYSTIS PNEUMONIA

- **Pre-test counselling should include discussion of:**
  - **The benefits of HIV testing:**
    - If the test is negative, the person can be reassured and given advice on reducing the risk of HIV infection.
    - If the test is positive, effective treatment can help prevent HIV-related illness and ensure that infection is not transmitted to contacts or in pregnancy.
  - **Details of how the result will be given:**
    - Check you have correct contact details, the preferred method of contact, and whether there may be any problems with leaving messages or talking.
    - If possible, arrange an appointment for the result to be given — negative results may be discussed by telephone.
  - **The significance of the window period** as a repeat test may be necessary.
    - The window period is the time between becoming infected and antibodies appearing — HIV antibodies usually appear 4–6 weeks after infection but can take up to 12 weeks.
  - **The importance of confidentiality** and that the result will need to be recorded in medical records.

HIV testing

- Fourth generation assays testing for HIV antibodies and p24 antigen in venous samples will detect the majority of people infected with HIV at 4 weeks after exposure.
    - A negative result at 4 weeks post exposure is highly likely to exclude HIV infection — an additional HIV test at 8 weeks need only be considered following an exposure assessed as high risk for infection.
    - Testing for other sexually transmitted infections should be advised and people at risk of hepatitis B virus who are not immune should be immunized.

When should I screen for HIV infection?

When should I offer a test for HIV?

- **All pregnant women should be offered screening for HIV as part of routine antenatal care.**
- **All children at risk of HIV infection should have an HIV test.**
- **For information on what to include in the pre-test and post-test discussion.**
- **If a venous blood sample is declined:**
    - Offer a less invasive form of specimen collection, such as a mouth swab or finger prick.
- **If the person chooses not to take up an immediate offer of a test:**
    - Advise them on nearby testing services and how to get self-sampling kits.
        - Self-sampling HIV kits allow people to collect their own sample of blood or saliva.

Management:

From birth onwards

How should I manage a person with a new diagnosis of HIV?

- **If the person is well and there is no indication for** urgent referral:
    - Discuss where they prefer to be treated and refer urgently to a specialist HIV clinic, to be seen preferably within 48 hours and at the latest within 2 weeks of testing positive.
        - Give the person written details of any appointment arranged.
    - Advise the person on risk reduction strategies to protect contacts such as:
        - Promoting safer sexual practices if risk is through sexual contact — ensure that the person has a supply of condoms and lubricant.
        - Directing those with risk through injecting drug use to local services for advice on safer injecting practices.
    - Arrange a follow-up appointment within a few days to:
        - Provide further support if needed.
        - Ensure the referral to specialist services has been received.

Scenario: How should I manage acute HIV-related problems?

From birth onwards

When should I admit a person with HIV?

- **Arrange admission if:**
  - The person is acutely unwell.
  - Respiratory symptoms are suggestive of a serious bacterial lower respiratory tract infection such as:
    - Pneumonia, *or*
    - Tuberculosis.
  - Visual symptoms are suggestive of a serious underlying condition such as cytomegalovirus (CMV) retinitis.
  - Progressive or acute neurological symptoms or delirium are present:
    - This may indicate a serious HIV-related condition such as cryptococcal meningitis, cerebral toxoplasmosis, or cerebral lymphoma.
  - Kaposi's sarcoma of the lung or gut is suspected — this can be life threatening.
  - Serious or life-threatening adverse effect of antiretroviral therapy are suspected, for example:
    - Drug hypersensitivity,
    - Lactic acidosis,
    - Liver toxicity,
    - Bone marrow suppression, *or*
    - Pancreatitis.
- **If unsure whether the person needs to be admitted or not, discuss with a specialist.**

What tests are used to monitor the progress of HIV infection?

Monitoring of HIV is usually carried out in specialist clinics using the CD4 lymphocyte cell (CD4) count and viral load.

- **CD4 count**
  - The CD4 count reflects the degree of immunosuppression in people infected with HIV.
  - In a healthy person not infected with HIV, the CD4 count is usually greater than 500 cells per microlitre — some people may have naturally lower counts.
  - People with CD4 counts below 200 cells per microlitre are most at risk of HIV-related opportunistic infections and cancers. If treatment is started at CD4 counts above 500 cells per microlitre, rather than later, prognosis is improved.
- **Viral load**
  - Viral load reflects rates of viral replication and is measured using a polymerase chain reaction (PCR) test.
  - A rising viral load may indicate non-adherence to ART, resistance to one or more antiretroviral drugs, or an interaction with another medication.
  - Viral load ranges from undetectable (less than 20–50 copies of viral genome/mL blood) to over a million copies/mL.
  - The degree of viral replication is linked to the rate of CD4 decline and therefore disease progression — when viral load is suppressed through ART, CD4 counts recover and risk of HIV-related opportunistic infections and cancers declines.

Antiretroviral therapy

Antiretroviral therapy (ART) has had a huge positive impact on HIV-related morbidity and mortality and aims to reduce viral load to undetectable levels by limiting viral replication.

- HIV mutates as it replicates so drugs are usually used in combinations of three or more to reduce drug resistance.
- Antiretroviral drugs are classified into five groups depending on how they act:
    - Nucleoside/tide reverse transcriptase inhibitors (NRTIs).
    - Non-nucleoside reverse transcriptase inhibitors (NNRTIs).
    - Protease inhibitors (PIs).
    - Integrase inhibitors (IIs).
    - Entry inhibitors (EIs).
- Specialist blood tests may be used to identify drug resistance and tailor ART regimens accordingly.
- ART suppresses viral replication but does not eliminate HIV therefore treatment is lifelong.

Adverse effects of antiretroviral therapy

Antiretroviral therapy (ART) can have multiple adverse effects, some of which can be serious or life threatening — adverse effects are most common with older classes of drugs but all have potential short- and long-term toxic effects.

- **Consider whether specialist advice or hospital admission is required.**
- **Adverse effects include:**
    - Hypersensitivity.
    - Neurological and psychiatric conditions.
        - Nightmares, sleep disturbance, mood or behaviour changes may occur; psychosis and suicidal ideation have also been reported.
        - Peripheral neuropathy.
    - Hyperlipidaemia (common).
        - Rises in cholesterol or triglycerides can be extreme, lipids need to be regularly monitored and managed.
        - Drug interactions with statins and fibrates occur frequently and can be serious — seek specialist advice before prescribing.
    - Lipodystrophy (changes in the distribution of body fat) and lipoatrophy (loss of subcutaneous fat).
        - These may be associated with diabetes, and hyperlipidaemia.
    - Type 2 diabetes mellitus.
        - This most likely occurs through insulin resistance and can be associated with ART.
    - Bone density loss.
    - Renal problems.
        - A decline in renal function may indicate Fanconi's syndrome (dysfunction of the proximal tubule) — seek specialist advice.
        - Ureteric colic, renal and ureteric stones may also occur.
    - Lactic acidosis and hepatic toxicity.
        - May present with non-specific symptoms such as nausea, anorexia, or abdominal pain — potentially life threatening.
    - Peripheral neuropathy.
    - Bone marrow suppression.
    - Pancreatitis.
        - Most often associated with older ART.

Health promotion

Cardiovascular disease (CVD) prevention

- Preventative measures such as lifestyle advice on diet, exercise, alcohol reduction and smoking cessation, blood pressure monitoring, and screening for diabetes and hyperlipidaemia should be offered.
- CVD risk assessment and initiation of treatment is usually carried out by specialist services but may be indicated in primary care.
    - Be aware that statins (especially simvastatin) and fibrates can have serious drug interactions with some antiretroviral therapy (ART).
    - Check the HIV drug interaction and seek expert advice before initiating any treatment.

Cervical screening

- HIV infection increases the risk of cervical cancer.
- All women with newly diagnosed HIV should have performed by, or in conjunction with, their HIV specialist team:
    - Initial colposcopy following diagnosis.
    - Annual cytology.

Immunizations

International travel

- Entry restrictions
    - Some countries restrict entry for people with HIV.
    - Advise the person to check with the relevant embassies before travel.
- Vaccinations
- Malaria prophylaxis
    - Advise the person that malaria prophylaxis can be used but may interact with ART.
- Advise the person:
    - On safer sexual practice where appropriate.
    - To take a summary of past medical history and prescribed drugs in case they become unwell abroad.

Sexual and reproductive health

Use of condoms

- Advise the person on consistent and correct use of condoms and lubricant to reduce the risk of HIV transmission and other sexually transmitted infections (STIs).
    - For optimal contraception condoms should always be used with an additional contraceptive method.
- Condoms should still be used even if:
    - Antiretroviral therapy (ART) is taken regularly and viral load suppressed.
    - The person's partner is also HIV positive (to prevent drug-resistant strains from being transferred).

Sexual health screening

Pre-exposure prophylaxis (PrEP)

- PrEP is a combination of two antiretroviral drugs that are taken before and after sex to reduce the risk of HIV acquisition – it is offered on a case by case, basis to HIV-negative people who are at high risk of HIV and who do not always use condoms.
    - Be aware that some medication regimens may have serious interactions with other drugs.

Post-exposure prophylaxis

- Be aware that PEP following sexual exposure is no longer recommended if a HIV-negative partner has unprotected sexual intercourse with an HIV-positive person who has a sustained undetectable viral load.
    - PEP *is* needed following sexual intercourse if a HIV-negative partner is exposed to HIV and there are any ART adherence issues, or if viral loads are unsuppressed or not being actively monitored.

Contraception

- Contraceptive choice for women with HIV may be complicated by potential drug interactions with ART — seek specialist advice.
    - Oral contraceptives and patches may have reduced effectiveness with some ART combinations.
    - Long-acting reversible contraceptives such as the copper intrauterine device (Cu-IUD), the levonorgestrel-releasing intrauterine system (IUS), or depot medroxyprogesterone acetate (DMPA) injection do not appear to be affected by enzyme-inducing drugs.
- Emergency contraception.
    - Seek specialist advice — for women on liver enzyme-inducing ART:
        - A Cu-IUD is the recommended method of emergency contraception except in women with a CD4 less than 200 cells per microlitre and detectable HIV RNA (owing to potential higher risk of infection post procedure).
        - If progesterone-only emergency contraception is used, a doubling of the standard dose may be advised (off-label).

Fertility and assisted conception

- Specialist fertility services input is recommended for all couples who wish to conceive where one or both partners are living with HIV.
- Options vary depending on concordance, treatment, and viral suppression and may include donor insemination, sperm washing, or self-insemination.

Antenatal and postnatal care

- Care should be provided through antenatal multi-disciplinary team including an HIV specialist, obstetrician, specialist midwife, and paediatrician.
- Risk of transmitting HIV from mother to child in pregnancy can be reduced from around 20% to less than 1% by:
    - Taking ART during pregnancy.

    - Using ART at the time of delivery, together with a short course of ART for the newborn baby.
    - Vaginal delivery for women on ART who have an undetectable viral load or elective Caesarean section for others.
    - Avoidance of breastfeeding

Menopause

- Management of menopause in women living with HIV should be in line with NICE guidance, however, specialist input is indicated as there may be drug interactions with some ART and commonly used menopause treatments.

Mental health

- **People with HIV are at greater risk of mental health problems** (such as depression, anxiety, and substance abuse) than the general population.
- Mental health conditions should be assessed and managed as usual, referring to specialist mental health services where appropriate.
    - Many drug treatments used to treat mental health conditions have serious interactions with antiretroviral therapy (ART) — seek specialist advice before prescribing.

Scenario: Post-exposure prophylaxis for HIV

From birth onwards

How should I manage a potential occupational exposure to HIV?

- **Occupational exposure to potentially HIV-infected bodily fluids is a medical emergency requiring prompt assessment and treatment.**
    - The immediate treatment of a site exposed to potentially HIV-infected bodily fluids is to clean it thoroughly:
        - Skin and skin wounds should be washed with soap and water, and mucous membranes irrigated with water. Squeezing the wound to express blood is not recommended.
        - Eyes should be irrigated with saline or water before and after removing contact lenses.
        - Puncture wounds should be encouraged to bleed freely, but should not be sucked. Small wounds and punctures may also be cleansed with an antiseptic, for example an alcohol-based hand hygiene solution.
    - Local guidelines should be followed and primary healthcare providers should ensure that they have arrangements in place for rapid 24-hour access to urgent advice and post-exposure prophylaxis (PEP) — usually via the local Accident and Emergency department.
- **Occupational exposures to HIV must be reported to the Health and Safety Executive (HSE)**
- **Primary healthcare professionals should advise any carers of a person with HIV about:**
    - Infection control measures, *and*
    - The importance of promptly attending the Accident and Emergency department should they be potentially exposed.

How should I manage a potential sexual exposure to HIV?

- **Exposure to HIV is a medical emergency and requires prompt treatment with a course of antiretroviral therapy (ART).**
- **Direct the person immediately to an HIV/sexual health clinic or an Accident and Emergency department for consideration of post-exposure prophylaxis following sexual exposure (PEPSE) if:**
    - They are an uninfected sexual partner of a person known to have HIV, to prevent infection after sex without a condom or where the condom has split.
        - PEP is generally no longer recommended if the HIV-positive partner is adherent to ART and has been confirmed as having a sustained (more than 6 months) undetectable plasma HIV viral load (less than 200 copies/ml).
    - They have had unprotected sex with a person in a high-risk group for HIV whose infection status is not known.
    - PEPSE may be offered to victims of sexual assault depending on risk assessment.
- **PEPSE can be initiated if the person presents within 72 hours of exposure and should be given as early as possible** (ideally within 24 hours of exposure).
- **Follow up should be arranged by the prescribing team** — HIV testing is recommended 8–12 weeks after exposure.
- **All people seeking PEPSE should be advised to attend for future regular sexual health checks.**

- It can take 48–72 hours before HIV can be detected within regional lymph nodes and up to 5 days before HIV can be detected in blood after HIV crosses a mucosal barrier. Early initiation of antiretroviral therapy (ART) following an exposure reduces and inhibits viral replication.

TOPIC: HAND FOOT AND MOUTH

- **Hand, foot, and mouth disease (HFMD) is an acute viral illness characterized by vesicular eruptions in the mouth and papulovesicular lesions of the distal limbs.** It is not to be confused with foot and mouth disease of animals, which is caused by a different virus.
- **HFMD is usually mild and self-limiting.** The incubation period is 3–7 days. The mucocutaneous lesions appear 1–2 days after a prodromal period of fever, malaise, and myalgia; and last approximately 7–10 days with spontaneous resolution.
- **HFMD is caused by enteroviruses from the picornavirus family** (small, non-enveloped, single stranded RNA viruses). The main enterovirus serotypes responsible for HFMD are the Coxsackie viruses.
- **In Europe and the US, HFMD is most commonly associated with group A coxsackie viruses:**
    - Coxsackie A16 is the most common cause, although outbreaks have been associated with the A6 and A10 strains.
    - The Coxsackie virus A6 strain causes an atypical HFMD, with a more widespread and extensive skin disease, nail shedding, and higher risk for adult infection.
    - Group B Coxsackie viruses may also be causative.
    - Less commonly, but more seriously, HFMD can be caused by enterovirus 71 (EV71), which has caused epidemics in East and Southeast Asian countries.

How is it transmitted?

- **Hand, foot, and mouth disease (HFMD) is spread by:**
    - Direct contact with the nasal and throat secretions of an infected person, for example via coughing and sneezing.
    - Direct contact with fluid from the blisters.
    - Faeco-oral transmission. This can also occur from infected people who are asymptomatic but who may still excrete the virus in their faeces.
- **Vertical spread from mother to fetus can occur.**
- **It is transmittable immediately before and during the acute stage of illness, and for longer in some people, due to its presence in faeces.**
    - Faecal shedding can persist for approximately 4–8 weeks after the onset of illness.
- **HFMD is not transmitted to or from animals. It is not related to foot and mouth disease of animals.**

What are the complications?

- **Complications of hand, foot, and mouth disease (HFMD) are rare.**
    - Dehydration is the most common complication, due to oral pain causing reduced fluid intake.
    - Secondary bacterial infection of lesions is rare, but can occur.
    - HFMD caused by the enterovirus 71 (EV71) strain is associated with rare serious, life-threatening neurological sequelae such as brainstem encephalitis, aseptic meningitis, acute flaccid paralysis, and autonomic dysregulation leading to pulmonary oedema and myocardial impairment.
    - HFMD acquired around the time of delivery can lead to neonatal infection. This is mostly mild and self-limiting; however, neonatal complications including severe hepatic, respiratory, and neurological disease have been documented. Reassuringly, none of these appear to have been associated with the most common Coxsackie A16 strain.

What is the prognosis?

- **The prognosis of hand, foot, and mouth disease (HFMD) is generally excellent as it is most commonly a self-limiting disease.**
    - HFMD caused by Coxsackie A16 is a mild disease and the vast majority of people fully recover.
        - Oral and skin lesions nearly always fully resolve without treatment in 7–10 days.

Diagnosis:

When should I suspect hand, foot, and mouth disease?

The diagnosis of hand, foot, and mouth disease (HFMD) is usually clinical, based on typical symptoms and signs.

- **HFMD typically presents as a mild illness with:**
    - Sore throat, with or without a low-grade fever.
    - Tender lesions in the mouth (enanthem) and/or a rash on the body (exanthem).
- **Ask about the presence of prodromal symptoms.**
    - The prodromal period may last 12–36 hours.
    - Early symptoms are fever (typically 38–39°C), malaise, loss of appetite, cough, abdominal pain, and sore mouth. Rarely, vomiting occurs if HFMD is caused by enterovirus 71.
- **Examine the mouth for characteristic features.**
    - Scattered ulcerative lesions of the oral cavity occur within 1–2 days.
        - Typically, they begin as 2–8 mm erythematous macules and papules, and appear most commonly on the hard palate, tongue, and buccal mucosa, but also on the lips and pharynx.
        - Lesions progress rapidly to vesicles, which readily erode leaving shallow yellow-grey ulcers surrounded by an erythematous halo.
        - In severe cases, the lesions may coalesce leaving the tongue red and oedematous.
- **Examine the rash for characteristic features.**
    - Macules and papules of the hands and feet usually soon follow the oral lesions.
        - Typically, they are 2–5 mm sparse erythematous macules and papules, often with a central greyish vesicle.
        - The sides of the fingers, dorsum of the hands, and margins of the heels are more frequently affected than the palms or the soles.
        - Lesions are frequently elliptical with the long axis running parallel to the skin lines.
        - The buttocks and groin area may also be affected.
        - Lesions may be asymptomatic or painful.
    - HFMD caused by Coxsackie virus A6 may present with a more widespread rash that extends beyond the palms and soles and may preferentially occur in areas prone to atopic dermatitis (the antecubital and popliteal fossae). Petechiae and severe blistering may be observed. There may be subsequent skin peeling and/or nail shedding.
- **Laboratory investigations (such as oral, skin, or rectal swabs) are not usually necessary in primary care owing to the self-limiting nature of the illness.**

What else might it be?

It may be difficult to differentiate viral illness from more serious causes of fever and rash in young children.

- Conditions that may resemble hand, foot, and mouth disease (HFMD) include:
  - Herpangina — is caused by Coxsackie A viruses including A16, which presents with high fever, malaise, headache, and oral lesions, but has no associated exanthema (rash).
  - Herpes stomatitis — common in children and in immunocompromised people.
  - Aphthous ulcers — can become herpetiform and lead to stomatitis.
  - Chickenpox — suggested by very itchy rash mainly limited to trunk and extremities.
  - Erythema multiforme/Stevens–Johnson syndrome — typically presents with target lesions on palms, soles, and distal limbs. Vesicles coalesce and ulcerate, and mucous membrane involvement can progress to life-threatening Stevens–Johnson syndrome.
  - Viral pharyngitis.
  - Kawasaki disease — a systemic vasculitis that primarily affects infants and young children; it causes fever, conjunctivitis, 'strawberry' tongue, peeling skin, polymorphous rash, and cervical lymphadenopathy. Rare complications include coronary artery dilatation and aneurysm formation.
  - Pompholyx eczema — blisters may develop on hands and feet.
  - Lichen planus — may present in the mouth with painful erosive lesions (Whickman striae).
  - Traumatic ulcers — such as from ill-fitting dentures, thermal trauma, or buccal aspirin.
  - Gingivitis.
  - Behçet's disease — typically recurrent aphthous stomatitis with genital ulceration, eye involvement, and skin lesions.
  - Pemphigus vulgaris or oral bullous pemphigoid — painful mouth ulcers are present, and skin lesions are usually evident elsewhere.

Management:

Scenario: Management of hand, foot, and mouth disease

From birth onwards

How should I manage suspected hand, foot, and mouth disease?

- **Arrange urgent hospital admission if there are symptoms or signs of central nervous system involvement, such as:**
  - Persistent or severe headache or fever.
  - Myoclonus with sleep disturbances.
  - Confusion, weakness, lethargy, drowsiness, irritability, generalized seizures, and coma (may suggest encephalitis).
- **Where features are typical, reassure the person, parent, or guardian that hand, foot, and mouth disease is usually a mild, self-limiting illness.**
- **Advise on self-care measures:**
  - Ensure fluid intake is adequate.
  - Advise that a soft diet may be necessary if mouth ulcers are painful, and that foods that are hot, spicy, salty, or acidic may cause oral pain.

    - Consider advising use of paracetamol or ibuprofen, if required, to reduce fever and pain.
- **Advise on measures to reduce the risk of transmission.**
- **Do not prescribe antibiotics** either orally or topically unless secondary infection is suspected.
- **Do not prescribe antiviral medication.**
- **Women who are pregnant and have suspected hand, foot, and mouth disease (HFMD) (or who have had contact with a person with HFMD) should be generally managed as for non-pregnant women.** However, consider seeking advice from an obstetrician if the woman is immunocompromised, or is within 3 weeks of delivery, as investigations to confirm the diagnosis may be necessary, and observation of the neonate may be indicated.
- **Follow up is not routinely required,** but advise the person to seek medical advice if they become dehydrated or more unwell, or if oral ulcers persist for more than 3 weeks.
    - **If oral ulcers are persistent, use clinical judgement to determine whether to refer using the 2-week wait pathway.**

Measures to reduce the risk of transmission

- **Advise the person on general hygiene measures to reduce the risk of transmission:**
    - Ensure that hands are washed and dried thoroughly after using the toilet and before eating.
    - Ensure the mouth and nose are covered when coughing and sneezing. The nose and mouth should be wiped with disposable tissues, and then hands washed.
    - Particular care should be taken when handling nappies and tissues.
    - Soiled clothes, bedding, and towels should be washed on a hot cycle of the washing machine.
    - Cups, eating utensils, towels, and clothing should not be shared.
    - Blisters should not be deliberately pierced, as the fluid is infectious.
- **Pregnant women should avoid close contact with any person with hand, foot, and mouth disease.**
- **Children do not need to be excluded from nursery, childcare, or school for infection control purposes.**
    - The child should not attend nursery, childcare, or school if they are too unwell to attend.
    - There is no need to isolate the child (for example excluding visitors or household contacts) or investigate close contacts.
    - Notification is not required in the UK but required in kuwait

TOPIC: PARVOVIRUS B19 INFECTION

- Parvovirus B19 is a small single-stranded DNA virus and is a member of the Parvoviridae family (genus *Erythrovirus*).
    - The virus is usually transmitted by droplet spread through respiratory secretions, more rarely by vertical transplacental transmission from a pregnant woman with acute infection, or by blood transfusion.
    - The virus targets rapidly growing erythroid progenitor cells which form red blood cells found in bone marrow, fetal liver, umbilical cord, and peripheral blood.
    - The incubation period is about 14–21 days.
- Parvovirus B19 infection can cause slapped cheek syndrome (erythema infectiosum or fifth disease), commonly in school-age children. It is usually a mild, self-limiting illness.
    - It was named fifth disease as it was the fifth identified childhood exanthema.

What are the complications?

Complications of parvovirus B19 infection are rare in healthy children and adults. Pregnant women, and people who are immunocompromised or have haematological disorders such as hereditary anaemias, are at increased risk of serious and potentially life-threatening complications.

- **Pregnancy** — in the majority of cases of maternal infection the fetus is unaffected, however it can result in:
    - Non-immune hydrops fetalis.
    - Fetal death.
    - Maternal pre-eclampsia-like syndrome ('mirror syndrome').
        - This is rare, and is the development of maternal oedema secondary to fetal hydrops. It is usually associated with severe pre-eclampsia.
        - It is so-called because the maternal signs reflect those present in the fetus.
    - Note: universal screening of pregnant women for past parvovirus B19 infection is not recommended, as there is no vaccine or prophylaxis treatment available.
- **People who are immunocompromised (for example leukaemia, lymphoma, HIV infection, receiving chemotherapy, or bone marrow or solid organ transplantation) or have a haematological disorder (for example sickle cell disease, thalassaemia, hereditary spherocytosis, or autoimmune haemolytic anaemia) are at increased risk of complications, including:**
    - Transient aplastic crisis/anaemia.
        - People with an underlying haemolytic disorder or who have increased red blood cell turnover are at increased risk.
        - This usually causes a severe, transient anaemia, however potentially life-threatening aplastic crisis may occur when there is severe anaemia associated with absent reticulocytes (which may cause angina, heart failure, and bone marrow necrosis).
        - Affected people may initially present with severe anaemia associated with pallor, weakness, lethargy, fever, tachycardia, and tachypnoea.
    - Pure red cell aplasia.
        - People who are immunocompromised are more at risk.
        - Persistent parvovirus B19 infection may cause chronic bone marrow suppression with chronic, severe anaemia and potentially life-threatening pancytopenias.
    - Neurological complications.
        - These include encephalitis, meningitis, stroke, and peripheral neuropathy.

- Other complications (rare).
    - These include myocarditis, cardiomyopathy in children, nephritic syndrome, hepatitis, and vasculitis.

What is the prognosis?

- **For most healthy people,** parvovirus B19 infection causes a mild, self-limiting illness which is followed by life-long immunity.
    - In children, the erythematous rash, which can develop on the torso and extremities after the initial facial rash, usually resolves after 1–3 weeks, but may recur following exposure to sunlight, heat, or stress.
    - In adults, a symmetrical polyarthropathy may persist for several weeks, but rarely may persist for months, or may recur several months after the acute infection.
- **In pregnant women** with confirmed parvovirus B19 infection the prognosis partially depends on the gestational age at infection. The risk of transmission to the fetus is increased at 9–20 weeks' gestation.
- **People who are immunocompromised** may develop persistent infection and are at increased risk of haematological complications.

Diagnosis:

When should I suspect parvovirus B19 infection?

In healthy children the diagnosis of parvovirus B19 infection is usually made on clinical grounds.

- **Suspect infection in children if they present with an acute biphasic illness with a characteristic macular erythematous facial rash, that appears on one or both cheeks (resembling a 'slapped cheek').**
    - Prodromal symptoms are common and occur 1 week after exposure and 1–2 weeks prior to onset of the exanthema. These can include low-grade fever, headache, nasal discharge, abdominal pain, pharyngitis, and coryza.
    - A maculopapular rash may develop 1–4 days after the initial facial rash, spread to the extremities and torso, and evolve into a lacy reticular appearance. The rash usually resolves after 1–3 weeks, but may persist for longer.
    - Symmetrical polyarthropathy affecting the small joints of the hands and occasionally the knees, ankles, and wrists may occur although this is uncommon in children, and rarely the rash can present as papular-purpuric gloves and socks syndrome.

In healthy adults, it is difficult to make a diagnosis of parvovirus B19 infection on clinical grounds, as symptoms may be atypical, it may mimic other conditions**, and 25–50% of infections (in adults and children) are asymptomatic.**

- Prodromal symptoms are more common in adults and may appear about a week after contact with the infection.
- Self-limited symmetrical polyarthropathy of the small joints is often the most common presenting symptom, occurring in around 50% of adults and is more common in women.
    - This typically affects the small joints of the hands, wrists, knees, and ankles, and lasts several weeks, although rarely it may persist for months.

- The characteristic facial rash seen in children may be absent in adults, but 2–3 weeks after the first prodromal symptoms, a maculopapular rash on the trunk, back, and limbs may develop, which fades to produce a lace-like, reticular rash. This usually resolves within a week.

Note: consider parvovirus B19 infection in people who are immunocompromised or who have an underlying haematological disorder, who present with sudden-onset or severe anaemia.

How should I investigate suspected parvovirus B19 infection?

In healthy children and adults, the diagnosis of parvovirus B19 infection should be based on clinical features — laboratory investigation to confirm the diagnosis is not required.

If a pregnant woman has suspected parvovirus **B19 infection, contact the local virology, microbiology, or infectious diseases department immediately for further advice.**

This is likely to include urgent blood testing for:

- Parvovirus B19-specific immunoglobulin M (IgM). Note: serum from antenatal booking bloods or earlier samples may be available to help confirm the diagnosis, but initial blood testing should not be delayed.
  - **If parvovirus B19-specific IgM is not detected,** this excludes infection in the 4 weeks prior to the blood test. Infection cannot be excluded if the blood test is taken more than 4 weeks after the onset of rash or associated illness. Seek specialist advice on any additional testing needed, for example, of antenatal booking bloods.
  - **If parvovirus B19-specific IgM is detected** in the first 20 weeks of pregnancy, this suggests recent infection with parvovirus B19, and the diagnosis is confirmed by testing for high-titre viral DNA using polymerase chain reaction (PCR) analysis, or parvovirus B19-specific IgG in antenatal booking blood samples.
- Rubella infection, irrespective of the woman's previous testing or immunization status.

If parvovirus B19 infection is suspected in a person who is immunocompromised, has a haematological disorder, or is in early transient aplastic crisis, arrange urgent blood testing for:

- High-titre viral DNA using PCR analysis, to confirm the diagnosis of parvovirus B19 infection. Contact the local virology, microbiology, or infectious diseases department if there is any uncertainty about blood tests to arrange.
- Full blood count (including reticulocyte count).

What else might it be?

Conditions that may present similarly to parvovirus B19 infection include:

- **Rubella**
- **Measles**
- **Scarlet fever**
- **Roseola infantum**
- **Enterovirus and adenovirus**
  - These are common causes of rubella-like rash.
- **Rheumatoid arthritis**
  - This often causes a symmetrical polyarthropathy, and people with parvovirus B19 infection may test positive for rheumatoid factor.

Other causes include:

- Tropical viruses, including alphaviruses and flaviviruses (for example Dengue fever) — should be considered if the person has recently travelled in the tropics.
- Cytomegalovirus (CMV), HIV infection, glandular fever (infectious mononucleosis), Kawasaki disease, syphilis, toxoplasmosis — may all cause a rash with associated lymphadenopathy.
- Drug reaction, for example to anti-epileptic drugs.

Management:

Scenario: Children and adults (not pregnant)

From birth to 12 years

For children and adults (who are not pregnant) with suspected parvovirus B19 infection:

- **Take a medical history and ask about:**
  - The date of onset, clinical features, and type and distribution of any rash and associated illness.
  - Previous relevant history of infection, antibody testing, and vaccine administration (with dates/places).
  - Known contact with any person with a potentially infectious rash or illness (with dates of contact).
- **Give advice on:**
  - **Self-management strategies for symptom relief,** including:
    - Using simple analgesia, such as paracetamol or a nonsteroidal anti-inflammatory drug (NSAID) such as ibuprofen, if needed and appropriate.
- **For healthy children who have the typical rash of slapped cheek syndrome,** advise that:
  - The child should no longer be infectious once the rash develops, and it is not necessary for the child to stay off school or nursery.
  - School or nursery should be informed of the child's suspected parvovirus B19 infection, so that the following groups of people can be managed effectively:
    - People at risk of complications (such as pregnant women, and children and adults who are immunocompromised or have a haematological disorder) who have been in contact with the child during the prodromal period.
    - Other nursery or school children who develop possible prodromal symptoms.
- **For healthy adults with suspected parvovirus B19 infection:**
  - Advise that it is not usually necessary to stay off work if symptoms are controlled.
  - Check the person's rubella status if they have not been fully immunized against rubella or got a documented history of previous rubella infection. It may be sensible to avoid contact with pregnant women while any rash is present, until the person's rubella status is known.
- **For people who are at risk of complications,** ensure that appropriate laboratory investigations have been arranged.
- **Arrange follow up in primary care, referral, or admission, if appropriate.**
  - Consider arranging an urgent full blood count including reticulocyte count (if not already done) or emergency admission for people with symptoms of severe or aplastic anaemia (such as breathlessness, lightheadedness, lethargy, or confusion), depending on clinical judgement.

Scenario: Pregnant women

From age 13 years onwards (Female)

How should I manage suspected parvovirus B19 infection in pregnancy?

Take a full history and conduct an examination in all pregnant women with suspected parvovirus B19 infection.

- **Ask the woman about:**
  - The gestation of pregnancy.
  - Date of onset, clinical features, and type and distribution of any rash and associated illness.
  - Previous relevant history of infection, antibody testing, and vaccine administration (with dates/places).
  - Recent travel history and relevant dates.
  - Known contact with any person with a rash illness, or recent travel (with dates of contact).
- **Contact the local virology, microbiology, or infectious diseases department immediately for further advice on what** laboratory investigations **and monitoring should be arranged.** This should include:
  - Confirming the diagnosis of parvovirus B19 infection.
  - Checking for rubella infection at the same time, irrespective of the woman's previous testing or immunization status to rubella.
- **Give advice on self-management strategies for symptom relief,** including:
  - Using simple analgesia, such as paracetamol, if needed and appropriate.
- **Advise the woman that:**
  - It is not usually necessary to stay off work if symptoms are controlled, as the infection is no longer contagious by the time the rash or arthropathy develops.
  - If the woman has not been fully immunized against rubella or got a documented history of previous rubella infection, it may be sensible to avoid contact with other pregnant women while any rash is present, until her rubella status is known.

How should I manage confirmed parvovirus B19 infection in pregnancy?

If parvovirus B19 infection is confirmed in a pregnant woman following laboratory investigations**:**

- **Arrange urgent referral to a specialist in fetal medicine** for ongoing monitoring and mahagement.
  - Advise the woman that there is a risk of transmission to the fetus (approximately 30%), and specialist fetal monitoring will be arranged. If fetal transmission occurs, there may be effective specialist treatment options available.

TOPIC: MUMPS

- Mumps is an acute infectious disease caused by a paramyxovirus characterised by bilateral parotid swelling.
- It is spread by respiratory droplets, fomites or saliva, and replicates mainly in the upper respiratory mucosa.
- The incubation period of mumps is 16-18 days (range 12–25 days)
- It is most infectious from around 1–2 days before onset of symptoms, to about 9 days afterwards, although it may be asymptomatic in 15–20% of people.
    - Asymptomatic mumps infection is common in children.
- Nearly all people develop life-long immunity to mumps after one episode of infection.
    - However, 1–2% of cases are thought to be re-infections.

What is the prognosis?

- Mumps is usually a self-limiting disease that resolves within 1–2 weeks, and most people recover without any long-term complications.
- Although generally rare, complications occur more frequently among adults than children.

What are the complications?

- **Parotitis** (inflammation of the parotid glands) usually resolves without complications.
    - The submandibular and sublingual salivary glands are also affected in about 10% of people with mumps, usually in conjunction with bilateral parotitis.
- **Epididymo-orchitis** — the most common complication affecting up to 38% of men.
    - Mumps rarely leads to infertility, but can contribute to sub-fertility.
    - Unilateral mumps epididymo-orchitis can significantly, but only transiently, diminish the sperm count, mobility, and morphology.
        - Impairment of fertility is estimated to occur in about 13% of men.
    - Epididymo-orchitis is bilateral in 15–30% of affected men, and tends to have a worse prognosis.
        - Bilateral mumps epididymo-orchitis causes infertility in 30–87% of affected men.
    - if affected testicles, 30–50% show a degree of testicular atrophy.
    - Most men recover after 2 weeks, although in about 20% of affected men it may persist for longer.
- **Encephalitis**
- **Oophoritis** — reported in about 7% of women with mumps, but rarely causes infertility or premature menopause.
- **Aseptic meningitis** — occurs in up to 25% of people with mumps, and is three times more common in males. Mumps meningitis is usually benign, and almost all patients have a complete recovery with no residual neurological deficits and no risk of mortality.
- **Transient hearing loss** — affects about 4% of people, but permanent deafness is much less common (about 1 in 20,000 people). Bilateral hearing loss is very rare.
- **Myocardial complications** — ECG changes, including ST segment depression, T wave inversion, and prolonged PR intervals, are seen in 15% of mumps infections. Fatal myocarditis with dilated cardiomyopathy has been reported but is extremely rare.
- **Pancreatitis** — affects about 4% of people, but is usually mild.
- **Spontaneous abortion**
- **Rarer complications** of mumps that have been reported include other central nervous system disorders (such as cerebellar ataxia, facial palsy, transverse myelitis, and Guillain–Barre syndrome), thyroiditis, mastitis, prostatitis, hepatitis, and thrombocytopenia.

Diagnosis:

When should I suspect a diagnosis of mumps?

- **The diagnosis of mumps is usually clinical and is confirmed by laboratory analysis of a saliva sample to detect presence of immunoglobulin (Ig)M mumps antibody.**
- **Consider a diagnosis of mumps in people presenting with parotitis (swollen parotid glands) — this is present in 95% of symptomatic cases.**
    - Typically, one parotid gland is affected first, reaching a maximal size after 2–3 days, with the other gland closely following it. About a quarter of affected people have unilateral parotitis.
    - The ear lobe over the affected gland may be deflected upward and outward, and the angle of the mandible may be obscured (this does not occur with cervical adenopathy).
    - The affected gland may be tender to touch.
    - During the period of gland enlargement, the person may complain of earache, and have difficulty with pronunciation of words or chewing.
- **Other features that are consistent with a diagnosis of mumps include:**
    - **Non-specific symptoms (which may precede parotitis),** such as low-grade fever, headache, earache, malaise, muscle ache, and loss of appetite — typically occur 1 day before overt signs of parotitis and peak around the time the parotid glands are most swollen.
- **Assess the likelihood of mumps, by considering:**
    - **Immunization history.**
        - Mumps is unlikely in people who have been fully immunized.
        - Younger people who have not received two doses of the combined measles, mumps, and rubella (MMR) vaccine are most at risk.
    - **History of mumps** — mumps is unlikely in people who have previously had mumps.
    - **Contact with someone with mumps** — significant contact is considered as being in the same room for 15 minutes or more, or face-to-face contact. Mumps may have been contracted up to 4 weeks previously.
    - **Recent outbreaks**
    - **Age** — mumps is unlikely in infants aged under 1 year.
- Consider a different cause for the symptoms if the person is likely to have immunity to mumps, clinical features are atypical, there is no history of contact with mumps, and there are no local outbreaks.

Saliva sample

- Detection of specific IgM in saliva samples (ideally between one and six weeks) after the onset of rash or parotid swelling, has been shown to be highly sensitive and specific for confirmation of mumps. It is recommended that oral fluid samples should be obtained from all notified cases, other than during a large epidemic.

What else may cause parotitis?

- **Other infectious causes that may present with parotitis include:**
    - Viral infections, such as Epstein–Barr (the virus that causes mononucleosis), parainfluenza, adenovirus, influenza type A, coxsackievirus, parvovirus B19 (the virus that causes erythema infectiosum, also known as slapped cheek syndrome), lymphocytic choriomeningitis virus, and HIV.

    - Acute suppurative parotitis — acute bacterial infection, most commonly caused by *Staphylococcal aureus.* It may also be caused *by* atypical mycobacteria (for example, tuberculosis).
- **Non-infectious causes of parotitis include:**
    - Parotid duct obstruction – for example, salivary stones, cysts or tumours.
    - Prescription drugs (for example, thiazide diuretics, phenothiazines, thiouracil, iodide contrast media).
    - Metabolic disorders (for example, diabetes mellitus, cirrhosis, uraemia).
    - Autoimmune disease (for example, sarcoidosis, Sjogren's syndrome, granulomatosis with polyangitis [previously known as Wegener's granulomatosis]).

Management:

From birth onwards

How should I manage people with mumps?

- **Mumps is a notifiable disease.**
- **Advise the person (or their carer):**
    - That mumps is usually a self-limiting condition. It will usually resolve over 1–2 weeks, with no long-term consequences and antibiotic treatment is not required.
    - To rest, drink adequate fluids, and take paracetamol or ibuprofen for symptomatic relief (aspirin should be avoided in children younger than 16 years of age).
    - To apply warm or cold packs to the parotid gland as it may ease discomfort.
    - To stay off school or work for 5 days after the initial development of parotitis.
- **Consider follow-up about 1 week after the onset of parotitis.**
    - Check that symptoms have resolved or are resolving adequately.
    - Ensure the person is up-to-date with their vaccinations, where applicable.
- **Advise the person to seek medical advice if they develop symptoms of:**
    - Meningitis — for example, severe headache, vomiting, neck stiffness (urgent attention should be sought if altered consciousness or convulsions develop).
    - Epididymo-orchitis — characterized by swollen and painful testicles.
- Treatments, such as Human Normal Immunoglobulin (HNIG), antibiotics, or corticosteroids are *not* recommended for mumps.

How should I manage a person who has been in contact with possible mumps?

- Offer immunization with the combined measles, mumps, and rubella (MMR) vaccine to people who are not fully immunized (that is, people who have not had two doses of the MMR vaccine) as soon as it is convenient.
- Advise the person to seek medical advice if they develop symptoms of mumps.

- Antibody response to the mumps component of the combined measles, mumps, and rubella (MMR) vaccine does not develop soon enough to provide effective prophylaxis following contact with suspected mumps.
    - However, the consultation provides a good opportunity to offer the MMR vaccine for future prevention of mumps (if it is not contracted) as well as measles and rubella.
    - The MMR vaccine will not exacerbate the effect of naturally-acquired mumps.

How should I manage a pregnant woman with suspected mumps, or possible exposure to mumps?

- Manage pregnant women who are suspected of having mumps in the same way as otherwise healthy people.
- The combined measles, mumps, and rubella (MMR) vaccine is contraindicated in pregnancy.

- There is no evidence that pregnant women are at any higher risk of complications of mumps than other adults and no specific treatment other than supportive is recommended.
- The evidence base for an increased risk of fetal loss due to mumps in pregnancy is weak. There are no available interventions to reduce any possible risk to the fetus.

How should I manage an immunocompromised person with suspected mumps, or possible exposure to mumps?

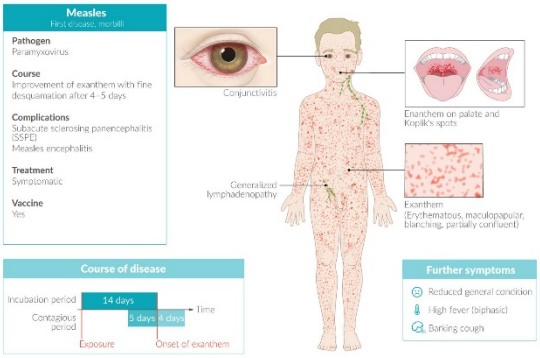
- Manage immunocompromised people who are suspected of having mumps in the same way as otherwise healthy people.
- For immunocompromised people who may have been exposed to mumps, but who are not fully immunized (that is, they have not had two doses of the MMR vaccine):
    - Do not routinely give the combined measles, mumps, and rubella (MMR) vaccine.
    - Follow local protocols and take specialist advice if required.
    - Advise the person to seek medical advice if they develop symptoms of mumps.

How should I manage suspected mumps epididymo-orchitis?

- **Offer advice and reassurance if mumps epididymo-orchitis is diagnosed**.
    - Advise the man on symptomatic relief, such as:
        - Bed rest.
        - Scrotal support.
        - Application of warm or cold topical packs.
        - Paracetamol or ibuprofen.
    - Inform the man that, in most cases, the symptoms will completely resolve within 2 weeks and there are unlikely to be long-term problems with fertility.
        - If the man is concerned about fertility, offer semen analysis 3 months after the mumps has resolved, particularly if there was severe or bilateral epididymo-orchitis.
- **There is no specific treatment for mumps epididymo-orchitis** — oral corticosteroids and antibiotics are not routinely recommended.

TOPIC: Measles

Measles is now rarely seen in the developed world following the adoption of immunization programmes. Outbreaks are occasionally seen, particularly when vaccinations rates drop, for example after the MMR controversy of the early 2000s.



- **Overview:**
- RNA paramyxovirus.
- One of the most infectious known viruses.
- Spread by aerosol transmission.
- Infective from prodrome until 4 days after rash starts.
- Incubation period = 10-14 days.

- **Features:**
- Prodromal phase.
  - Irritable.
  - Conjunctivitis.
  - Fever.
- Kolpik spots.
  - Typically develop before the rash.
  - White spots ('grain of salt') on the buccal mucosa.
- Rash.
  - Starts behind ears then to the whole body.
  - Discrete maculopapular rash becoming blotchy & confluent.
  - Desquamation that typically spares the palms and soles may occur after a week.
- Diarrhea occurs in around 10% of patients.

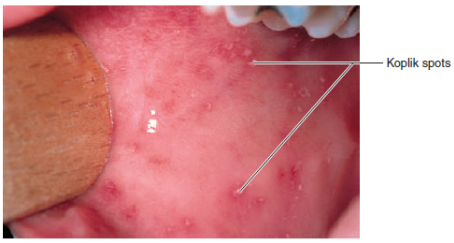


FIGURE 22.19 Koplik Spots, Characteristic of Measles (Rubeola) These spots resemble grains of salt on a red base.

- **Investigations:**
- IgM antibodies can be detected within a few days of rash onset.
- **Management:**
- Mainly supportive.
- Admission may be considered in immunosuppressed or pregnant patients.
- Notifiable disease → inform public health.
- **Complications:**
- Otitis media: the most common complication.
- Pneumonia: the most common cause of death.
- Encephalitis: typically occurs 1-2 weeks following the onset of the illness).
- Subacute sclerosing panencephalitis: very rare, may present 5-10 years following the illness.
- Febrile convulsions.
- Keratoconjunctivitis, corneal ulceration.
- Diarrhea.
- Increased incidence of appendicitis.
- Myocarditis.
- **Management of contacts:**
- MMR vaccine should be given within 72 hours.

TOPIC: RUBELLA

- Rubella (also known as German measles) is a viral infection spread by direct contact with an infected person or droplet spread from respiratory secretions — it is preventable with vaccination.
- Rubella is generally a mild infection, but it can cause serious complications in pregnancy (congenital rubella syndrome).
    - The risk of serious congenital defects is highest in the first trimester of pregnancy.
- Once infected or immunized, most people develop life-long immunity to rubella. On rare occasions, reinfection can occur, but it is usually sub-clinical and difficult to diagnose.
- Rubella is a notifiable disease.

What causes it?

- Rubella is caused by a togavirus, which is the only member of the genus *Rubivirus.*
    - Transmission occurs through direct contact with an infected person or droplet spread from nasopharyngeal secretions.
    - The virus replicates in the nasopharynx and local lymph nodes and is then spread haematogenously to the rest of the body (including the placenta and foetus in pregnant women).
    - Following exposure to the rubella virus, the incubation period is 14-21 days, with most people developing a rash 14-17 days after exposure — people with rubella are infectious from 7 days before symptoms appear to 4-10 days after the onset of rash.
        - People with rubella are most infectious when the rash is erupting.

What are the complications?

- **Rubella can cause serious complications in pregnancy including miscarriage, stillbirth, and severe birth defects known as congenital rubella syndrome (CRS).**
    - The severity and type of congenital defects likely to develop vary according to the stage of pregnancy when infection occurs.
        - Infection in the first 8-10 weeks of pregnancy results in damage in up to 90% of surviving infants.
        - The risk of damage declines to about 10-20% if infection occurs in weeks 11-16, and is rare with infection after 16 weeks, with only deafness reported following infections up to 20 weeks of pregnancy.
    - CRS can lead to the development of one or more abnormalities including eye defects (such as cataracts), hearing impairment, cardiac abnormalities (such as patent ductus arteriosus and pulmonary artery stenosis), microcephaly, developmental delay, intra-uterine growth restriction, autism, inflammatory lesions of brain, liver, lungs and bone marrow, and endocrine abnormalities (such as diabetes mellitus and thyroid dysfunction).
- **Rubella rarely causes complications in otherwise healthy people.**
    - Arthritis and arthralgia (most commonly in adult women) can occur and may persist for weeks or months, or recur.
    - Bleeding disorders (thrombocytopenia) have been reported in about 1 in 3000 infections.
    - Encephalitis has been reported in about 1 in 5000 cases — other rarely occurring neurological complications include myelitis, optic neuritis, peripheral neuritis, and Guillain-Barre syndrome.

What is the prognosis?

- Most cases of rubella infection are mild (with transient rash and lymphadenopathy), and resolve spontaneously within a week — occasionally, joint inflammation and pain can occur (most often in adults).
- However, maternal infection in non-immune women during pregnancy can cause serious congenital abnormalities (known as congenital rubella syndrome), lifelong disability, and foetal loss.

Diagnosis:

When should I suspect rubella?

- **There are no clinical features specific to rubella infection** — diagnosis cannot be made on clinical features alone. laboratory confirmation is required.
    - Symptoms and signs of rubella infection are similar to many other conditions, including other viral illnesses (such as parvovirus B19), toxoplasmosis, and allergic reactions. Rubella infection may be asymptomatic in up to 50% of people.
- **When present, clinical features develop 2–3 weeks after exposure and include:**
    - **Rash** — present in 50-80% of cases, it typically starts on the face and neck before spreading down the body and becoming generalized. The rash is erythematous (pink or light-red), discrete, maculopapular, sometimes mildly pruritic, and transient (usually present for 3–4 days).
    - **Lymphadenopathy** — may precede the rash by 5-10 days and last for 2 weeks after the rash resolves. The suboccipital (lower part of the back of the skull), postauricular (behind the ears), and cervical (neck) lymph nodes are most often affected.
    - **Arthritis and arthralgia** — arthralgia or arthritis is more common in adults and may occur in up to 70% of women with rubella.
    - **Non-specific symptoms** tend to affect adults more than children and in adults are sometimes prodromal — they include low-grade fever (less than 39°C), malaise, mild upper respiratory tract symptoms and non-purulent conjunctivitis.
- **Rubella should be suspected in people with clinical features consistent with infection or with risk factors:**
    - It is essential to have a low threshold of suspicion for rubella in pregnant women, especially before 20 weeks gestation.
    - Risk factors include:
        - Incomplete immunization and no evidence of previous infection — rubella is unlikely in people who have previously had rubella (although reinfection can occur).
        - History of exposure to contacts with rubella within the last 3 weeks.
        - Travel to an area where rubella is endemic.

How should I assess a person with suspected rubella?

- **Take a history asking about:**
    - Clinical features, including onset and progression.
    - The possibility of pregnancy.
    - Past medical history (including immunosuppression) and drug history.
    - Other potential causes of symptoms.
- **Examine the person looking for clinical signs of rubella infection and complications:**
    - Check vital signs including temperature.

    - Look for a rash; lymphadenopathy; conjunctivitis; and respiratory, musculoskeletal, and neurological signs.
- **Consider the need for investigations:**
    - If there is any suspicion of rubella infection, immediately notify the local Health Protection Team (HPT) — rubella is a notifiable disease.
    - Rubella cannot be accurately diagnosed through history and clinical features alone.
        - Arrange serology testing for all pregnant women.
        - For all other people, the Health Protection Team (HPT) will provide advice on testing (for example they may request an oral fluid sample).
    - Note: when rubella testing is requested in women with a rash illness, they will be also tested for parvovirus B19 infection.

What else might it be?

Other conditions that can present with similar clinical features to rubella include:

- **Parvovirus B19**
- **Measles**
- **Other viral infections including:**
    - Herpesvirus type 6 (roseola infantum), enterovirus (for example coxsackievirus and echovirus), and cytomegalovirus.
    - Tropical viruses including alphaviruses and flaviviruses (for example dengue virus, West Nile virus, chikungunya virus, and Zika virus) — consider tropical viruses if the person has recently travelled to an endemic area.
- **Other infections including:**
    - Scarlet fever.
    - Kawasaki disease, syphilis, and toxoplasmosis.
- **Drug reactions,** for example, mononucleosis reaction with some penicillins (such as amoxicillin).

Management:

Scenario: Non-pregnant - suspected rubella or possible exposure

From birth onwards

Management of suspected rubella or possible exposure in a non-pregnant person

If there is clinical suspicion of rubella infection:

- **Contact the local Health Protection Team (HPT) immediately.**
    - Rubella is a notifiable disease:
        - A notification form should be completed immediately and sent to the HPT within 3 days.
- **Advise the person (or their carer):**
    - That rubella is usually a mild, self-limiting condition that typically resolves within a week.
    - That there is no specific treatment for rubella — they should rest, drink adequate fluids, and take paracetamol or ibuprofen (if appropriate) for symptomatic relief (aspirin should be avoided in children younger than 16 years of age).
    - To stay away from school or work for at least 5 days after the initial development of the rash.

- To avoid contact with pregnant women.
- To get up-to-date with their vaccinations, if applicable, once they have recovered from the acute symptoms.
    - Adult women of childbearing age should avoid getting pregnant for at least 4 weeks after receiving MMR vaccine.

Scenario: Pregnant - suspected/confirmed rubella or possible exposure

From age 13 years onwards (Female)

How should I manage suspected or confirmed rubella in a pregnant woman?

If rubella is suspected in a pregnant woman:

- **Contact the local Health Protection Team immediately.**
    - Rubella is a notifiable disease:
        - A notification form should be completed immediately and sent to the HPT within 3 days.
    - Laboratory investigation is necessary for all pregnant women regardless of previous testing, immunization status, or stage of pregnancy.
- **If rubella infection is confirmed and the woman is in the first 20 weeks of pregnancy, or there is any doubt about the gestational age:**
    - Refer urgently to obstetrics (fetal medicine) for risk assessment, counselling, and management. Risk to the foetus is dependent on the stage of pregnancy:
        - Beyond 20 weeks gestation there have been no published case reports of congenital rubella syndrome (CRS).
        - Between 16–20 weeks gestation there is a low chance of deafness occurring.
        - Between 11–16 weeks gestation there is a 10-20% risk of CRS.
        - Before 8–10 weeks gestation there is a 90% risk of CRS and a high likelihood of multiple defects.
    - There are no effective treatments to prevent CRS.

How should I manage contact with suspected or confirmed rubella in pregnancy?

- **Advise the woman to inform their midwife, GP, or obstetrician urgently if they develop a rash at any time in pregnancy.** Until they are assessed and confirmed to be uninfected or non-infective, they should:
    - Avoid contact with other pregnant women (including in antenatal clinics or maternity settings).
    - Inform clinical staff of suspected infection prior to attending any medical area.
- **Arrange follow-up for unvaccinated women to be immunized with MMR vaccine after delivery** — rubella immunization should *not* be administered in pregnancy but may be given post-partum.

TOPIC: MENINGITIS

- **Meningitis** is inflammation of the two inner meninges (the pia and arachnoid mater) of the brain and spinal cord.
    - This is different from encephalitis, which is inflammation of the brain tissue itself.
- Causes of meningitis can be infective (bacterial, viral, and fungal) and non-infective (certain cancers, autoimmune disorders, and drugs).
- **Bacterial meningitis** is a life-threatening condition that affects all ages, but is most common in babies and children.

    - **Meningococcal disease** is infection with *Neisseria meningitidis*.
    - **Pneumococcal disease** is infection with *Streptococcus pneumoniae* (also called pneumococcal).
- **Bacterial meningitis and meningococcal disease are notifiable diseases.**

What causes it?

- Overall, in the UK, *Neisseria meningitidis* (meningococcus), *Streptococcus pneumoniae* (pneumococcus), and *Haemophilus influenzae* type b (Hib) are the most common causative organisms of acute bacterial meningitis in children aged 3 months or older and adults.
- Meningococcal disease is the leading infectious cause of death in early childhood.
    - There are 12 capsular groups of meningococci — B, C, W, and Y were historically the most common in the UK, however, after the introduction of the meningococcal C vaccination programme, group B (MenB) now accounts for the majority of cases.
    - MenB is responsible for the majority of invasive meningococcal disease cases in people aged under 25 years.
- In neonates (younger than 1 month of age), *Streptococcus agalactiae*, *Escherichia coli, S. pneumoniae*, and *Listeria monocytogenes* are the most common causative organisms.
- In children aged 3 months or older, *N. meningitidis*, *S. pneumoniae*, and Hib are the most common causative organisms.
- In adults, *S. pneumoniae* is the most common cause of bacterial meningitis.

How is it transmitted?

- **Bacterial meningitis and meningococcal disease are transmitted by aerosol, droplets, or direct contact with secretions from the upper respiratory tract.** Transmission usually requires either frequent or prolonged close contact.
- ***Neisseria meningitidis*** is usually commensal.
- ***Streptococcus pneumoniae*** — some serotypes of pneumococcus may be carried in the nasopharynx without symptoms, with disease occurring in a small proportion of infected people.

What are the risk factors for bacterial meningitis and meningococcal disease?

- **Young age** is the most significant risk factor.

    - Although bacterial meningitis can affect all ages, incidence of bacterial meningitis and meningococcal disease is highest in children aged under 2 years and declines during childhood.
    - There is also an increased risk in adolescence and early adulthood.

- **Other risk factors include:**
    - Winter season — bacterial meningitis is more prevalent during the winter.
    - An absent or non-functioning spleen.
    - Older age (more than 65 years).
    - Immunocompromised state (for example, HIV infection or chemotherapy).
    - Incomplete immunization.
    - Cancer – people with leukaemia and lymphoma are more susceptible to bacterial meningitis.
    - Organ dysfunction — for example liver or kidney disease.
    - Smoking.
    - Living in overcrowded households, college dormitories, or military barracks.
    - Cranial anatomical defects — congenital or acquired.
    - Cochlear implants.
    - Contiguous infection — for example, otitis media, sinusitis, pneumonia, mastoiditis.
    - Sickle cell disease.

What are the complications of bacterial meningitis?

- Although overall mortality from acute bacterial meningitis has fallen in recent years, there has been no change in the rate of complications.
    - Complications are more common following pneumococcal meningitis and occur in about 30% of people compared with 7% with meningococcal meningitis.
    - The frequency of complications is much higher in meningococcal septicaemia (up to 57%).
- Cerebral infarction occurs in one in four people with bacterial meningitis
    - However, neurological complications may also be caused by other pathologies, such as subdural empyema, cerebral abscess, or intracerebral bleeding.
- **Neurological complications include:**
    - Hearing loss (34%).
    - Seizures (13%).
    - Cognitive impairment (9%).
    - Motor deficits (12%).
    - Visual impairment (6%).
- **Physical complications include:**
    - Amputations.
    - Skin scars.
- **Other complications include:**
    - Hydrocephalus (7%).
    - Reduced quality of life.
    - Anxiety.
    - Learning difficulties.
    - Emotional and behavioural difficulties.

What is the prognosis of acute bacterial meningitis?

- With prompt and adequate antimicrobial treatment and supportive therapy, the outcome of acute bacterial meningitis is excellent.
- Features associated with a poor prognosis in adults include:

    - Low Glasgow Coma Scale score on admission (low level of consciousness).
    - Tachycardia.
    - Absence of rash.

    - Thrombocytopenia.
    - Elevated erythrocyte sedimentation rate (ESR).
    - Positive blood culture.
    - Cerebrospinal fluid leucocyte counts of less than 1000 x 10^9 cells per mL.
- Complications of bacterial meningitis occur in up to 30% of children and up to one-third of adults and are more common in pneumococcal meningitis than in meningococcal meningitis.

Diagnosis:

When should I suspect bacterial meningitis or meningococcal disease?

- **Consider a diagnosis of bacterial meningitis and meningococcal septicaemia in people presenting with the following clinical features:**
    - **Common non-specific symptoms/signs:**
        - Fever.
        - Vomiting/nausea.
        - Lethargy.
        - Irritability/unsettled behaviour.
        - Ill appearance.
        - Refusing food/drink.
        - Headache.
        - Muscle ache/joint pain.
        - Respiratory symptoms/signs or breathing difficulty.
    - **Less common non-specific symptoms/signs include:**
        - Chills/shivering.
        - Diarrhoea, abdominal pain/distension.
        - Sore throat/coryza or other ear, nose, and throat symptoms/signs.
    - **More specific symptoms/signs include:**
        - Non blanching rash
        - Stiff neck.
        - Capillary refill time of more than 2 seconds, cold hands and feet.
        - Unusual skin colour.
        - Shock and hypotension.
        - Leg pain.
        - Back rigidity.
        - Bulging fontanelle.
        - Photophobia.
        - Kernig's sign (person unable to fully extend at the knee when hip is flexed).
        - Brudzinski's sign (person's knees and hips flex when neck is flexed).
        - Unconsciousness or toxic/moribund state.
        - Paresis.
        - Focal neurological deficit including cranial nerve involvement and abnormal pupils.
        - Seizures.
- Be alert to the possibility of bacterial meningitis or meningococcal septicaemia when assessing children or young people with acute febrile illness.
- Be aware that:
    - Children and young people with bacterial meningitis commonly present with non-specific symptoms and signs, including fever, vomiting, irritability, and upper respiratory tract symptoms. These may be difficult to distinguish from other viral infections.
        - Some children with bacterial meningitis present with seizures.

    - Children and young people with the more specific symptoms and signs are more likely to have bacterial meningitis or meningococcal septicaemia, and the symptoms and signs may become more severe and more specific over time.
    - Classical signs of meningitis (neck stiffness, bulging fontanelle, high pitched cry) are often absent in infants with bacterial meningitis.
- Transfer all suspected cases urgently to secondary care for a second opinion if there is any doubt.

How should I assess a rash that is suspicious of meningococcal disease?

- **When a non-blanching rash is present,** it may appear as a:
    - **Petechial rash** (red or purple non-blanching macules smaller than 2 mm in diameter).
    - **Purpuric (haemorrhagic) rash** (spots larger than 2 mm in diameter) — this may be absent in the early phase of the illness and may initially be blanching or macular in nature.
- **Examine the whole body systematically for rash and unusual skin colour** — particularly for non-blanching rashes and petechiae.
    - Ask the parent or person about any new rashes or marks (such as a bruise, spot, blister, or stain — the person may use other words to describe a rash).
    - Consider checking for non-blanching rashes using the 'glass test'.
        - This involves pressing the side of a glass or tumbler firmly against the rash to see if the rash fades or loses colour under pressure. A petechial or purpuric rash does not fade.
        - Be aware that the glass test should *not* be used solely for diagnosing bacterial meningitis and meningococcal septicaemia.
    - For people with darker skin, a rash may be less visible (even when using the glass test). Check paler areas of the body (such as the soles of the feet or palms of the hands) or the conjunctivae or palate.
- **Be aware that:**
    - The disease process can be advanced before the rash starts to appear.
    - A blanching rash may develop into a non-blanching rash as the illness progresses.
    - A rapidly evolving petechial or purpuric rash is a sign of very severe disease and is a risk factor for a fatal outcome.
- **The risk of meningococcal disease is high in a child or young person with petechiae if:**
    - The petechiae start to spread.
    - The rash becomes purpuric.
    - There are signs of bacterial meningitis.
    - There are signs of meningococcal septicaemia.
    - The child or young person appears unwell.

The glass test:

- Although the glass test is widely promoted in patient information leaflets, Prodigy found no evidence on its use. The glass test is not recommended in the NICE guideline or the UK joint specialist societies guideline.
- The recommendation that the glass test should not be used solely for diagnosing bacterial meningitis and meningococcal septicaemia is pragmatic and what Prodigy considers good medical practice.

Assessing vital signs in a person with suspected bacterial meningitis or meningococcal disease

- **In all people with suspected bacterial meningitis or meningococcal disease perform a risk stratification and assess:**
  - Conscious level (for example using the Glasgow Coma Scale [GCS], or the Alert, Voice, Pain, Unresponsive [AVPU] scale).
  - Heart rate and blood pressure.
  - Respiratory rate, oxygen saturation (if a pulse oximeter is available).
  - Temperature.
  - Capillary refill time.
- **Features of shock include:**
  - Capillary refill time of more than 2 seconds, cold hands and feet.
  - Unusual skin colour.
  - Tachycardia and/or hypotension.
  - Respiratory symptoms or breathing difficulty.
  - Leg pain.
  - Toxic/moribund state.
  - Altered mental state/decreased conscious level.
  - Poor urine output.

What else might it be?

The diagnosis of acute bacterial meningitis can be challenging, as it can present with common, non-specific clinical features. Differential diagnoses include:

- Other, infective meningitis and meningoencephalitis, for example:
  - Viral meningitis — common and can occur at any age, although most commonly occurs in children. Management is generally conservative and most cases are self-limiting with a very good prognosis.
    - Enteroviruses (for example Coxsackie A and B viruses, and echoviruses) are the most common cause in children and adults.
    - Herpes simplex virus (HSV) — this is usually a complication of primary genital herpes, especially with HSV-2.
  - Fungal meningitis — life-threatening, but rare. Immunocompromised people are at increased risk. Presentation is often insidious with onset over weeks or months.
    - Cryptococcus is the most common cause of fungal meningitis in immunocompromised people.
  - Tuberculous meningitis — history of contact, or resident in endemic area. Signs and symptoms of pulmonary and extra-neural disease. Some features are indistinguishable from bacterial meningitis.
  - Drug-induced meningitis — history of responsible drug (such as nonsteroidal anti-inflammatory drug, trimethoprim/sulfamethoxazole, amoxicillin, ranitidine).
- Sepsis.
- Pneumonia.
- Encephalitis — suggested by abnormal cerebral function, such as altered behaviour and speech or motor function, particularly if associated with fever.
- Malignancy.
- Central nervous system abscess.
- HIV infection.
- Subarachnoid haemorrhage.
- Other non-infective causes of meningitis, for example:

- Autoimmune disorders such as systemic lupus erythematosus (can cause aseptic meningitis) and Behçet's syndrome.

Petechiae may have other causes, such as:

- Other infections — for example, enterovirus, Epstein-Barr virus, rubella, adenoviruses, and respiratory viruses.
- Non-infective causes — for example, clotting factor or platelet deficiencies. Petechiae in the eyes or on the face may also result from non-infective causes (such as sneezing, coughing, vomiting, and trauma).

Management:

Scenario: Non-blanching rash or meningococcal septicaemia

From birth onwards

How should I manage suspected bacterial meningitis with a non-blanching rash or meningococcal septicaemia?

- **Arrange emergency medical transfer to hospital.**
- **Administer a single dose of parenteral benzylpenicillin (intravenously or intramuscularly) at the earliest opportunity, but do not delay urgent transfer to hospital.**
    - If given intramuscularly, it should be given as proximally as possible — preferably into a part of the limb that is still warm (as cold areas will be less well perfused).
    - Dosage:
        - Children younger than 1 year of age — 300 mg.
        - Children 1–9 years of age — 600 mg.
        - Adults and children 10 years of age or older — 1200 mg.
    - Benzylpenicillin should be carried in GP emergency bags.
    - Paramedics also have the mandate to give benzylpenicillin for suspected meningococcal disease.
- **Withhold benzylpenicillin if the person has a clear history of penicillin anaphylaxis** (a history of a rash following penicillin is not a contraindication).

Alternative pre-hospital parenteral antibiotic in people with penicillin allergy

- The *UK joint specialist societies guideline on the diagnosis and management of acute meningitis and meningococcal sepsis in immunocompetent adults* recommend that if an antibiotic is given in primary care, it should be benzylpenicillin or a third-generation cephalosporin such as cefotaxime or ceftriaxone and that in the case of known anaphylaxis to penicillins or cephalosporins, antibiotics should not be given until the person is admitted to hospital.

Scenario: Without non-blanching rash

From birth onwards

How should I manage suspected bacterial meningitis without a non-blanching rash?

- **Arrange emergency medical transfer to hospital.**
- **Do not give parenteral antibiotic treatment unless urgent transfer to hospital is not possible** (for example in remote locations or owing to adverse weather conditions).
    - Choice of empiric antibiotic will depend on availability or local policy.
        - Options include benzylpenicillin, cefotaxime, or chloramphenicol.
- If the person has no specific clinical features of bacterial meningitis, but it is not possible to exclude the person being in the early stages of the disease, admit to secondary care for a full assessment.

Scenario: Managing close contacts

From birth onwards

How should I manage close contacts?

- Following confirmation of bacterial meningitis in secondary care, primary healthcare professionals may be involved in:
    - Identifying contacts who are at risk.
    - Reassuring people who may be in contact with the index case but do not require prophylactic treatment.
    - Prescribing or supplying prophylactic treatments for at-risk groups.
- Be aware that bacterial meningitis and meningococcal disease are notifiable diseases.
- Inform close contacts that the risk to a contact is low with meningococcal disease. The risk is highest in the first 7 days after a case is diagnosed and falls sharply thereafter. Prophylaxis or vaccination for close contacts is usually arranged by secondary care.
    - Prophylaxis against meningococcal disease should be considered for the following close contacts, regardless of meningococcal vaccination status:
        - People who have had prolonged close contact with the case in a household-type setting during the 7 days before onset of illness (for example, people who are living or sleeping in the same household, pupils in the same dormitory, boy/girlfriends, or university students sharing a kitchen in a hall of residence).
        - People who have had transient close contact with a case only if they have been directly exposed to large particle droplets/secretions from the respiratory tract of a case around the time of admission to hospital.
    - Antibiotic prophylaxis ciprofloxacin for one day or rifampcin for 2 days should be given as soon as possible (ideally within 24 hours) after the diagnosis of the index case but pneumococcal meningitis no need for antibiotic prophylaxis.

Scenario: Follow up after hospital discharge

- **Ensure that appropriate hospital follow up has been arranged** (people discharged from hospital who are recovering from bacterial meningitis or meningococcal disease require specialist follow up).
    - All children should have a review with a paediatrician 4–6 weeks after hospital discharge to assess their recovery.
- **Be alert for possible late-onset complications such as sensory, neurological, orthopaedic, and psychosocial effects of bacterial meningitis and meningococcal disease.**
- **Give advice on driving** if appropriate.

TOPIC: SCARLET FEVER

- **Scarlet fever (or 'scarlatina') is a notifiable infectious disease caused by the bacterium *Streptococcus pyogenes,* also known as group A streptococcus (GAS).**
    - It typically causes a sore throat, fever, and extensive red, sandpaper-like rash.
- **An outbreak of scarlet fever is defined as** 'a credible report of two or more probable or confirmed scarlet fever cases attending the same school or nursery or other childcare setting, notified within 10 days of each other (two maximum incubation periods), with an epidemiological link between cases, for example they are in the same class or year group'.

How is it transmitted?

- **Scarlet fever is caused by toxin-producing strains of *Streptococcus pyogenes*, also known as Group A streptococcus (GAS).**
    - GAS bacteria can colonise the throat or skin. Asymptomatic pharyngeal carriage of *S. pyogenes* occurs in 3–26% of healthy children.
    - The incubation period is usually 2–3 days but can range from 1–6 days.
    - When GAS cause infection, the primary site is usually the throat resulting in pharyngitis.
    - The rash and fever associated with scarlet fever are due to exotoxin or superantigen release by the bacteria.
- **Scarlet fever is highly contagious** and is transmitted when a person's mouth, throat, or nose comes into contact with infected saliva or mucus by aerosol transmission (by breathing infected airborne droplets produced through coughing or sneezing) or by direct contact with contaminated surfaces (including cups, utensils, taps, and handles).

What are the risk factors?

- **People who are at increased risk of invasive Group A Streptococcal infection (iGAS) and complications include people:**
    - At extremes of the age range, such as the very young and old, or postpartum women.
    - Who are immunocompromised or immunosuppressed.
    - With comorbidities such as skin breakdown, diabetes mellitus, or underlying malignancy.
    - With concurrent chickenpox or influenza.
    - Who inject drugs or are alcohol dependent.

What are the complications?

Serious complications of scarlet fever are generally rare.

- **Suppurative complications are due to local spread of the infection and tend to occur early in the course of infection,** and include:
    - Otitis media.
    - Throat infection and abscess (peritonsillar cellulitis, peritonsillar abscess, retropharyngeal abscess).
    - Acute sinusitis and mastoiditis.

- **Non-suppurative (immune-mediated) complications tend to occur later in the course of infection,** particularly in untreated people, and include:
    - Acute rheumatic fever — this is an immunologically-mediated response and can cause carditis and endocarditis (leading to valvular heart disease) and reactive arthritis.
    - Acute post-streptococcal glomerulonephritis — this typically occurs 2 weeks or more after the acute initial infection, and may present with haematuria, reduced urine output, peripheral oedema, proteinuria, and hypertension. Permanent kidney damage is rare.
- **Invasive Group A streptococcal (iGAS) infections may be life-threatening and include:**
    - Streptococcal pneumonia.
    - Meningitis and cerebral abscess.
    - Endocarditis, septic arthritis, and liver abscess.
    - Cellulitis, necrotizing fasciitis, and streptococcal toxic shock syndrome.
    - Sepsis.
    - Risk of death — severe iGAS infections carry a substantial risk of mortality (around 15–25%).

What is the prognosis?

Scarlet fever is usually a mild and self-limiting illness.

- In most cases, the clinical features of infection including rash resolve over about 1 week.
- Once a person has had scarlet fever, a recurrence of infection is unlikely, but they are still susceptible to other forms of streptococcal infection.

Diagnosis:

Suspect a diagnosis of scarlet fever if there are typical clinical features present.

- **Initial clinical features may be non-specific and include:**
    - Sore throat.
    - Fever (typically greater than 38.3°C).
    - Headache, fatigue, nausea, and vomiting.
- **A blanching rash usually develops on the trunk 12–48 hours after initial symptoms, before spreading to the rest of the body and flexures.**
    - The rash is characteristically red, generalized and pinpoint (punctate) with a rough, sandpaper-like texture.
    - It may be accentuated in the skin folds of the neck, axillae, groin, elbows, and knees with a deep red, linear appearance (so-called 'Pastia's lines').
    - The palms and soles are typically spared.
    - The skin may peel after the rash resolves, particularly at the tips of the fingers and toes, and less commonly on the trunk and limbs.
- **Examination may also reveal:**
    - Strawberry tongue — initially the tongue is covered with a white coat through which red papillae may be seen. Later, the white covering disappears, leaving the tongue with a beefy red appearance.
    - Associated cervical lymphadenopathy.
    - Flushed face, with marked circumoral pallor.
    - Pharyngitis and small red spots (petechiae) on the hard and soft palate ('Forchheimer spots').

When is laboratory testing needed to confirm scarlet fever?

Throat swabs and blood tests are not routinely indicated for the diagnosis of scarlet fever.

- **Consider taking a throat swab for culture of Group A streptococcus (GAS) (prior to starting treatment) if:**
  - There is uncertainty about the clinical diagnosis.
  - A case is suspected to be part of an outbreak — the local health protection team should advise primary care if a local outbreak is suspected and when testing is appropriate.
  - There is a true allergy to penicillin, to determine antimicrobial susceptibility, depending on clinical judgement.
  - A case is in regular contact with vulnerable people who are at high risk of complications, such as healthcare workers.
- **Measurement of serum anti-streptolysin O (ASO) antibody titres are not useful in acute infection,** but may be helpful in the diagnosis of postinfection complications, such as acute rheumatic fever or glomerulonephritis.

What else might it be?

Alternative diagnoses that may present similarly to scarlet fever include:

- Rubella
- Parvovirus B19 infection
- Measles.
- Roseola infantum (herpesvirus type 6)
- Kawasaki disease
- Staphylococcal toxic shock syndrome (TSS) — this is a potentially life-threatening condition, and symptoms may include fever, headache, skin rash, confusion, vomiting, or diarrhoea. Risk factors include tampon use, the postpartum period, staphylococcal skin or soft tissue infections, and comorbid influenza infection.
- Brucellosis, cytomegalovirus, toxoplasmosis, HIV, syphilis, and mononucleosis reaction to amoxicillin — these may all cause rash accompanied with lymphadenopathy.
- Other adverse drug reactions, echovirus, Coxsackievirus, and viral hepatitis — although these cause a rash they are not usually associated with lymphadenopathy.
- Tropical viruses, including alphaviruses and flaviviruses (for example Dengue fever), rare — these should be considered if the person has recently travelled to affected areas.

Management:

Scenario: Management of scarlet fever

From age 1 month onwards

How should I manage suspected or confirmed scarlet fever?

- **Arrange urgent hospital admission if a person has:**
  - Severe symptoms that cannot be managed in primary care.
  - A suspected serious or life-threatening complication.
  - A high risk of developing complications, depending on clinical judgement.

- **If the person does not need hospital admission, prescribe appropriate antibiotics promptly, regardless of the severity of illness.**
    - Prescribe a 10-day course of phenoxymethylpenicillin (penicillin V) first-line.
    - Second-line options for people with penicillin allergy are:
        - Birth to 6 months — clarithromycin for 10 days.
        - Non-pregnant adults and children aged 6 months to 17 years — azithromycin for 5 days, or clarithromycin for 10 days.
        - Pregnant or postpartum (within 28 days of childbirth) — erythromycin for 10 days.
- **Notify the local health protection team promptly within 3 days by completing a notification form if a diagnosis of scarlet fever is suspected.**
- **Advise the person or family/carers about appropriate self-care measures.**
    - Advise on the use of over-the-counter paracetamol or ibuprofen for symptom relief if there are no contraindications.
    - Encourage the person to rest and drink adequate fluids.
    - Advise to clean and cover any skin breaks including bites, cuts, grazes, or wounds.
- **Advise the person or family/carers on measures to reduce the risk of cross-infection:**
    - Exclusion from nursery, school, or work is needed for at least 24 hours after starting appropriate antibiotic treatment.
    - Effective and frequent handwashing is needed (for example before preparing and eating food, after using the toilet, after play, and after sneezing and disposing of tissues).
    - Avoid sharing eating utensils and towels, and dispose of tissues promptly.
    - Avoid contact with people at high risk of complications of scarlet fever.
- **Advise the person to arrange follow up if symptoms worsen or have not improved after 7 days.**
- **Consider seeking local health protection team advice if a person is a** high-risk **contact.**
    - The specialist team may advise whether exclusion from work, school, or nursery and/or antibiotic prophylaxis is necessary (for example in people with severe immunosuppression).
    - Advise that high-risk contacts should seek immediate medical advice if they develop any clinical features of Group A streptococcal (GAS) infection or its complications.
    - Routine testing of asymptomatic household contacts is not recommended.

TOPIC: SEPSIS

1. **T**ake blood tests and microbiology samples including:
    - Blood gas including glucose and lactate measurement — hypoglycaemia may result from depleted glycogen stores; hyperglycaemia may result from the stress response to sepsis; hyperlactataemia is a non-specific indicator of cellular or metabolic stress and is a marker of illness severity, with a higher level predictive of higher mortality rates.
    - Blood culture — ideally done before antibiotic administration, to identify a primary bacteraemia.
    - Full blood count — white cell count may be high or low; thrombocytopenia may indicate disseminated intravascular coagulation (DIC).
    - C-reactive protein (CRP) — may indicate infection and/or inflammation/ procalcitonin marker used to distinguish bacterial sepsis from other inflammatory conditions that cause similar symptoms.
    - Creatinine, urea and electrolytes — may indicate dehydration and/or acute kidney injury.
    - Liver function tests — increased bilirubin or alanine aminotransferase (ALT) levels may indicate cholestasis or other liver dysfunction.
    - Clotting screen — if abnormal may indicate coagulopathy/DIC.
    - Urine analysis and culture, chest X-ray, and additional investigations depending on the person's clinical presentation — this may allow identification of the source of infection, pathogen(s) and sensitivities, and subsequent tailoring and/or de-escalation of antibiotic therapy if appropriate. Source control to eliminate a focus of infection may be possible, such as abscess drainage, debridement of infected tissue, removal of infected devices or foreign bodies, or surgery.

TOPIC: LYME DISEASE

- Lyme disease, or Lyme borreliosis, is an infection caused by a group of spiral-shaped bacteria (spirochetes) which are transmitted to humans following a bite from an infected tick.

What causes it?

- Lyme disease is caused by a group of bacteria termed *Borrelia burgdorferi*s sensu lato (sl).

What are the risk factors?

- Risk factors for Lyme disease include:
  - Occupational and recreational exposure to woodland, and fields — particularly in areas with a higher incidence of infection.
  - Increased duration of tick attachment.
    - Transmission of *Borrelia burgdorferi* from infected nymphal ticks generally occurs after 36–48 hours of attachment.
    - Transmission from adult ticks occurs after 48 hours or more.

What is the prognosis?

- **Most people recover completely with appropriate treatment**.
- **Antibiotic treatment in people with early Lyme disease is highly effective** and reduces the risk of further symptoms developing, and increases the chance of complete recovery**.**

What are the complications?

- **Long-term complications are uncommon in people who are treated appropriately.** However, even after appropriate treatment it may take several months for some people to recover fully and some may have residual symptoms, for example:
  - **Severe neurological symptoms** — incomplete recovery from severe neurological symptoms may occur if treatment has been started late in the disease.
    - People may develop long-lasting neurological conditions such as chronic meningitis, encephalomyelitis, radiculomyelitis or peripheral neuropathy.
    - Most people with meningoradiculoneuritis or facial palsy recover fully over time but a small number may have residual paraesthesias or facial paresis.
  - **Acrodermatitis chronica atrophicans** — in people who have severe tissue damage following acrodermatitis chronica atrophicans, atrophic lesions, peripheral neuropathy and joint deformities may remain.
  - **Lyme arthritis** — most people recover completely over a period of months, but about 10% of people with Lyme arthritis have a more prolonged recovery.
  - **Persisting non-specific symptoms**, such as fatigue, aches and pains and cognitive difficulties occur in a minority of people. This is sometimes called post-treatment Lyme disease syndrome or chronic Lyme disease. It is difficult to ascertain how common this is as these are common symptoms in the general population.

Diagnosis:

When should I suspect a diagnosis of Lyme disease?

- **Make a clinical diagnosis of Lyme disease** in people with erythema migrans. No laboratory testing is required.
- **For people *without* erythema migrans, clinical presentation and laboratory testing is used to guide diagnosis and treatment.**
- **Consider** the possibility of Lyme disease in people presenting with several of the following symptoms, because Lyme disease is a possible (but uncommon) cause of:
    - Cognitive impairment, such as memory problems and difficulty concentrating.
    - Fatigue.
    - Fever and sweats.
    - Headache.
    - Malaise.
    - Migratory joint or muscle aches and pain.
    - Neck pain or stiffness.
    - Paraesthesia.
    - Swollen glands.
- **Consider** the possibility of Lyme disease in people presenting with symptoms and signs relating to one or more organ systems (focal symptoms), because Lyme disease is a possible (but uncommon) cause of:
    - Eye symptoms — such as uveitis or keratitis.
    - Inflammatory arthritis affecting one or more joints that may be fluctuating and migratory.
    - Cardiac problems — such as heart block or pericarditis.
    - Neurological symptoms — such as facial palsy or other unexplained cranial nerve palsies, meningitis, mononeuritis multiplex or other unexplained radiculopathy; or rarely encephalitis, neuropsychiatric presentations or unexplained white matter changes on brain imaging.
    - Skin rashes — such as acrodermatitis chronica atrophicans or lymphocytoma.
        - Borrelial lymphocytoma is a rare manifestation, mainly seen in children. It presents as a painless blue-red nodule, often on the ear lobe, nipple or scrotum. It may present with, before, or after erythema migrans.
        - Acrodermatitis chronica atrophicans is a late manifestation, presenting months to years after untreated infection, usually in distal extremities, and starting as a slowly progressive red or blueish lesion which can eventually become atrophic and eventually be associated with joint and bone deformity and polyneuropathy.
- If symptoms suggest the possibility of Lyme disease, ask:
    - How long the person has had symptoms.
    - About their history of possible tick exposure — for example, activities that might have exposed them to ticks, such as travel to areas where Lyme disease is known to be highly prevalent.
- **Do not:**
    - Rule out the possibility of Lyme disease in people with symptoms but no clear history of tick exposure.
    - Diagnose Lyme disease in people without symptoms, even if they have had a tick bite.

Erythema migrans

- **Primary (or solitary) erythema migrans** has the following features:
  - **Appearance**
    - A spreading erythema with a well-defined edge.
    - Usually round or oval in shape, but may be triangular or linear.
    - Red, purple or bluish-red in colour.
    - Usually flat, but may be palpable, and there may be a central punctum, spot, vesicle or pustule is present at the centre.
    - As the rash expands, there may be clearing behind the leading edge, giving the classical annular or bulls-eye appearance.
  - **Size**
    - Usually expands over days to weeks.
    - Diameter is usually larger than 5 cm (typically around 15 cm), and can be as large as 70 cm.
  - **Location**
    - At the site of a tick bite.
    - Often on the legs, at flexor creases (knees, axillae, and groins), around the waistband, under the breasts, near to straps (which impede the forward progress of ticks), or (particularly in children) at the hairline or the upper parts of the body.
  - **Timing**
    - It usually becomes visible from 1–2 weeks (but can appear from 1 day to 36 days) after a tick bite and lasts for several weeks untreated.
  - **Associated symptoms**
    - The rash is not usually itchy hot, or painful.
    - In Europe, around a third of people with erythema migrans experience flu-like symptoms, including fever, headache, tiredness, nausea, vomiting, arthralgia
    - (joint pain), and myalgia (muscle pain). Flu-like symptoms are more common in the US.
    - Rarely, neurological features may also be present, but these usually occur later in the disease.
  - **Be aware that a rash which is not erythema migrans can develop as a reaction to a tick bite, but it:**
    - Usually develops and recedes within 48 hours from the time of the tick bite.
    - Is more likely than erythema migrans to be hot, itchy or painful.
    - May be caused by an inflammatory reaction or infection with a common skin pathogen.
- **Secondary (or disseminated) erythema migrans** may occur following haematogenous dissemination of infection, resulting in multiple lesions, usually smaller than 5 cm in diameter.
- **Erythema migrans may be confused with a number of other skin conditions**.

What else might it be?

- **Erythema migrans may be confused with:**
  - **Tick bite hypersensitivity reaction.**
    - An erythematous skin lesion presenting while the tick is still attached or which has developed within 48 hours of detachment is most likely to be a tick bite hypersensitivity reaction rather than erythema migrans.
    - Tick bite hypersensitivity reactions are usually less than 5 cm in diameter, and itchy; they sometimes have an urticarial appearance and typically begin to disappear within 24–48 hours.

- **Hypersensitivity to an insect bite** — develops rapidly, is often itchy, and is usually smaller than erythema migrans.
- **Other skin conditions including**:
  - **Cellulitis** — usually occurs on a limb or at site of skin trauma. It presents as uniform erythema, which is warm and painful, and has a faster onset than erythema migrans. It may be present at the same time as erythema migrans.
  - **Erythema multiforme** — appears as multiple target lesions (sometimes blisters). The lesions are small (less than 2 cm in diameter), diffuse, and symmetrical, with slow enlargement; palmar and mucosal involvement is common and there is often an obvious precipitant.
  - **Granuloma annulare** — is usually smaller than erythema migrans. The periphery is usually papular, and it can have central clearing.
  - **Nummular eczema** — lesions are usually smaller and less erythematous than erythema migrans. Does not enlarge rapidly, pruritic, well-demarcated, skin may be thickened or weepy.
  - **Tinea (ringworm**) — these lesions are characterized by scale and itch.
  - **Urticaria** — multiple raised lesions are nettle-like or blotchy, and very itchy. It usually resolves within 24–36 hours.

- **Neurological symptoms and syndromes** (can be a feature of early or late disseminated Lyme disease.)
  - Early neurological disease — other causes of nerve palsies, radiculopathy, meningoencephalitis or myelitis.
  - There are a wide range of peripheral and central presentations of late neurological Lyme disease. Assessment by an appropriate specialist is recommended to ensure conditions such as malignancy, multiple sclerosis and motor neurone disease are not missed.
- **Persisting non-specific systemic symptoms**. The UK Health Security Agency (UKHSA) website points out the differential diagnosis for this is extremely wide, and depending on predominant symptoms and presentation, some of the many possibilities to consider include:
  - Cytomegalovirus (CMV).
  - Epstein-Barr virus (EBV).
  - Hepatitis B or C.
  - HIV.
  - Syphilis.
  - Toxoplasmosis.
  - Other rare (and acquired abroad) infections which can be tick-borne such as anaplasmosis, rickettsioses, Q fever.
  - Tick-borne encephalitis.
  - Auto-immune conditions including rheumatoid arthritis.
  - Malignancy.
  - Psychiatric disorders.
  - Fibromyalgia.
  - Chronic Fatigue Syndrome.

Scenario: Management of Lyme disease

From birth onwards

- **Discuss the diagnosis and management of Lyme disease in children and young people aged under 18 years with a specialist, unless they have a single erythema migrans lesion and no other symptoms.**

- **For all other people with erythema migrans who have no evidence of focal symptoms (for example, neurological, cardiac, or joint involvement), start treatment with oral antibiotics.**
- **If Lyme disease is suspected in people without erythema migrans**, offer an enzyme-linked immunosorbent assay (ELISA) test for Lyme disease.
    - If there is a high clinical suspicion of Lyme disease, consider starting treatment with antibiotics while waiting for the results.
- **If the ELISA test (in people without erythema migrans) is:**
    - **Positive or equivocal** — perform an immunoblot test for Lyme disease.
        - Consider starting treatment with antibiotics while waiting for the results if there is a high clinical suspicion of Lyme disease.
    - **Negative** and the person still has symptoms — review their history and symptoms, and consider an alternative diagnosis.
        - If Lyme disease is still suspected in people tested within 4 weeks from symptom onset, repeat the ELISA 4–6 weeks after the first ELISA test.
        - If Lyme disease is still suspected in people who have had symptoms for 12 weeks or more, perform an immunoblot test.
- **If the immunoblot test is:**
    - **Positive** — diagnose Lyme disease. Start treatment with oral antibiotics.
    - **Negative,** but symptoms persist — consider a discussion with or referral to a specialist (regardless of the ELISA result) to review whether further tests are required or to consider alternative diagnoses.
    - **Negative**, and symptoms have resolved — no treatment is required.
- **Be cautious** about diagnosing Lyme disease in people without a supportive history or positive serological testing because of the risk of:
    - Missing an alternative diagnosis.
    - Providing inappropriate treatment.
- **Refer people with focal symptoms to the appropriate specialist, or consider a discussion with a specialist, but do not delay treatment.**

What should I prescribe for people with Lyme disease?

- **For people diagnosed with Lyme disease (or with a high suspicion of Lyme disease awaiting test results), prescribe oral antibiotics:**
    - **Adults and children aged 12 years or older:**
        - Doxycycline 100 mg twice daily (or 200 mg once daily) for 21 days.
        - Amoxicillin 1000 mg three times daily for 21 days (if doxycycline is not suitable).
        - Azithromycin 500 mg daily for 17 days (if doxycycline and amoxicillin are not suitable).
    - **Children aged 9–12 years:**
        - Doxycycline — 5 mg/kg in two divided doses on day one, followed by 2.5 mg/kg daily in one or two divided doses for a total of 21 days, for children under 45 kg. For severe infections, up to 5 mg/kg daily for 21 days.
        - Amoxicillin — 30 mg/kg three times daily for 21 days, for children 33 kg and under (if doxycycline is not suitable).
        - Azithromycin — 10 mg/kg daily for 17 days, for children 50 kg and under (if doxycycline and amoxicillin are not suitable).
    - **Children aged under 9 years:**
        - Amoxicillin — 30 mg/kg three times daily for 21 days, for children 33 kg and under. Children over 33 kg, 1000 mg three times daily for 21 days.
        - Azithromycin — 10 mg/kg daily for 17 days, for children 50 kg and under (if amoxicillin is unsuitable).

- Manage symptoms (for example, headache, muscle pain) as required.

How should I manage people with ongoing symptoms of Lyme disease?

- If treatment for Lyme disease has:
    - **Started** and symptoms worsen despite treatment for Lyme disease — assess for an allergic reaction to the antibiotic.
        - A Jarisch-Herxheimer reaction may cause an exacerbation of symptoms but does not usually warrant stopping treatment.
    - **Finished** and symptoms have not improved, or have worsened — review the person's history and symptoms to explore:
        - Possible alternative causes of the symptoms.
        - If re-infection may have occurred — if the person's history suggests reinfection, offer antibiotic treatment.
        - If treatment may have failed.
        - Details of any previous treatment, including whether the course of antibiotics was completed without interruption.
        - If symptoms may be related to organ damage caused by Lyme disease, for example, nerve palsy.
- Consider a second course of antibiotics for people with ongoing symptoms if treatment may have failed — use an alternative antibiotic to the initial course.
- **If symptoms persist despite two completed courses of antibiotics:**
    - Do not routinely offer further antibiotics.
    - Consider discussion with a national reference laboratory, or discussion, or referral to a specialist.
- **Explain to people with ongoing symptoms that:**
    - Continuing symptoms may not mean they still have an active infection.
    - Symptoms of Lyme disease may take months or years to resolve even after treatment.
    - Some symptoms may be a consequence of permanent damage from infection.
    - There is no test to assess for active infection and an alternative diagnosis may explain their symptoms.
- **Longer term ongoing symptoms** - be alert to the possibility of symptoms related to Lyme disease that may need assessment, management and support, such as:
    - Chronic pain.
    - Depression and anxiety.
    - Fatigue.
    - Sleep disturbance.

What information and advice should I give people with Lyme disease?

- Explain how to minimise the risk of being bitten by a tick.
- Explain to people diagnosed with Lyme disease that:
    - It is a bacterial infection treated with antibiotics.
    - Most people recover completely.
    - Prompt antibiotic treatment reduces the risk of further symptoms developing and increases the chance of complete recovery.
    - It may take time to get better, but their symptoms should continue to improve in the months after antibiotic treatment.
    - They may need additional treatment for symptom relief.
- Tell people who are starting antibiotics for Lyme disease that some people may have a Jarisch-Herxheimer reaction to treatment. Explain that:

    - This causes a worsening of symptoms early in treatment.
    - It can happen when large numbers of bacteria in the body are killed.
    - It does not happen to everyone treated for Lyme disease.
    - They should contact their doctor and keep taking their antibiotics if their symptoms worsen.
- Advise people with Lyme disease to talk to their doctor if their symptoms have not improved or if symptoms return after completing treatment.
- Explain to people with Lyme disease that infection does not give them lifelong immunity and that it is possible for them to be re-infected and develop Lyme disease again.

What advice should I give someone to help prevent tick bites?

- **To minimize the risk of being bitten by an infected tick, give people advice about:**
    - Where ticks are commonly found (such as grassy and wooded areas, including urban gardens and parks).
        - Keep to paths and away from long grass or overgrown vegetation, as ticks crawl up long grass in their search for a feed.
        - Regularly check clothing and exposed skin for ticks and brush off immediately.
    - Covering exposed skin and using insect repellents that protect against ticks.
        - Consider using an insect repellent containing N, N-diethyl-m-toluamide (DEET).
        - Wear appropriate clothing (long sleeved shirt and long trousers tucked into socks) in tick-infested areas. Light coloured fabrics are useful, as it is easier to see ticks against a light background.
    - How to check themselves and their children for ticks on the skin:
        - Inspect skin frequently and remove any attached ticks as soon as possible after noticing them.
        - At the end of the day, check again for ticks, especially in skin folds.
        - Make sure that children's head and neck areas, including scalps, are properly checked.
        - Check that ticks are not brought home on clothes.
        - Check that pets do not bring ticks into the home on their fur.
    - The importance of prompt, correct tick removal and how to do this.

What advice should I give to a person bitten by a tick?

- **If a tick is still attached to the skin, remove it as soon as possible** using the following method:
    - Grasp the tick as close to the point of attachment to the skin as possible using fine-tipped tweezers or a tick removal device.
    - Pull upwards slowly and firmly, away from the skin, without twisting — there is a possibility that fragments of the mouthparts may be left in the skin, which can cause a local infection.
    - After removal:
        - Clean the skin with water and antibacterial soap, and wash hands.
        - Keep an eye on the bite area for several weeks for changes.
- **If the person has a history of a recent tick bite but is otherwise well:**
    - Antibiotic prophylaxis following a tick bite is *not* routinely recommended.
    - Advise the person to seek immediate medical advice if they develop any symptoms of Lyme disease or other tick-borne diseases (such as a rash or flu-like illness).

TOPIC: MALARIA

- Malaria is a life-threatening illness caused by infection of red blood cells by plasmodium parasites.
- Several species of Plasmodium (*P. vivax*, *P. falciparum*, *P. ovale*, and *P. malariae*) cause malaria in humans. Human infection with *P. knowlesi*, a malaria parasite of monkeys in South-East-Asia, can also occur.
- Transmission of malaria to humans occurs through the bite of infected female *Anopheles* mosquitoes.
- Malaria can also occur (rarely) in the UK as a result of:
    - "Airport malaria" or "baggage malaria" — malaria imported to non-endemic areas by infected *Anopheles* mosquitoes hiding in planes or in baggage.
    - Person-to-person transmission during:
        - Blood transfusion or implantation of infected tissues.
        - Pregnancy.
        - Injecting drug use.

- **What is the prognosis?**

- If malaria is identified promptly, appropriate treatment is given, and no organ dysfunction has occurred, most people make a rapid and complete recovery. If malaria treatment is delayed or inappropriate, severe or fatal malaria can develop.
    - Prognostic indicators of poor outcome include neurological dysfunction (coma or seizures), acidosis, hypoglycaemia, and respiratory distress.

What are the complications?

- **Complications of malaria include:**
    - Acute kidney injury — a common complication in the short-term due to hypovolaemia and dehydration. Renal tubular necrosis can also occur in severe disease.
    - Cerebral malaria — severe malaria due to *P. falciparum* with coma, or malaria with coma persisting for more than 30 minutes after a seizure.
    - Acute respiratory distress syndrome.
    - Spontaneous bleeding and coagulopathy.
    - Sepsis.
    - Severe anaemia — more likely in young children due to direct red cell lysis, autoimmune haemolysis, and disturbed bone marrow function.
    - Hypoglycaemia — more likely in pregnant women and may be worsened by quinine.
    - Metabolic acidosis — secondary to tissue hypoxia from hypovolaemia, hypotension, and anaemia.
    - Nephrotic syndrome.
    - Jaundice.
    - Splenic rupture — seen in *P. vivax* infection.

- **In pregnancy:**
    - Severe malaria is more likely, particularly in the second and third trimesters, and is often associated with pulmonary oedema and hypoglycaemia.
    - Cerebral malaria has a mortality rate of 50% (higher than non-pregnant adults).
    - Intrauterine growth retardation, premature delivery, and fetal death (stillbirth) are possible complications in pregnant women infected with malaria.

The Glasgow Coma Scale

The Blantyre Coma Scale

- The Blantyre Coma Scale (BCS) is a modification of the GCS for use with children who are too young to talk.
- A score of 2 or less indicates 'unrousable coma'. The maximum score is 5.

Table 2 The BCS for preverbal children.

Type of response	Response	Score
Best eye movement	Undirected — fails to watch or follow	0
	Directed — watches or follows	1
Best verbal response	No vocal response to painful stimulus	0
	Moans or abnormal cry	1
	Appropriate cry	2
Best motor response	No response or inappropriate response	0
	Withdraws limb from painful stimulus	1
	Localizes painful stimulus	2

Diagnosis:

When should I suspect malaria?

- Suspect malaria in anyone who has returned from or previously visited an area endemic for malaria, who is unwell, or has a fever or history of fever, regardless of malaria chemoprophylaxis
    - Almost all cases of *P. falciparum* malaria present within 6 months of exposure and most within 2–3 months.

    - Infection with other malaria species, such as *P. ovale* and *P. vivax* may present months or years after exposure due to reactivation of hypnozoites (a dormant liver stage).
    - Presentation may be delayed in people who have taken chemoprophylaxis.
    - Most missed cases of malaria are wrongly diagnosed as non-specific viral infections, influenza, gastroenteritis, and hepatitis.
- **Clinical features of malaria include:**
    - Fever (often 39°C or higher), sweats, and/or chills — absence of fever should *not* remove the suspicion of malaria.
    - Headache.
    - General malaise, lethargy, and fatigue — somnolence is more common in children than in adults.
    - Anorexia, gastrointestinal disturbance (such as nausea, abdominal pain, vomiting, diarrhoea), and jaundice.
    - Poor feeding in children.
    - Myalgia and arthralgia.
    - Sore throat, cough, lower respiratory tract symptoms, and respiratory distress.
    - Confusion.
    - Hepatomegaly, splenomegaly, and somnolence are more common on examination in children.
- **Features of severe or complicated malaria in adults include:**
    - Cerebral malaria — impaired conscious level (glascow coma score less than 11) or seizures.
    - Renal impairment (may present with oliguria).
    - Acidosis (may present with acidotic breathing).
    - Hypoglycaemia (< 2.2 mmol/L) — common in pregnant women.
    - Respiratory distress which may be due to pulmonary oedema or acute respiratory distress syndrome (ARDS) — common in pregnant women.
    - Severe anaemia (may present with pallor).
    - Spontaneous bleeding/disseminated intravascular coagulation.
    - Shock (BP < 90/60 mmHg).
    - Sepsis — more common in pregnant women.
    - Haemoglobinuria — *P. falciparum* can cause severe haemolysis with dark red urine ('blackwater fever').
    - Parasitaemia > 10%.
- **Features of severe or complicated malaria in children include:**
    - Cerebral malaria — impaired conscious level (Glasgow coma score less than 11 or Blantyre coma score less than 3), seizures, altered respiration, and posturing (decorticate or decerebrate).
    - Severe anaemia (may present with pallor).
    - Respiratory distress or acidosis (may present with acidotic breathing).
    - Hypoglycaemia (< 2.2 mmol/L).
    - Prostration (inability to stand or sit).
    - Parasitaemia > 2% red blood cells parasitized.

How should I assess a person with suspected malaria?

- **Take a history asking about:**
    - Symptoms of malaria (such as fever, sweats, chills, malaise, myalgia, headache, vomiting, diarrhoea, and cough) and timing of onset.
    - Travel history including:
        - Country and area of travel.

        - Stopovers and other countries transited through.
        - Date of return.
        - Type of travel and activities while abroad — people returning from visiting friends and family in endemic areas are more at risk of malaria than tourists.
    - Possible differential diagnosis:
        - Immediately isolate anyone suspected of having a viral haemorrhagic fever (VHF) such as Ebola, Lassa fever, or Marburg in a side room.
    - Preventative measures such as:
        - Malaria chemoprophylaxis (drug, dose, adherence, and cessation) — full adherence to appropriate prophylaxis does not guarantee protection against malaria.
        - Precautions taken against biting insects for example insecticide-impregnated bednets and repellent.
        - Travel immunizations against other travel related infections such as yellow fever.
    - Risk of severe malaria:
        - Severe malaria is more likely in children, pregnant women, older people, and immunocompromised people (for example those with splenectomy or HIV/AIDS).
- **Examine the person, looking for:**
    - Signs of severe malaria such as impaired consciousness, confusion, hypotension, respiratory distress, or jaundice.
        - Check vital signs such as blood pressure, pulse, temperature (fever may not be present), respiration rate, and oxygen saturation.
        - The Glascow and Blantyre come score can be used to assess level of consciousness.
    - Pallor — suggestive of anaemia.
    - Hepatomegaly or splenomegaly (more common in children).
- **Arrange appropriate investigations:**
    - Malaria is a medical emergency and if suspected, a blood test to confirm the diagnosis must be carried out without delay.
    - Diagnosis of malaria is only possible with microscopy of thick and thin blood films (the gold standard) or an antigen detection test.
        - If the person is well and local facilities can ensure return of a result to the requesting doctor within 2 hours, testing may be carried out from primary care after discussion with an Infectious Disease specialist.
        - EDTA should be used for blood samples.
        - If the first blood films are negative, further blood testing must be arranged 12–24 hours later and again after a further 24 hours to rule out infection.
        - In pregnancy, thick films can be negative, despite the presence of parasites in the placenta. Expert advice should always be sought if malaria is suspected.

Blood films are essential to confirm a diagnosis of malaria

Differential diagnosis

- Malaria is difficult to diagnose as it may mimic other illnesses encountered in the UK, including:
    - Meningitis or encephalitis.
    - Lower respiratory tract infection.
    - COVID-19.

    - Influenza and other viral infections such as Epstein-Barr virus or cytomegalovirus.
    - Gastroenteritis.
    - Urinary tract infection.
    - Lymphoma.
    - Sepsis.
    - Viral hepatitis.
    - HIV seroconversion.
    - Legionella.
    - Leptospirosis.
- Malaria may present with similar symptoms to other travel-related infections, including:
    - Viral haemorrhagic fevers such as Lassa fever, Crimean-Congo haemorrhagic fever, Marburg, and Ebola — patients suspected of having a viral haemorrhagic fever require strict isolation.
    - Enteric fevers such as typhoid or paratyphoid.
    - Arboviruses such as Dengue, West Nile virus, and Japanese encephalitis.
    - Rickettsial infection such as scrub typhus and relapsing fever.
    - Trypanosomiasis.
    - Rabies.

Management:

From birth onwards

Management of suspected malaria in primary care

Malaria is a medical emergency

- **Advise people who have been diagnosed with malaria:**
    - To warn other members of their family or group they travelled with about the risk of malaria and advise them to seek immediate medical help if symptoms develop.
    - That an acute episode of malaria will not protect them from future attacks — they should seek pre-travel advice about avoidance of mosquito bites. They should also be assessed for the need for malaria chemoprophylaxis for future trips to endemic areas.
    - That they will be notified to public health authorities as part of routine surveillance.
    - That relapses of malaria can occur (for example from failure of treatment due to drug resistance or poor absorption) and they should report any recurrence of symptoms such as fever.
    - That they will be excluded from blood donation for some time following malaria infection — blood donors should inform and discuss their specific situation with the National Blood Service.

What drugs are used to treat malaria?

Treatment of malaria depends on a variety of factors including species of Plasmodium parasite, severity of infection, tolerability of specific drugs, and patterns of drug resistance.

- Antimalarial drugs include:
  - **Artesunate**
    - Parenteral artesunate is used to treat severe or complicated malaria.
    - Intravenous artesunate can cause haemolysis and follow-up blood tests are required.
  - **Quinine**
    - Quinine may be used initially to treat severe or complicated malaria if artesunate is unavailable.
  - **Artemisinin combination therapy (ACT)**
    - ACT may be used to treat uncomplicated malaria and is the preferred treatment for mixed infection.
  - **Atovaquone-proguanil**
    - Atovaquone-proguanil may be used to treat uncomplicated falciparum malaria if ACT is unavailable.
  - **Quinine plus doxycycline**
    - A combination of quinine plus doxycycline may be used to treat uncomplicated falciparum malaria if ACT is unavailable.
    - Doxycycline should not be given to children younger than 12 years of age.
  - **Chloroquine**
    - Chloroquine may be used to treat uncomplicated *P. malariae*, *P. ovale*, *P. knowlesi,* and most cases of *P. vivax* malaria but use depends upon patterns of resistance and tolerance.
  - **Primaquine**
    - Primaquine is the only currently effective drug for the eradication of hypnozoites (dormant parasites which persist in the liver after treatment of *P. vivax* and *P. ovale*).
    - Screening for G6PD deficiency is essential before treatment with primaquine is started as it can cause haemolysis in G6PD deficient individuals, which can be fatal. It is contraindicated in pregnancy and breastfeeding.
- Anti-malarial medications are associated with neurological and psychiatric symptoms.
- Artesunate-type drugs are more efficacious and better tolerated in pregnancy than quinines.
- Treatment failures are rare but have been noted and clinicians should be aware of the possibility.

TOPIC: MALARIA PROPHYLAXIS

Assessment:

How should I assess a person intending to travel to a malaria endemic area?

Take a history, and ask about:

- The travel itinerary including:
  - Timing — date of travel and duration of trip.
  - Countries to be visited including specific regions (rural or urban), transfers and stopovers.
  - Reason for travel such as visiting friends and relatives, tourism, backpacking, or work.
  - Type of accommodation such as hotel, hostel, local home, or tent.
  - Access to appropriate medical care.
- Prior travel experience and health beliefs.
  - Some people may mistakenly believe they are immune to malaria.
- Past medical history including conditions which may increase risk of severe malaria or affect choice of chemoprophylaxis:
  - Pregnancy or breastfeeding.
    - Pregnant women have an increased risk of developing severe or fatal malaria compared to non-pregnant women.
  - Immunosuppression (such as HIV infection or dysfunctional/absent spleen):
    - HIV progression may be hastened by acute malaria and the risk of developing severe malaria is increased in HIV infection.
    - People with a dysfunctional or absent spleen are at increased risk of severe malaria.
  - Psychiatric conditions such as depression.
    - A personal or family history of a severe mental health disorder may affect choice of chemoprophylaxis.
  - Epilepsy and other seizure disorders.
    - Certain antimalarial drugs are contraindicated in people with seizure disorders.
  - Inherited haemoglobinopathies such as sickle cell and thalassemia.
    - Malaria causes further haemolysis which significantly increases morbidity and mortality.
  - Psoriasis.
    - Psoriasis may affect choice of chemoprophylaxis as some medications can exacerbate the condition.
  - Liver or renal impairment.
    - May affect choice and dosage of chemoprophylaxis.
- Drug and vaccine history, including:
  - All current medications (including those prescribed by secondary care).
    - Many drugs can potentially interact with antimalarial medication for example antiarrhythmics (such as amiodarone), anticonvulsants, anticoagulants, antiretrovirals, and bupropion.
  - Previous malaria chemoprophylaxis taken and any associated problems.
  - Immunizations — routine and travel related.

- Allergies.

Be aware that malaria prevention is only one aspect of pre-travel advice.

Management:

From age 16 years onwards

How should I manage a person intending to travel to a malaria endemic area?

For people intending to travel to a malaria-endemic area:

- **Advise avoidance of travel to malarious areas if the person is at high risk of severe or fatal malaria** (for example pregnant women, children, the elderly [aged over 70 years] and those with complex comorbidities or immunosuppression).
    - If travel is unavoidable, explain the risks malaria presents, and the risks and benefits of antimalarial prophylaxis.
    - Emphasise the importance of rigorous personal protective measures and adherence to chemoprophylaxis. Emergency medical attention should be sought immediately if malaria infection is suspected.
- **Recommend** personal protective measures to prevent mosquito (and other vector) bites.
- **Consider the need for** chemoprophylaxis based on the risk assessment.
    - If the person has HIV, discuss options for chemoprophylaxis with their HIV physician who will make the decision on choice of agent.
    - Ensure that parents/carers understand the importance of ensuring that children take chemoprophylaxis properly and complete the full course — maximum recommended doses must not be exceeded as antimalarials can be particularly toxic to children.
- **Give advice and provide written information about:**
    - **Risk factors** for acquiring malaria such as:
        - Exposure to infective bites — risk is increased by being outside between dusk and dawn (when Anopheles mosquitos are likely to bite), longer stays, and unscreened accommodation.
        - Misconceptions about immunity — immunity acquired by people born in malarious countries is rapidly lost after migration to the UK. Having had malaria in the past does not protect from future attacks — effective personal protective measures and chemoprophylaxis are needed for all visits to endemic areas.
    - **The clinical features** of malaria and the importance of prompt diagnosis and early treatment.
        - Malaria can progress from mild to life-threatening disease within hours — if clinical features of malaria develop while travelling or up to one year after return, early diagnosis and treatment is essential.
        - Children and pregnant women with malaria can deteriorate very rapidly and become critically ill — ensure that all involved in the care of children returning from malarious areas (such as family/carers, school, and healthcare contacts) are aware that urgent medical attention (including a blood test for malaria) is required if the child becomes unwell within a year of leaving the malarious area.

What personal protective measures should I recommend?

Advise the person that the first line of defence against malaria infection (and other vector-borne diseases) is effective bite prevention. Advise people on the use of:

- **Physical barriers**
    - Full-length, loose-fitting, light-coloured clothing with long sleeves, long trousers and socks if out of doors after sunset.
        - Clothing can be treated with insecticide (such as permethrin) or cotton clothing sprayed with DEET to reduce biting through clothing (duration of action will be short as DEET is volatile).
    - Screening on doors, windows, eaves and other gaps in the building.
    - Bed nets treated with insecticide.
        - Insecticide-treated mosquito nets should be used if the person is sleeping outdoors or in unscreened accommodation. Insecticide helps prevent mosquitos biting through the net if the body touches the mesh.
        - Long-lasting pyrethroid insecticide-impregnated nets have an expected useful life of at least 3 years.
        - Standard nets need to be re-impregnated every 6 to 12 months (depending on how frequently the net is washed) to remain effective.
        - Bed nets must be free of tears and tucked in under the mattress.
- **Chemical barriers**
    - Advise the person to use a 50% DEET (N, N-Diethyl-meta-toluamide) based insect repellent first line on exposed areas of skin.
        - This is suitable for all people aged over 2 months.
        - DEET can damage some types of plastic (such as watch straps, watch covers, and plastic jewellery) – the person should not allow these items to come into contact with DEET.
        - If both sunscreen and DEET are required, DEET should be applied after the sunscreen. As DEET may induce a reduction in SPF — 30 to 50 SPF sunscreen should be applied to compensate for this.
    - If DEET is not tolerated, or preferred, advise the person to use the highest strength available of either:
        - Icaridin — provides repellent properties similar to DEET 20% with a comparable duration if it is also used at 20%.
        - Eucalyptus citriodora oil, hydrated, cyclized — provides protection for several hours.
        - 3-ethlyaminopropionate — this has a shorter duration of protection than DEET, but is still an effective repellent.
- **Avoidance measures** — advise people to:
    - Plan activities at times when there is reduced risk (for example, during the daytime).
        - Dusk and dawn should be avoided.
    - Visit areas where transmission is less likely (for example, urban centres and highland areas).
- **Coils/electric vapourisers**
    - If electricity is available, an electric pyrethroid vapouriser can be used during the night.
    - Mosquito coils containing insecticide can be burnt as an alternative to repel and kill mosquitoes but this is not recommended for indoor use.
- **Insecticide sprays**
    - The room should be sprayed before dusk with a knockdown insecticide, usually a pyrethroid (such as permethrin) to kill any mosquitoes which may have entered the accommodation during the day.

What chemoprophylaxis should I prescribe?

- **If chemoprophylaxis is required (based on individual risk assessment):**
  - Explain that a combination of mosquito avoidance and chemoprophylaxis will give significant protection against malaria, but no regime is 100% effective.
  - Identify chemoprophylactic agents appropriate for the destination and person by
  - Discuss the advantages and disadvantages of appropriate chemoprophylaxis regimens with the person.
  - Explain the dosing schedule (including the need to take the drug before, during and for some time after visiting a malarious area), adverse effects and any precautions with specific drugs (for example, avoidance of sun exposure with doxycycline).
  - Explain the risks of discontinuation of anti-malarial drugs – fatal malaria has occurred in people who have stopped all chemoprophylaxis or changed to a less effective drug.
    - If significant adverse events are experienced because of the chemoprophylactic agent, advice should be sought from an appropriately experienced healthcare professional regarding an effective alternative.
- **If the person discontinues chemoprophylaxis on or after return to UK due to adverse effects:**
  - No additional prophylactic drug is needed if suppressive prophylaxis (chloroquine, doxycycline, mefloquine) is discontinued before completing 4 weeks' dosage post return.
  - The person *must* be warned that they are at increased risk of malaria compared with those who have completed the full regimen.
  - Advice the person to seek urgent medical attention if they become unwell in the first year after return — an urgent blood test for malaria is essential.
- Causal chemoprophylaxis (for example, primaquine) is directed at the liver stages of the malaria parasite, which takes approximately 7 days to develop — this prevents the parasite from infecting red blood cells.
  - These agents need to be continued for 7 days after leaving the malarious area.
- Suppressive chemoprophylaxis (for example chloroquine, doxycycline, and mefloquine) is directed at the red blood cell stage of the malaria parasite.
  - These agents should be continued for 4 weeks after leaving the malarious area.

When should I prescribe standby emergency medication?

- **Standby emergency medication is intended for travellers who believe that they may have malaria** — it is not a replacement for chemoprophylaxis.
  - It is not required for most travellers but should be considered if the person is taking chemoprophylaxis and visiting remote areas which are not within 24 hours of medical facilities.
  - Explain and provide written information on:
    - When to start the medication — standby emergency medication should be started within 24 hours of onset of fever and medical attention sought as soon as possible for a full assessment to exclude any other serious causes of fever.
    - How the medication should be taken (including dosage, interval, and duration of treatment). What to do if vomiting occurs on taking the medication — a second full treatment dose of the antimalarial should be taken if vomiting occurs within 30 minutes of taking it (half-dose if vomiting occurs after 30–60 minutes).
    - Possible adverse effects.

        - How and when to restart chemoprophylaxis — on completion of the standby treatment course, antimalarial chemoprophylaxis should be restarted 1 week after taking the first treatment dose. If quinine is used for standby treatment, mefloquine prophylaxis should be restarted at least 12 hours after the last treatment dose.
    - Explain that standby emergency medication is *not* a replacement for chemoprophylaxis and only a temporary measure.

Prescribing information:

What issues should I consider when prescribing in children?

    - Ensuring that young children take antimalarial drugs can be difficult due to unpleasant taste and a lack of paediatric formulations.
        - It is preferable to avoid breaking tablets but the taste may be masked by crushing tablets and mixing with formula, honey, jam, yoghurt or similar foods.
        - Consult a pharmacist if further advice is required.
    - The dose for children is dependent on the weight and age of the infant or child — weight is a better guide than age.
- Advise nursing mothers that breastfeeding children need their own prophylaxis appropriate for the country to be visited.

Chloroquine

Dose

- Chloroquine is the drug of choice for people travelling to areas with chloroquine-sensitive malaria (chloroquine-resistant malaria is widespread). Chloroquine:
    - Is available as a syrup or tablets and is usually taken as a single dose once a week, on the same day each week.
    - It should be started 1 week before entering an endemic area, continued throughout the time in the area and for 4 weeks after leaving the area.
    - Is considered safe for long-term use but eye examinations every 6 to 12 months should be considered after 6 years' use for malaria prophylaxis.
- **The dose for adults** is 310 mg (two 250 mg chloroquine phosphate tablets) once weekly.
    - Alternatively, the liquid dose is 300 mg once weekly — 30 ml of chloroquine syrup 50 mg/5 ml.
    - For adults who weigh less than 45 kg, the dose is 232.5 mg once weekly.
    - Note: each 250 mg tablet contains 155 mg chloroquine base and syrup contains 50 mg/5 ml chloroquine base.
- Chloroquine can be given from birth.
    - Chloroquine has a bitter taste and may need to be mixed with food to encourage the child to take it.
- **Chloroquine is highly toxic in overdose, particularly in children**.
    - Overdose is extremely hazardous and difficult to treat. Life-threatening features include arrhythmias (which can have a very rapid onset) and convulsions (which can be intractable).
    - If overdose occurs urgent advice from the National Poisons Information Service is essential.

Contraindications and cautions

- **Chloroquine is contraindicated in people:**
  - With epilepsy.
  - With severe gastrointestinal disorders.
  - Taking amiodarone.
- **Chloroquine should be used with caution in**:
  - Acute porphyrias — may precipitate severe symptoms.
  - Diabetes mellitus (may lower blood glucose).
  - G6PD deficiency — there may be a risk of haemolysis.
  - Hepatic impairment — particularly cirrhosis.
  - Neurological disorders.
  - People at risk of prolonged QT interval.
  - People with psoriasis or myasthenia gravis — may exacerbate these conditions; avoid if possible.
  - Renal impairment — chloroquine is only partially excreted by the kidneys and reduction of the dose is not needed except in severe impairment.

Adverse effects

- **Adverse effects include:**
  - Blood — bone marrow failure, aplastic anaemia, agranulocytosis, thrombocytopenia, neutropenia, pancytopenia.
  - Eye — blurred vision, diplopia, macular defects of colour vision, blindness.
  - Gastrointestinal — abdominal pain, constipation, diarrhoea, vomiting and nausea.
  - Psychiatric — insomnia, confusion, depression, aggression, delusion.
  - Skin and subcutaneous tissue — alopecia, erythema multiforme, drug reaction with eosinophilia and systemic symptoms syndrome (DRESS), Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), acute generalised exanthematous pustulosis (AGEP), precipitation of psoriasis, and pruritus.
- **Other adverse effects include:**
  - Hypoglycaemia.
  - Hypotension
  - Myopathy.
  - ECG changes.

Pregnancy

  - The UK Teratology Information Service (UKTIS) found no conclusive evidence (from limited available studies) that chloroquine use in pregnancy is associated with increased risk of structural malformation or other adverse outcomes such as spontaneous abortion, stillbirth or low infant birth weight. It advises that it should not be withheld on the account of pregnancy as the known risks of maternal malaria infection are likely to be greater than any potential risks to the fetus from the drug. This view is also held by the manufacturers.

Breastfeeding

  - Chloroquine is present in breastmilk, but the amount is probably too small to be harmful.

Prescribing mefloquine

Dose

- A checklist should be used prior to mefloquine administration to ensure all contraindications and cautions are identified.
- Mefloquine should be started 2 to 3 weeks before entering a malarious area (to ascertain if there are any adverse effects) and continued until 4 weeks after leaving the area.
- It can be safely used for up to 3 years, but the manufacturer suggests periodic checks on liver function and eye assessments should be taken if mefloquine is used for a prolonged period.
    - Any person taking mefloquine presenting with a visual disorder should be referred to a specialist as chemoprophylaxis may need to be stopped.
- **For adults, the dose is 250 mg once weekly.**
    - For adults who weigh less than 45 kg, the dose is 187.5 mg.

Contraindications and cautions

- **Do not prescribe mefloquine in people with:**
    - A current or previous history of depression, generalized anxiety disorder, psychosis, schizophrenia, suicide attempts, suicidal thoughts, self-endangering behaviour or any other psychiatric disorder, epilepsy or convulsions of any origin.
    - A history of convulsions (if used for standby treatment).
    - A history of Blackwater fever.
    - Severe liver impairment.
- **Also do not prescribe mefloquine to:**
    - Infants under 5 kg weight.
- **Prescribe mefloquine with caution in people with:**
    - Cardiac conduction abnormalities.
    - Epilepsy.
    - Infants aged less than 3 months.
    - Traumatic brain injury.
- **Also consider the following:**
    - Diving — some diving centres do not permit people taking mefloquine to dive. Mefloquine can lower the seizure threshold and its adverse effects could potentially be confused with decompression or narcosis events. It may be best to avoid for diving holidays.
    - Military personnel — the Ministry of Defence has guidelines for malaria prevention specific to military personnel and civilian practitioners asked to give malaria prevention advice for military travellers should liaise with the Defence Medical Services.
    - Pilots — the UK Civil Aviation Authority advises that mefloquine should not be administered to pilots, although there is no evidence that mefloquine impairs function.

Adverse effects

- **Gastrointestinal** — abdominal pain, diarrhoea, nausea, vomiting.
- **Nervous system** — dizziness, headache.
- **Psychiatric** — mefloquine may induce psychiatric symptoms such as anxiety disorders, paranoia, depression, hallucinations and psychosis.

    - Advise people that if these reactions or changes to their mental state occur during mefloquine use, to stop taking it and seek medical advice immediately.
    - Note: adverse reactions may also occur after discontinuation of mefloquine.
- **Other common adverse effects include:**
    - Pruritus.
    - Vertigo.
    - Visual impairment.

Pregnancy:

Public Health England advises that mefloquine can be used in all trimesters for travellers to high-risk areas, but urges caution in the first trimester, although due to the potential severity of falciparum malaria in pregnant women, advises that use is generally justified for areas with high risk of falciparum malaria (such as sub-Saharan Africa).

Breastfeeding

- Mefloquine is present in breastmilk, but there is minimal risk to the infant and it is safe to use in women who are breastfeeding.

Prescribing doxycycline

Dose

- Doxycycline should be started 1 or 2 days before entering an endemic area and be continued until 4 weeks after leaving the area.
- Doxycycline is available as 50 mg or 100 mg capsules and 100 mg dispersible tablets.
- The capsules and solution of dispersible tablets should be swallowed with plenty of fluid in either the resting or standing position. The person should be advised not to lie down for at least 1 hour after ingestion to reduce the likelihood of oesophageal irritation and ulceration.
    - If gastric irritation occurs, it is recommended that doxycycline is taken with food or milk.
- **The dose for adults and children aged over 12 years is 100 mg once daily.**
    - Doxycycline is contraindicated in children aged under 12 years irrespective of weight.

Pregnancy

- The manufacturer advises that use during pregnancy is contraindicated.

Breastfeeding

- The Advisory Committee on Malaria Prevention (ACMP) advises that doxycycline should not be used in breastfeeding unless other options are unsuitable, and its use is felt to be essential.

TOPIC: LEISHMANIASIS

Leishmaniasis is caused by the intracellular protozoa Leishmania, which are spread by the bites of sandflies. Cutaneous, mucocutaneous leishmaniasis and visceral forms are seen

Cutaneous leishmaniasis:

- Caused by Leishmania tropica or Leishmania mexicana
- Crusted lesion at the site of bite
- There may be an underlying ulcer
- It is typically diagnosed by doing a punch biopsy from the edge of the lesion allowing for both histology and culture
- Cutaneous leishmaniasis acquired in South or Central America merits treatment due to the risk of mucocutaneous leishmaniasis whereas disease acquired in Africa or India can be managed more conservatively

Mucocutaneous leishmaniasis:

- Caused by Leishmania braziliensis
- Skin lesions may spread to involve mucosae of nose, pharynx etc

Mostly caused by Leishmania donovani

- Occurs in the Mediterranean, Asia, South America, Africa
- Fever, sweats, rigors
- Massive splenomegaly. hepatomegaly
- Poor appetite*, weight loss
- Grey skin - 'kala-azar' means black sickness
- Pancytopaenia secondary to hypersplenism
- The gold standard for diagnosis is bone marrow or splenic aspirate

*Occasionally, patients may report increased appetite with paradoxical weight loss.

TOPIC: LEPTOSPIROSIS

Leptospirosis is caused by the spirochaete Leptospira interrogans (serogroup L. icterohaemorrhagiae), classically being spread by contact with infected rat urine.

Epidemiology:

- Leptospirosis is commonly seen in questions referring to sewage workers, farmers, vets or people who work in an abattoir
- However, on an international level, leptospirosis is far more common in the tropics so should be considered in the returning traveller

Weil's disease should always be considered in high-risk patients with hepatorenal failure

Features:

- The early phase is due to bacteraemia and lasts around a week
    - May be mild or subclinical
    - Fever
    - Flu-like symptoms
    - Subconjunctival suffusion (redness)/haemorrhage
- Second immune phase may lead to more severe disease (Weil's disease)
    - Acute kidney injury (seen in 50% of patients)
    - Hepatitis: jaundice, hepatomegaly
    - Aseptic meningitis

Investigation:

- Serology: antibodies to Leptospira develop after about 7 days
- PCR
- Culture
    - Growth may take several weeks so limits usefulness in diagnosis
    - Blood and CSF samples are generally positive for the first 10 days
    - Urine cultures become positive during the second week of illness

Management:

- High dose benzylpenicillin or doxycycline

TOPIC: SCHISTOSOMIASIS

Schistosomiasis, or bilharzia, is a parasitic flatworm infection. The three main species of schistosome are *S. mansoni*, *S. japonicum* and *S. haematobium*.

Acute infections

Acute symptoms typically only develop in people who travel to endemic areas, as they don't have any immunity to the worms.

Acute manifestations may include:

- Swimmers' itch
- Acute schistosomiasis syndrome (Katayama fever)
    - Fever
    - Urticaria/angioedema
    - Arthralgia/myalgia
    - Cough
    - Diarrhoea
    - Eosinophilia

Chronic infections

Schistosoma haematobium
These worms deposit egg clusters (pseudopapillomas) in the bladder, causing inflammation. The calcification seen on x-ray is actually calcification of the egg clusters, not the bladder itself. Depending on the site of these pseudopapillomas in the bladder, they can cause an obstructive uropathy and kidney damage.
This typically presents as a 'swimmer's itch' in patients who have recently returned from Africa.
Schistosoma haematobium is a risk factor for squamous cell bladder cancer.

Features:

- Frequency
- Haematuria
- Bladder calcification

Investigation:

- For asymptomatic patients, serum schistosome antibodies are generally preferred
- For symptomatic patients the gold standard for diagnosis is urine or stool microscopy looking for eggs.

Management

- Single oral dose of praziquantel

Schistosoma mansoni and Schistosoma japonicum

These worms mature in the liver and then travel through the portal system to inhabit the distal colon. Their presence in the portal system can lead to progressive hepatomegaly and splenomegaly due to portal vein congestion.

These species can also lead to complications of liver cirrhosis, variceal disease and cor pulmonale.

Schistosoma intercalatum and Schistosoma mekongi

These are less prevalent than the other three forms, but are both attributed to intestinal schistosomiasis.

Investigations:

- Microscopic examination of stool or urine is the gold standard for diagnosis but requires the adult worms to be producing eggs.
- Serology can diagnose less advanced infections
    1. Antigen detection is used in endemic areas and antibody tests elsewhere. It usually takes 4-8 weeks for seroconversion to occur, although it can be up to 22 weeks and serology remains positive for two years after eradication.
    2. The antibody test cannot be used to differentiate active and past illness (therefore, it is not useful in endemic areas) and does not allow quantification of the egg burden.
- FBC shows eosinophilia and anemia.
- Renal function may be impaired if the urinary tract is obstructed.

Management:

Emergency treatment may be required for acute complications - eg, acute intestinal bleeding.

Drugs:

In the early days the drugs were possibly more toxic than the disease but the 1970s saw the advent of safer drugs:

- Praziquantel is the drug of choice in most cases. Praziquantel is effective for urinary schistosomiasis and has few adverse events.
- The World Health Organization (WHO) believes that praziquantel is safe in pregnancy and lactation and in children under the age of 24 months.
- Praziquantel paralyses adult worms with great rapidity but it has no effect on eggs or immature worms. Follow-up at 4-6 weeks is recommended with repeat of treatment in 6-12 weeks.
- Oxamniquine is the only alternative; it is used in intestinal infections in Africa and South America, where praziquantel is less effective. The use of oxamniquine is declining but is effective.
- Metrifonate, which was effective for urinary infections, has been withdrawn from the market.
- In acute Katayama fever, corticosteroids are very important to subdue the hypersensitivity reaction.
- Corticosteroids and anticonvulsants may be needed as adjuvants to praziquantel in neuroschistosomiasis.

TOPIC: ORF

Orf is generally a condition found in sheep and goats although it can be transmitted to humans. It is caused by the parapox virus.

In animals:

- 'Scabby' lesions around the mouth and nose

In humans:

- Generally, affects the hands and arms
- Initially small, raised, red-blue papules
- Later may increase in size to 2-3 cm and become flat-topped and haemorrhagic

Solitary orf nodule on the finger.

Early, inflamed orf nodule mimicking paronychia.

Multiple orf nodules on the forearm.

Nodule with gray-white center and red outer halo on the hand.

TOPIC: BED BUGS

Bedbugs are blood-sucking insects which feed from humans. They often live in bedding, beds and in cracks and crevices of surrounding walls and furniture. They may cause skin reactions and psychological effects, but are not known to transmit disease.

Bed bugs describe a variety of clinical problems including itchy skin rashes, bites and allergic symptoms secondary to infestation with Cimex hemipteru (Tropical region) or *Cimex lectularius (Middle East)*

Presentation:

As bedbugs feed at night and inject an anaesthetic when they bite, the initial bite is usually not felt and most patients have no reaction to them. The biting process takes up to ten minutes but the person being bitten is usually unaware of it.

Three to four bites forming a line or a curve on the body (usually areas exposed during sleeping such as face, neck, arms) are very suggestive of bedbug bites.

Isolated pruritus, papules or nodules can occur.

The onset of symptoms may be delayed (up to 11 days later)

Management

- Washing at 60°C or dry-cleaning affected linen.
- Tumble drying affected linen on a hot cycle for at least half an hour.
- Disposal of highly infested items, together with physical removal of bedbugs and mattress covers and vacuuming, are usually recommended.
- Encasing mattresses in a protective covering (such as those used for people with allergies) may be an option; however, the seal needs to be tight, with no gaps in the encasement material.

- The symptoms of bedbug bites are often treated with topical steroids (such a hydrocortisone 1%) once or twice a day for up to seven days. Antihistamines may be beneficial for those with pruritus.
- Definitive management is through a pest management company and fumigation of the house.

TOPIC: BODY FLUID SPILLAGES

Generally speaking, **hypochlorite** (chlorine in oxidation state +1) is recommended for the management of body fluid spillage.

It is often used in the granular or powder form which is then spread over the affected area.

Chlorine-releasing agents can be a hazard especially if used in large volumes in confined spaces or if mixed with urine. Ensuring adequate ventilation to the area is therefore important.

Other points:

- Personal protective equipment (plastic aprons, gloves, mask, eye protection) is recommended for staff cleaning up body fluids
- Mops should not be used

In some NHS trusts **Virkon** is recommended. It is a multipurpose disinfectant which contains oxone, potassium peroxymonosulphate, sodium dodecylbenzenesulfonate, sulphamic acid; and inorganic buffers.

Virkon is thought to be effective against HIV, hepatitis and MRSA.

TOPIC: DENGUE FEVER

Dengue fever is a viral infection that can progress to viral haemorrhagic fever (other examples include yellow fever, Lassa fever, Ebola).

Aetiology

- dengue virus is a RNA virus of the genus Flavivirus
- transmitted by the **Aedes aegypti mosquito, incubation period of 7days.**

Clinical features

Patients with dengue fever can be classified as follows:

1) **dengue fever**
    a. Without warning signs
        - Fever, headache (often retro-orbital)
        - Myalgia, bone pain and arthralgia ('break-bone fever')
        - Facial flushing (dengue), maculopapular rash, pleuritic pain
        - Haemorrhagic manifestations e.g. positive tourniquet test, petechiae, purpura/ecchymosis, epistaxis

    b. With warning signs
        - Abdominal pain, hepatomegaly, persistent vomiting
        - Clinical fluid accumulation (ascites, pleural effusion)

2) **severe dengue (dengue haemorrhagic fever)**

- This is a form of disseminated intravascular coagulation (DIC) resulting in:
    - Thrombocytopenia, spontaneous bleeding
- Around 20-30% of these patients go on to develop dengue shock syndrome (DSS)

Investigations:

- Typically, blood results
    - Leukopenia, thrombocytopenia, raised Liver enzymes
    - Prolongation of APTT and PT
- Diagnostic tests
    - Serology, Nucleic acid amplification tests for viral RNA
    - NS1 antigen test

Management:

- No specific treatment, Dengue is typically a self-limiting flu-like disease.
- Rest, stay hydrated, and seek medical advice
- Supportive care; fluid resuscitation, painkillers, antipyretics
- NSAIDs should be avoided due to risk of haemorrhage
- Haemorrhage and shock require FFP and platelets.
- The WHO recommends that countries should consider introduction of the dengue vaccine CYD-TDV for people living in endemic areas, 9-45 years of age, and who have had at least one episode of dengue virus infection in the past.

TOPIC: CUTANEOUS LARVA MIGRANS

Cutaneous larva migrans is a roundworm, usually hookworm, infection (*Ancylostoma braziliense is the most common), the host are cats, dogs and other animals. Most commonly in tropical area*

Risk factors

- Sunbathing and walking on the beach barefoot.
- Children may have been playing in sandpits.
- Soil under housing and at construction sites may also be contaminated.

Features

- The feet, toe web spaces, hands, knees, abdomen, anogenital region and buttocks are the most common sites of hookworm penetration.
- Penetration can cause tingling or prickling of the skin at that site within 30 minutes.
- Erythematous skin eruption at the site of hookworm entry.
- Skin lesions can cause intense pruritis.
- Once the larvae begin migration, raised, pink or flesh-coloured, tortuous, snake-like tracks that are 2-3 mm wide form about 3-4 cm from the site of penetration. This is due to an allergic immune response to the larvae or its byproducts.

Investigations

- Diagnosis is usually clinical
- Skin biopsy ahead of the leading tract may show a larva in a burrow and inflammatory infiltrate
- Optical coherence tomography can identify the larvae in the epidermis and allow direct removal

Management

Self-limiting infection; Most cases will resolve **within 4-8 weeks** because the larvae cannot penetrate the dermis, cannot reproduce, and eventually die.

- Treatment with **antehelmintics** (**albendazole** or **ivermectin**) can shorten the disease course, ease pruritus and lead to resolution of skin tracts **within one week.**
- Liquid nitrogen cryotherapy for the progressive end of the larval burrow is another alternative treatment. However, it is painful and multiple treatments are usually needed
- Antihistamines and topical corticosteroids can be added to the treatment to help pruritus.

TOPIC: BRUCELLOSIS

Brucellosis (also known as 'undulant fever,' 'Mediterranean fever,' or 'Malta fever') is a zoonotic infection transmitted to humans from infected animals. More common in the Middle East and in farmers, vets and abattoir workers

Four major species cause infection in humans: B. melitensis (sheep), B. abortus (cattle), B. canis (dogs) and B. suis (pigs).

Transmission:

- Consumption of infected food (such as unpasteurized dairy products)
- Contact of skin with infected tissue or fluids
- Inhalation of infected aerosolized particles

Brucellosis has an incubation period **2 - 6 weeks**

Features:

- Non-specific: fever, malaise, night sweats
- Arthralgia/arthritis
- Hepatosplenomegaly
- Sacroiliitis: spinal tenderness
- Lab finding; leukopenia, lymphocytosis, elevated Liver enzymes

Complications: osteomyelitis, infective endocarditis, meningoencephalitis, orchitis

Diagnosis:

A definitive diagnosis of brucellosis may be made via either of the following:

- Culture of the organism from blood, body fluids (urine, cerebrospinal fluid, synovial fluid, and pleural fluid), or tissue (such as bone marrow or liver biopsy). However, this method is time consuming and hazardous
- A fourfold or greater rise in *Brucella* antibody titer between acute and convalescent phase serum specimens obtained ≥2 weeks apart

A presumptive diagnosis of brucellosis may be made via either of the following:

- *Brucella* total antibody titer ≥1:160 by standard tube agglutination test (SAT) in serum specimen obtained after onset of symptoms or Rose Bengal agglutination test (RBT)
- Detection of *Brucella* DNA in a clinical specimen by polymerase chain reaction assay

Management:

Doxycycline oral for 6 weeks plus streptomycin (parenteral) for the first 14 to 21 days

Or

Doxycycline plus rifampcin both for 6 weeks orally.

TOPIC: SPLENECTOMY

Following a splenectomy, patients are particularly at risk from pneumococcus, Haemophilus, meningococcus and Capnocytophaga canimorsus* infections (usually from dog bite)

Vaccination:

- If elective, should be done 2 weeks prior to operation
- Hib, meningitis A & C
- Annual influenza vaccination
- Pneumococcal vaccine every 5 years

Antibiotic prophylaxis:

- Penicillin V: unfortunately, clear guidelines do not exist of how long antibiotic prophylaxis should be continued. It is generally accepted though that penicillin should be continued for at least 2 years and at least until the patient is 16 years of age, although the majority of patients are usually put on antibiotic prophylaxis for life.

Surgical aspects:

Indications:

- Trauma: 1/4 are iatrogenic
- Spontaneous rupture: EBV
- Hypersplenism: hereditary spherocytosis or elliptocytosis etc
- Malignancy: lymphoma or leukaemia
- Splenic cysts, hydatid cysts, splenic abscesses

Elective splenectomy:

- Elective splenectomy is a very different operation from that performed in the emergency setting. The spleen is often large (sometimes massive)
- Most cases can be performed laparoscopically. The spleen will often be macerated inside a specimen bag to facilitate extraction.

Complications:

- Haemorrhage (may be early and either from short gastrics or splenic hilar vessels
- Pancreatic fistula (from iatrogenic damage to pancreatic tail)
- Thrombocytosis: prophylactic aspirin
- Encapsulated bacteria infection e.g. *Strep. pneumoniae*, *Haemophilus influenzae* and *Neisseria meningitidis*

Post-splenectomy changes:

- Platelets will rise first (therefore in ITP should be given after splenic artery clamped)
- Blood film will change over following weeks, Howell-Jolly bodies will appear
- Other blood film changes include target cells and Pappenheimer bodies
- Increased risk of post-splenectomy sepsis, therefore prophylactic antibiotics and pneumococcal vaccine should be given.

TOPIC: STRONGYLOIDES STERCORALIS

Strongyloides stercoralis is a human parasitic nematode worm. The larvae are present in soil and gain access to the body by penetrating the skin. Infection with *Strongyloides stercoralis* causes strongyloidiasis.

Features:

- Diarrhoea
- Abdominal pain/bloating
- Papulovesicular lesions where the skin has been penetrated by infective larvae e.g. soles of feet and buttocks
- Larva currens: pruritic, linear, urticarial rash
- If the larvae migrate to the lungs a pneumonitis similar to Loeffler's syndrome may be triggered

Treatment:

- Ivermectin and albendazole are used

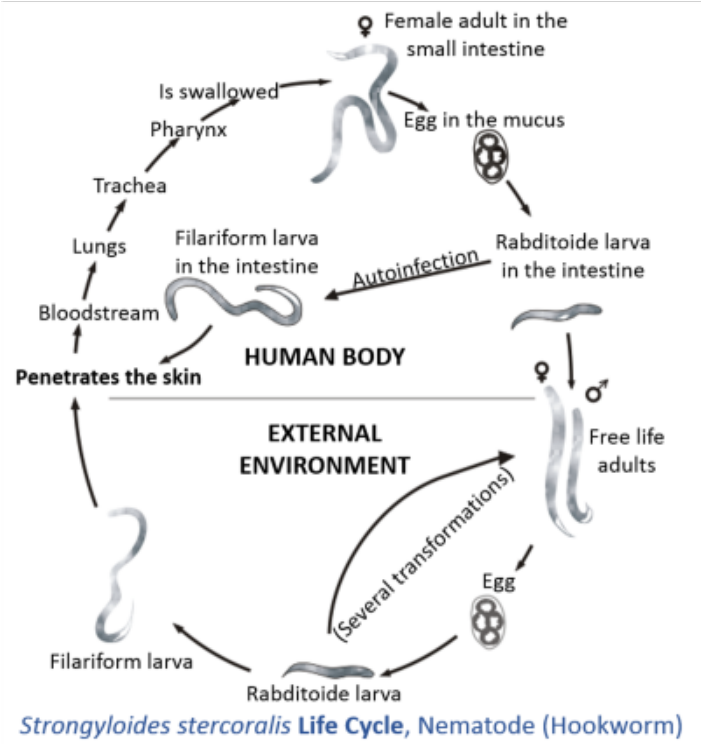


Diagram showing the lifecycle of *Strongyloides stercoralis*

TOPIC: TRYPANOSOMIASIS

Two main form of this protozoal disease are recognised - African trypanosomiasis (sleeping sickness) and American trypanosomiasis (Chagas' disease).

Two forms of **African trypanosomiasis**, or **sleeping sickness**, are seen - *Trypanosoma gambiense* in West Africa and *Trypanosoma rhodesiense* in East Africa. Both types are spread by the tsetse fly. *Trypanosoma rhodesiense* tends to follow a more acute course.

Clinical features include:

- Trypanosoma chancre - painless subcutaneous nodule at site of infection
- Intermittent fever
- Enlargement of posterior cervical lymph nodes
- Later: central nervous system involvement e.g. somnolence, headaches, mood changes, meningoencephalitis

Management:

- Early disease: IV pentamidine or suramin
- Later disease or central nervous system involvement: IV melarsoprol

American trypanosomiasis, or **Chagas' disease**, is caused by the protozoan *Trypanosoma cruzi*. The vast majority of patients (95%) are asymptomatic in the acute phase although a chagoma (an erythematous nodule at site of infection) and periorbital oedema are sometimes seen. Chronic Chagas' disease mainly affects the heart and gastrointestinal tract

- Myocarditis may lead to dilated cardiomyopathy (with apical atrophy) and arrhythmias
- Gastrointestinal features includes megaoesophagus and megacolon causing dysphagia and constipation

Management:

- Treatment is most effective in the acute phase using azole or nitro derivatives such as benznidazole or nifurtimox
- chronic disease management involves treating the complications e.g., heart failure

TOPIC: ZIKA VIRUS

Zika is a mosquito-borne infection caused by Zika virus, a member of the genus flavivirus and family Flaviviridae. It was first isolated from a monkey in the Zika forest in Uganda in 1947.

Transmission is usually via the bite of an infected Aedes mosquito, although a small number of cases of sexual transmission have been reported. There is increasing evidence of transmission via the placenta from mother to fetus

The majority of people infected with Zika virus have no symptoms. For those with symptoms, Zika virus tends to cause a mild, short-lived (2 to 7 days) febrile disease. Signs and symptoms suggestive of Zika virus infection may include a combination of the following:

- Fever
- Rash
- Arthralgia/arthritis
- Conjunctivitis
- Myalgia
- Headache
- Retro-orbital pain
- Pruritus

Serious complications in adults are not common, although the virus has been associated with Guillain-Barre syndrome. Scientific consensus however has linked Zika with microcephaly and other congenital abnormalities, which has led the World Health Organisation (WHO) to declare a Public Health Emergency of International Concern (PHEIC).

TOPIC: ANTIBIOTIC GUIDELINES

Based on British National Formulary (BNF) current guidelines

Respiratory System

Condition	Recommended treatment
Exacerbations of chronic bronchitis	Amoxicillin or tetracycline or clarithromycin
Uncomplicated community-acquired pneumonia	Amoxicillin (Doxycycline or clarithromycin in penicillin allergic, add flucloxacillin if staphylococci suspected e.g. In influenza)
Pneumonia possibly caused by atypical pathogens	Clarithromycin
Hospital-acquired pneumonia	Within 5 days of admission: co-amoxiclav or cefuroxime More than 5 days after admission: piperacillin with tazobactam OR a broad-spectrum cephalosporin (e.g. ceftazidime) OR a quinolone (e.g. ciprofloxacin)

Urinary Tract

Condition	Recommended treatment
Lower urinary tract infection	Trimethoprim or nitrofurantoin. Alternative: amoxicillin or cephalosporin
Acute pyelonephritis	Broad-spectrum cephalosporin or quinolone
Acute prostatitis	Quinolone or trimethoprim

Skin

Condition	Recommended treatment
Impetigo	Topical hydrogen peroxide, oral flucloxacillin or erythromycin if widespread
Cellulitis	Flucloxacillin (clarithromycin, erythromycin or doxycycline if penicillin-allergic)
Cellulitis (near the eyes or nose)	Co-amoxiclav (clarithromycin, + metronidazole if penicillin-allergic)
Erysipelas	Flucloxacillin (clarithromycin, erythromycin or doxycycline if penicillin-allergic)
Animal or human bite	Co-amoxiclav (doxycycline + metronidazole if penicillin-allergic)
Mastitis during breast-feeding	Flucloxacillin

Ear, Nose & Throat

Condition	Recommended treatment
Throat infections	Phenoxymethylpenicillin (erythromycin alone if penicillin-allergic)
Sinusitis	Phenoxymethylpenicillin
Otitis media	Amoxicillin (erythromycin if penicillin-allergic)
Otitis externa	Flucloxacillin (erythromycin if penicillin-allergic)
Periapical or periodontal abscess	Amoxicillin
Gingivitis: acute necrotising ulcerative	Metronidazole

Genital system

Condition	Recommended treatment
Gonorrhoea	Intramuscular ceftriaxone
Chlamydia	Doxycycline or azithromycin
Pelvic inflammatory disease	Oral ofloxacin + oral metronidazole or intramuscular ceftriaxone + oral doxycycline + oral metronidazole
Syphilis	Benzathine benzylpenicillin or doxycycline or erythromycin
Bacterial vaginosis	Oral or topical metronidazole or topical clindamycin

Gastrointestinal

Condition	Recommended treatment
Clostridioum difficile	First episode: oral vancomycin Second or subsequent episode of infection: oral fidaxomicin
Campylobacter enteritis	Clarithromycin
Salmonella (non-typhoid)	Ciprofloxacin
Shigellosis	Ciprofloxacin

TOPIC: CLASSIFICATION OF BACTERIA

Remember

- Gram-positive cocci = staphylococci + streptococci (including enterococci)
- Gram-negative cocci = Neisseria meningitidis + Neisseria gonorrhoeae, also Moraxella catarrhalis

Therefore, only a small list of Gram-positive rods (bacilli) need to be memorised to categorise all bacteria - mnemonic = ABCD L

- Actinomyces
- Bacillus anthracis (anthrax)
- Clostridium
- Diphtheria: Corynebacterium diphtheriae
- Listeria monocytogenes

Remaining organisms are Gram-negative rods, e.g.:

- Escherichia coli
- Haemophilus influenzae
- Pseudomonas aeruginosa
- Salmonella sp.
- Shigella sp.
- Campylobacter jejuni